Technical Supplement: The Structure of Admissible Physics

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Companion to Paper 2: "The Structure of Admissible Physics"

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Abstract

This supplement provides self-contained proofs for every theorem stated in Paper 2 whose derivation depends on material from companion papers or requires detail beyond what the main text presents. The goal is that a reader with access to Paper 1, its Technical Supplement, and this document can verify the structural chain from the *Principle of Least Enforcement Cost* (PLEC) — the canonical variational formulation of the enforceability-of-distinction foundation, decomposed into four structurally necessary components (A1 finite capacity, MD positive floor, A2 argmin, BW non-degeneracy) — to the Standard Model gauge group and fermion content without consulting any other paper in the series. The cosmological density fractions are derived conditionally: the structural partition $42+3+16 = 61$ is proved here; its identification with the Friedmann-equation density fractions is conditional on the horizon identification established in Paper 6. The one-loop β-function sign is verified here using an imported formula; its derivation from capacity arguments is in Paper 4A.

Part I establishes the core derivation chain: cost function uniqueness (L_{Cauchy}), the bridge from superadditivity to gauge structure ($L_\Delta \to T_{\text{alg}} \to$ gauge identification), the three carrier prerequisites that close the Theorem R dependency on formerly external lemmas (T_{7B}, $L_{\text{irr,uniform}}$, $B1'$), and the derivation of Lorentzian spacetime from the cost kernel: causal ordering from irreversibility, $d = 4$ from Lovelock uniqueness and DOF minimality, and the Einstein field equations (Δ_{sig}, T_8, $T_{9,\text{grav}}$). Part II derives the gauge template and matter content: the exhaustive Lie algebra classification ($L_{\text{gauge,unique}}$), the 1,680-template fermion scan with seven formally specified filters (T_{field}), anomaly cancellation, and hypercharge uniqueness. Part III covers capacity counting and cosmology: the MECE partition, horizon equipartition (L_{equip}), the bridge to the Friedmann equations, saturation independence, and uncertainty analysis. Appendix A compares the framework with existing reconstruction and "SM from principles" programs.

All upstream results (the four PLEC components A1, MD, A2, BW; constitutive semantics SP, K3, FD1–FD6; and the full Paper 1 derivation chain) are imported from the Paper 1 Technical Supplement and are restated in the glossary below. Appendix C provides complete, self-contained proofs of all ten imported results, so that no external document is needed to verify any theorem in this supplement. No result depends on Papers 3–7; forward references to those papers appear only as scope pointers and are flagged as such.

What Is Assumed, What Is Proved, What Is Deferred

Imported from Paper 1 Supplement (proved there; re-proved in Appendix C for self-containment). The first four items are the structurally necessary components of PLEC (none derivable from the others; essentiality proofs in Paper 1 Supplement, §1).

1. **A1** (PLEC upper bound): enforcement capacity is finite and positive.
2. **MD** (PLEC positive floor): uniform positive cost per unit of enforcement complexity.
3. **A2** (PLEC argmin selection): the realized configuration is the arg min of total enforcement cost over the admissible set.
4. **BW** (PLEC non-degeneracy): the cost spectrum witnesses a budget-window triple.

5. **SP, K3, FD1–FD6**: constitutive admissibility semantics.
6. T_{form}: finite-dimensional substrate.
7. T_{embed}: canonical vector-space embedding.
8. T_{sep}: sector decomposition $S_\Gamma = \bigoplus_d M_d \oplus \Pi$.
9. L_{ε^*}: uniform cost floor $\varepsilon \geq \varepsilon^* > 0$.
10. L_Δ: superadditivity of co-located enforcement costs.
11. L_{blk}, L_Π: direct-sum failure and codespace rotation.
12. T_{alg}: finite-dimensional noncommutative $*$-algebra.
13. T_{GNS}: GNS Hilbert-space representation.
14. T2a–T2c: Wedderburn–Artin decomposition, field selection $\mathbb{F} = \mathbb{C}$.
15. T_{Born}: Born rule. T_M: interface monogamy.
16. L_{loc}: budget additivity for independent interfaces.
17. L_{irr}: irreversibility from A1 + L_{nc} + L_{loc}.
18. T3: locality $\to$ gauge structure (Skolem–Noether + Doplicher–Roberts).
19. L_ω: sector-orthogonal bilinear form.
20. T_{adj}: self-adjoint enforcement projections.

Core results proved in this supplement (self-contained, no downstream dependence).

1. L_{Cauchy}: cost function uniqueness (§1).
2. Local support and commuting algebras from disjoint substrates (Definition 2.1, Lemma 2.2).
3. Gauge identification theorem: internal vs. kinematic automorphisms; $G = \text{Inn}(\mathcal{A}) = \prod_k \text{PU}(n_k)$ (§2.2).
4. Exact sequence $1 \to Z \to \text{U}(\mathcal{A}) \to \text{Inn}(\mathcal{A}) \to 1$ and the source exclusivity of abelian factors from T_{rel} (Proposition 2.7, Lemma 2.13).
5. Unique abelian factor: structural theorem (template-independent, Proposition 2.12) and SM specialization (Theorem 2.18).
6. Lift from $\text{PU}(n)$ to $\text{SU}(n)$ and global structure $[\text{SU}(3)\times\text{SU}(2)\times\text{U}(1)_Y]/\mathbb{Z}_6$ (Propositions 2.19, 2.20).
7. T_{7B}: positive-definite cost kernel at a single interface (§3.1).
8. $L_{\text{irr,uniform}}$: gauge-sector irreversibility at saturated interfaces (§3.2).
9. $B1'$: complex carrier from oriented composite robustness (§3.3).
10. $L_{\text{gauge,unique}}$: gauge template uniqueness (§6).
11. T_{field}: fermion content by exhaustive 1,680-template scan with seven filters (§7).
12. $L_{\text{hypercharge}}$: hypercharge uniqueness from anomaly system (§8).
13. L_{count}, P_{exhaust}: capacity counting and MECE partition (§10, §11).

Downstream results: proved here conditional on inter-paper bridges.
These results use the core chain above together with imported mathematical theorems (Hawking–King–McCarthy, Malament, Kolmogorov, Lovelock) whose exact hypotheses are verified in the text. They are included to show the derivation path, but their full rigor depends on structure established in companion papers; the epistemic status of each is flagged explicitly.

1. Δ_{sig}: Lorentzian signature from causal structure (§5.3). *Status:* hypothesis verification for HKM/Malament provided; the causal partial order $\prec$ is derived from L_{irr}; the construction of the spacetime metric from the cost kernel is presented (§5.4) but the full dynamical treatment requires Paper 6.
2. T_8: spacetime dimension $d = 4$ (§5.5). *Status:* four-condition proof complete; the argument uses the linearized GR DOF formula (imported) and Lovelock's theorem (hypotheses verified in Lemma 5.14).
3. $T_{9,\text{grav}}$: Einstein field equations (§5.6). *Status:* Lovelock hypotheses verified; the

five admissibility conditions are derived; the full coupling to matter stress-energy requires Paper 6.

4. L_{equip}: horizon equipartition (§12). *Status:* the maximum-entropy argument is self-contained; the identification of the enforcement horizon with the cosmological de Sitter horizon is established in Paper 6. Without that bridge, the cosmological prediction $\Omega_\Lambda = 42/61$ is conditional.
5. $L_{\text{sat,part}}$: saturation independence (§13). *Status:* self-contained, but its cosmological application requires L_{equip}'s horizon bridge.

Not proved here (honest flags).

1. $L_{\text{irr,uniform}}$ is a *sufficient-condition* theorem: it guarantees gauge-sector irreversibility at interfaces satisfying the saturation bound (7). It does not prove irreversibility at *all* interfaces; the corollary shows it holds generically at near-capacity interfaces.
2. Quantitative β-function derivation: structural necessity of asymptotic freedom is proved here; the one-loop coefficient is derived in Paper 4A ($L_{\text{beta,capacity}}$). A schematic sign calculation is included (§4).
3. Cosmological horizon identification: the structural partition $42 + 3 + 16 = 61$ is proved here; the identification of the enforcement horizon with the cosmological de Sitter horizon is established in Paper 6.
4. Mass hierarchy, CKM/PMNS mixing, CP violation: require the generation structure derived in Papers 4A/4B.
5. Confinement scale Λ_{QCD}, string tension, hadronization dynamics: require quantitative RG treatment (Paper 4A).

Framing convention: "forces" and "selects" denote uniqueness, not processes

The supplement's theorems and proofs use the verbs "forces," "selects," and "rejects" as standard mathematical shorthand for uniqueness results, arg min properties, and inadmissibility results respectively. Under the APF ontology, these verbs do not denote processes: nothing is forced by an agent, nothing is selected by a mechanism, nothing is rejected by a filter. Each theorem asserts a property of the set of admissible configurations—that it has a unique member, a minimum member, or no member satisfying some condition. For example, "capacity minimality selects $N_c = 3$" is shorthand for "$N_c = 3$ is the arg min of dim(G) over R1–R3-admissible color groups," and the alternative values (SU(4), SU(5), E_6, ...) are *admissible* in the sense that each has cost below the capacity bound, but have higher cost than SU(3) and are therefore not what is. The companion note *What Is, Is Minimum-Cost Admissible* spells this out in full. Readers may substitute the descriptive reading throughout the supplement without any change to the proofs.

Self-contained Standard Model datum. Section 8 derives the unique anomaly-free hypercharge assignments for one generation of the Standard Model fermion template purely from the tools assembled in this supplement (gauge template from §6, fermion content from §7, anomaly conditions from §8). The result ($Y_Q = 1/6$, $Y_u = 2/3$, $Y_d = -1/3$, $Y_L = -1/2$, $Y_e = -1$) with charge quantization as a derived consequence constitutes a nontrivial, experimentally verified prediction obtained without any input beyond A1 and the derivation chain.

Glossary of imported objects. Each entry gives the object's name, one-sentence definition, source in the Paper 1 Supplement (P1S), and where it is first used in this supplement.

Symbol	Definition	Source	Used at
Γ	Enforcement interface (physical boundary)	P1S FD1	Throughout
S_Γ	Substrate: set of configurations at Γ	P1S FD1	Throughout
$\mathcal{D}(\Gamma)$	Enforceable distinctions (binary predicates, $\varepsilon > 0$)	P1S FD1	§1
$\varepsilon(d)$	Enforcement cost: $\inf_{\mathcal{P}(d)} \kappa$	P1S FD4	§1
$C(\Gamma)$	Total enforcement budget at Γ	P1S A1	Throughout
ε^*	Uniform cost floor: $\varepsilon(d) \geq \varepsilon^*$ for all d	P1S L_{ε^*}	§1
M_d	Anchor set: minimal perturbation-support subspace	P1S FD3	§2
Π	Pool: $S_\Gamma \ominus \bigoplus_d M_d$	P1S T_{sep}	§2
E_d	Enforcement projection: B-orth. proj. onto M_d	P1S FD5/T_{adj}	§2
$B(\cdot,\cdot)$	Sector-orthogonal bilinear form on V	P1S L_ω	§3.1
$\mathcal{A}$	Enforcement algebra: $\text{Alg}_{\mathbb{R}}(\{E_d, E_{\{d_i,d_j\}}\})$	P1S O3	§2.2
ω	Faithful positive state on $\mathcal{A}$	P1S O4	§2.2
$\mathcal{H}_\omega$	GNS Hilbert space from $(\mathcal{A}, \omega)$	P1S T_{GNS}	§2.2
$\Delta(d_1, d_2)$	Superadditive surplus: $\varepsilon(\{d_1, d_2\}) - \varepsilon(d_1) - \varepsilon(d_2)$	P1S L_Δ	§2

APF $\leftrightarrow$ standard terminology.

APF term	Standard / GPT equivalent
Distinction d	Effect / binary test / observable outcome
Enforcement projection E_d	Sharp effect / compression / filter
Pool Π	Shared latent support / coupling sector
Superadditivity surplus Δ	Nonclassicality witness / coupling measure
Sector decomposition (T_{sep})	Face decomposition of state space
Enforcement algebra $\mathcal{A}$	Observable algebra / C^*-algebra of observables
GNS representation	Gelfand–Naimark–Segal construction

Dependency table. Each row gives a theorem proved in this supplement, its inputs (all previously proved or imported), and the downstream results that depend on it.

Theorem	Inputs	Used by
L_{Cauchy} (§1)	A1, L_{loc}, L_{ε^*}	$L_{\text{gauge,unique}}$ (cost ranking)
Gauge ident. (§2.2)	T_{alg}, T_{sep}, L_{loc}, T3	Theorem R (gauge = internal Aut)
T_{7B} (§3.1)	A1, K2, K3, SP, L_ω, T_{adj}	$L_{\text{irr,uniform}}$
$L_{\text{irr,uniform}}$ (§3.2)	L_{loc}, L_{nc}, L_{irr}, T_{7B}, saturation bound	Theorem R (R2 chain)
$B1'$ (§3.3)	A1 (minimality), T_M, rep. theory	Theorem R (R1 chain)
$L_{\text{gauge,unique}}$ (§6)	Theorem R, L_{Cauchy}, Lie classif.	T_{field}, L_{count}
T_{field} (§7)	$L_{\text{gauge,unique}}$, AF, anomalies	L_{count}, P_{exhaust}
$L_{\text{hypercharge}}$ (§8)	T_{field}, anomaly conditions	L_{count} (charge quantization)
L_{count} (§10)	T_{field}, T_{Higgs}, $L_{\text{gauge,unique}}$	P_{exhaust}, L_{equip}
P_{exhaust} (§11)	L_{count}, T_3, T_{conf}	L_{equip}
L_{equip} (§12)	P_{exhaust}, L_{irr}, L_{ε^*}	Ω_Λ, Ω_m predictions
$L_{\text{sat,part}}$ (§13)	P_{exhaust}, L_{equip}	Robustness of predictions

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Part I
Core Derivations

1 Cost function uniqueness (L_{Cauchy})

The enforcement cost function F maps channel count to capacity committed. Paper 2 §4 states the uniqueness result; this section provides the full formal treatment, specifying domain, regularity, and operational meaning of each condition.

1.1 Domain and operational definitions

The enforcement cost function F maps the *number of enforcement channels* consumed by a distinction to the capacity committed. We now specify its domain, its operational meaning, and the three conditions that determine it uniquely.

Definition 1.1 (Enforcement channel count)**.** At an interface Γ with finite capacity $C(\Gamma)$ (A1), each irreducible distinction $d \in \mathcal{D}_{\text{irr}}(\Gamma)$ costs ε^* to enforce (L_{ε^*}, Paper 1 Supplement). A composite distinction occupying n independent irreducible channels has cost $\varepsilon(\{d_1, \ldots, d_n\}) = n \cdot \varepsilon^*$ when the n channels are at *separate interfaces* (L_{loc}), or $n \cdot \varepsilon^* + \Delta$ when co-located (L_Δ, with $\Delta \geq 0$). The *channel count* is the integer $n(d) := \varepsilon(d)/\varepsilon^*$ evaluated at separate interfaces, where the surplus $\Delta = 0$.

The domain of F is therefore the positive integers: $F : \mathbb{N} \to \mathbb{R}_{>0}$, where $F(n)$ gives the capacity cost (in units of ε^*) of enforcing n independent channels at separate interfaces.

Remark 1.2 (Why the domain is $\mathbb{N}$, not $\mathbb{R}^+$)**.** Enforcement channels are discrete. This discreteness is not assumed — it is derived from A1 + MD:

(i) A1 provides finite capacity $C(\Gamma) < \infty$.
(ii) L_{ε^*} (from MD) provides a positive cost floor $\varepsilon^* > 0$ for each irreducible channel.
(iii) Therefore the maximum number of independently enforceable channels at any interface is $\lfloor C(\Gamma)/\varepsilon^* \rfloor < \infty$, an integer.

The domain $\mathbb{N}$ is a *consequence*, not a choice. The classical Cauchy problem on $\mathbb{R}^+$ — with its pathological non-measurable solutions — does not arise because the domain is already discrete.

1.2 The three conditions and their derivation

Theorem 1.3 (L_{Cauchy}: Cost Function Uniqueness)**.** *Let $F : \mathbb{N} \to \mathbb{R}_{>0}$ satisfy:*

$$\textbf{\textit{C1 (Additivity):}} \quad F(a+b) = F(a) + F(b) \quad \textit{for all } a, b \in \mathbb{N}, \tag{1}$$

$$\textbf{\textit{C2 (Monotonicity):}} \quad a > b \;\Rightarrow\; F(a) > F(b) \quad \textit{for all } a, b \in \mathbb{N}, \tag{2}$$

$$\textbf{\textit{C3 (Unit):}} \quad F(1) = 1. \tag{3}$$

Then $F(n) = n$ for all $n \in \mathbb{N}$.

Proof. By C1 (induction): $F(n) = F(1 + 1 + \cdots + 1) = n \cdot F(1)$. By C3: $F(1) = 1$. Therefore $F(n) = n$ for all $n \in \mathbb{N}$. C2 is satisfied automatically: $n > m \Rightarrow F(n) = n > m = F(m)$. □

The proof on $\mathbb{N}$ is trivial once the three conditions are established. The substantive content is the *derivation of the conditions from A1*:

Proposition 1.4 (Derivation of C1–C3 from admissibility)**.**

***C1* (Additivity).** *Let d_1, d_2 be distinctions at* separate *interfaces Γ_1, Γ_2. By L_{loc} (Paper 1 Supplement [2], §4): enforcement at separate interfaces is independent, with independent budgets. The total enforcement cost is $\varepsilon(d_1) + \varepsilon(d_2)$ — costs add exactly when the enforcement regions are disjoint. In channel-count language: $F(n(d_1) + n(d_2)) = F(n(d_1)) + F(n(d_2))$.*
The operational meaning of "additivity" is therefore: independently enforceable channels at separate interfaces contribute independently to the total capacity commitment. *This is the content of L_{loc} applied to the cost functional.*
What C1 does not claim: C1 does not assert additivity at a shared *interface. At a shared interface, the superadditive surplus $\Delta > 0$ (L_Δ) produces non-additive costs. C1 applies only to the separated-interface regime, which is precisely the regime relevant to the cost function F (Definition 1.1).*

***C2* (Monotonicity).** *More enforcement channels require more capacity. This is immediate from A1: each channel costs $\varepsilon^* > 0$ (L_{ε^*}), so adding a channel increases the total cost.*

***C3* (Unit).** *$F(1) = 1$ is a choice of capacity units: we measure cost in units of the single-channel cost ε^*. This is a normalization convention, analogous to $c = 1$ in natural units. It introduces no physical content.*

Connection to resource theories and GPTs. The conditions C1–C3 have close analogues in resource-theoretic frameworks: C1 corresponds to the tensor-product structure of independent resources; C2 to monotonicity under free operations; C3 to the choice of a gold standard (unit resource). The operational stance — deriving algebraic structure from resource constraints — is shared with GPT-style reconstructions (Hardy [11], CDP [12]) and the "accessible fragments" approach. The difference is scope: those programs reconstruct the state space; this program uses the same starting point to derive gauge content and matter spectrum.

1.3 Rational extension (for Paper 4A)

Paper 2's results ($L_{\text{gauge,unique}}$, T_{field}, L_{count}, P_{exhaust}, L_{equip}) use F only on $\mathbb{N}$. The Weinberg angle derivation (Paper 4A) evaluates F at the rational argument $1/d$ (the reciprocal channel count), producing $\gamma = F(d) + F(1/d) = d + 1/d$. We record the rational extension here for completeness.

Proposition 1.5 (Rational extension of F, with operational premise)**.** ***Premise (Sharing-admissibility, S)****: A single enforcement channel evaluated at fractional occupancy $1/q$ ($q \in \mathbb{N}$) contributes $F(1/q)$ to each of q separated-interface sectors that jointly share the channel, and the q shared contributions combine additively under C1. Equivalently, F is extended to $\mathbb{Q}_{>0}$ as the unique additive extension consistent with C1 applied q times across sharing.*

Conclusion. *Under premise S, $F(p/q) = p/q$ for all $p, q \in \mathbb{N}$.*

Proof. For any $q \in \mathbb{N}$: by S, F is defined at p/q; by C1 applied q times, $q \cdot F(p/q) = F(p/q + \cdots + p/q) = F(p) = p$. Therefore $F(p/q) = p/q$. □

Logical status of the sharing premise. Premise S is not derivable from C1 alone: C1 as stated in the integer domain ($a, b \in \mathbb{N}$) is vacuous at fractional arguments, so "C1 extended to $\mathbb{Q}_{>0}$" is definitionally a new domain assignment. S supplies the operational content: it interprets $F(1/q)$ as the per-sector share of a channel decomposed across q separated interfaces. The interpretation is consistent with L_{loc} (locality of capacity) and T_M (interface monogamy)—sharing at separated interfaces does not incur superadditive surplus—but it is a reading beyond C1's native domain. The honest flag is that the Weinberg-angle derivation in Paper 4A rests on C1 + S, not on C1 alone. S is therefore listed alongside MD, BW, and the constitutive admissibility semantics (SP, K3, FD1–FD6) in the assumption inventory when Paper 4A uses the rational extension.

Operational justification for the rational extension. The rational argument $1/d$ arises when the cost functional is evaluated on a *fractional* channel occupancy: a single enforcement channel shared among d sectors contributes $1/d$ of a full channel to each sector. This is the capacity-accounting interpretation of the Weinberg angle formula's "inverse channel count" term. The extension from $\mathbb{N}$ to $\mathbb{Q}_{>0}$ requires no additional regularity assumption (no measurability, no continuity): it follows from C1 applied q times, as the proof above shows.

Exact status of the rational extension in the axiomatics. The rational extension is a *theorem* of C1, not an additional axiom. The proof ($q \cdot F(p/q) = F(p) = p$, therefore $F(p/q) = p/q$) uses only C1 (additivity) applied q times. No "sharing axiom" is introduced. The physical interpretation — that $F(1/d)$ describes the cost of a channel shared among d separated interfaces — is a *reading* of the mathematical result, not a premise. Sharing is admissible only in the separated-interface regime (the domain of F): at a *shared* interface, the superadditive surplus $\Delta > 0$ prevents clean fractional decomposition, which is precisely why F is defined on separated-interface costs and the surplus is tracked separately by L_Δ. The rational extension evaluates $F(1/d)$ at configurations where the sharing cost is accounted for at the pool level, not at the channel level.

No real extension needed. The classical Cauchy problem on $\mathbb{R}^+$ admits pathological (non-measurable, non-monotone, additive) solutions constructible via the Axiom of Choice. These are excluded by C2 (Darboux, 1875): any Lebesgue-measurable (or merely bounded on an interval) additive function $f : \mathbb{R}^+ \to \mathbb{R}$ satisfying $f(x + y) = f(x) + f(y)$ and $f(1) = 1$ is $f(x) = x$. However, no result in this paper or in Paper 4A requires the domain to extend beyond $\mathbb{Q}_{>0}$. The Darboux theorem is available as a safety net but is never load-bearing.

Code reference: `check_L_Cauchy_uniqueness` in `L_Cauchy_uniqueness.py`.

1.4 What $\mathbf{L}_{\text{Cauchy}}$ eliminates

Without L_{Cauchy}, the cost function F is a free parameter. Any monotone function $F : \mathbb{N} \to \mathbb{R}_{>0}$ with $F(1) = 1$ would be compatible with A1. With L_{Cauchy}, the class collapses to a single element:

Candidate	**Condition violated**	**Consequence if allowed**
$F(n) = n^2$	C1 ($9 = F(3) \neq F(1) + F(2) = 5$)	Weinberg angle shifts
$F(n) = \sqrt{n}$	C1 ($\sqrt{3} \neq 1 + \sqrt{2}$)	Capacity non-additive
$F(n) = \log n$	C1 ($\log 3 \neq 0 + \log 2$); C3 ($F(1) = 0$)	No unit cost
$F(n) = c$ (constant)	C2 ($F(2) = F(1)$); C1 ($c \neq 2c$)	Cost insensitive
$F(n) = n^{3/2}$	C1 ($3\sqrt{3} \neq 1 + 2\sqrt{2}$)	Weinberg angle shifts

Each alternative is excluded by C1 alone (additivity). The uniqueness is not a consequence of fine-tuning or accidental cancellation: it is a theorem of algebra (the only additive function on $\mathbb{N}$ with $F(1) = 1$ is the identity). C2 is logically redundant on $\mathbb{N}$ once C1 and C3 are established, but it is stated explicitly because it is the physically motivated condition (more channels cost more) and because it becomes non-redundant on $\mathbb{R}^+$ where it excludes pathological solutions.

2 From superadditivity to gauge structure

This section restates the derivation chain $L_\Delta \to L_{\text{blk}} \to L_\Pi \to T_{\text{alg}}$ (proved in the Paper 1 Supplement) in condensed form, and then proves the gauge identification theorem that separates internal (gauge) automorphisms from kinematic (diffeomorphism) automorphisms.

The chain is summarized first; the proofs are in the Paper 1 Supplement and are not reproduced. The gauge identification theorem (§2.2) is new to this supplement.

2.1 The non-commutativity chain (summary)

The chain from superadditivity to non-commutativity has four links, each proved in the Paper 1 Supplement (P1S). We restate the key result of each without re-proof.

Link 1: L_Δ (Superadditivity). For co-located distinctions d_1, d_2 at an interface Γ with non-trivial pool $\Pi \neq \{0\}$: $\Delta(d_1, d_2) := \varepsilon(\{d_1, d_2\}) - \varepsilon(d_1) - \varepsilon(d_2) > 0$. *Source:* P1S Theorem 5.1. *Inputs:* A1, K2, K3, T_{sep}, SP.

Link 2: L_{blk} (Direct-Sum Failure). If $\Delta > 0$, the minimum-cost joint defense cannot be a direct sum of individual defenses: $E_{\{d_1,d_2\}} \neq E_{d_1} + E_{d_2}$. *Source:* P1S Lemma 5.3.

Link 3: L_Π (Codespace Rotation). The minimum-cost joint defender π_{W_*} has a codespace W_* rotated into the pool Π by angle $\theta \neq 0$. The off-diagonal component $F_\Pi := \pi_{W_*} - E_{d_1} - E_{d_2} \neq 0$ is a new generator supported on the pool. *Source:* P1S Proposition 5.5.

Link 4: T_{alg} (Non-Commutativity). The commutator $[E_{d_1}, F_\Pi] \neq 0$ witnesses non-commutativity. The algebra $\mathcal{A} = \text{Alg}_{\mathbb{R}}(\{E_d\} \cup \{E_{\{d_i,d_j\}}\})$ admits no faithful commutative $*$-representation. *Source:* P1S Theorem 5.7. *Inputs:* L_Δ, L_{blk}, L_Π.

The chain is pre-Hilbert: no GNS construction, no spectral theory, no Hilbert space is invoked. The non-commutativity is an algebraic fact about $\text{End}(V)$, established before $\mathcal{H}_\omega$ is constructed.

2.2 Gauge identification theorem

The central question is: how is the gauge group identified with the automorphism group of the enforcement algebra, and how are spatial/kinematic automorphisms (e.g., diffeomorphisms) excluded?

The answer rests on a structural separation: the enforcement algebra $\mathcal{A}$ acts on the *internal* sector decomposition at a single interface Γ, which is finite-dimensional and defined by T_{sep}. Diffeomorphisms are *inter-interface* maps (L_{loc}) and therefore commute with the internal algebra by construction.

Definition 2.1 (Local support of enforcement operations)**.** An enforcement operation A at interface Γ has *local support* in S_Γ if:

(LS1) A maps S_Γ to itself: $A(S_\Gamma) \subseteq S_\Gamma$.

(LS2) A acts as the identity on the B-orthogonal complement: $A|_{S_\Gamma^\perp} = \text{id}$.

Derivation of local support from FD3. FD3 (anchor-set locality) states that the anchor set M_d of any distinction $d \in \mathcal{D}(\Gamma)$ lies within S_Γ. The enforcement projection E_d has $\text{im}(E_d) = M_d \subseteq S_\Gamma$ and $\ker(E_d) \supseteq S_\Gamma^\perp$ (by T_{adj}: E_d is the B-orthogonal projection onto M_d, hence $E_d|_{M_d^\perp} = 0$, and $S_\Gamma^\perp \subseteq M_d^\perp$). Therefore $E_d(v) = v$ for $v \in M_d$ and $E_d(v) = 0$ for $v \in S_\Gamma^\perp$. The enforcement algebra $\mathcal{A}(\Gamma) = \text{Alg}(\{E_d : d \in \mathcal{D}(\Gamma)\})$ is generated by operators satisfying (LS1)–(LS2), and products and sums of such operators also satisfy (LS1)–(LS2). Therefore every $A \in \mathcal{A}(\Gamma)$ has local support in S_Γ.

Lemma 2.2 (Commuting algebras from disjoint support)**.** *Let Γ_1, Γ_2 be interfaces with B-orthogonal substrates: $B(u, v) = 0$ for all $u \in S_{\Gamma_1}$, $v \in S_{\Gamma_2}$ (from L_ω applied to disjoint interfaces via L_{loc}). If every $A \in \mathcal{A}(\Gamma_1)$ has local support in S_{Γ_1} and every $B \in \mathcal{A}(\Gamma_2)$ has local support in S_{Γ_2}, then $[A, B] = 0$ for all $A \in \mathcal{A}(\Gamma_1)$, $B \in \mathcal{A}(\Gamma_2)$.*

Proof. Decompose $V = S_{\Gamma_1} \oplus S_{\Gamma_2} \oplus W$, where $W = (S_{\Gamma_1} \oplus S_{\Gamma_2})^\perp$ (exists because V is finite-dimensional and B is positive-definite).

By local support (LS1–LS2):

$$A = A_1 \oplus \mathrm{id}_{S_{\Gamma_2}} \oplus \mathrm{id}_W,$$
$$B = \mathrm{id}_{S_{\Gamma_1}} \oplus B_2 \oplus \mathrm{id}_W,$$

where $A_1 := A|_{S_{\Gamma_1}} \in \mathrm{End}(S_{\Gamma_1})$ and $B_2 := B|_{S_{\Gamma_2}} \in \mathrm{End}(S_{\Gamma_2})$.

For any $v = v_1 + v_2 + w$ with $v_i \in S_{\Gamma_i}$ and $w \in W$:

$$AB(v) = A(v_1 + B_2 v_2 + w) = A_1 v_1 + B_2 v_2 + w,$$
$$BA(v) = B(A_1 v_1 + v_2 + w) = A_1 v_1 + B_2 v_2 + w.$$

Therefore $AB = BA$.

The second line uses: $B_2 v_2 \in S_{\Gamma_2}$ (LS1 for B), so A acts on it as id (LS2 for A). Similarly, $A_1 v_1 \in S_{\Gamma_1}$ (LS1 for A), so B acts on it as id (LS2 for B). □

Remark 2.3 (Two bilinear structures: B and $g_{\mu\nu}$)**.** The bilinear form B used in Definition 2.1 and Lemma 2.2 is a *positive-definite* inner product on the sector space S_Γ at a single interface. Later in this supplement (§5.4), a *Lorentzian* metric $g_{\mu\nu}$ is constructed on the spacetime manifold M. These are two distinct bilinear structures at two distinct structural levels; they must not be conflated.

Where they live: B is defined on the *internal* sector space $S_\Gamma = \bigoplus_d M_d \oplus \Pi$ at a single interface Γ. Its domain is the finite-dimensional vector space of perturbation directions within the enforcement algebra at Γ. The spacetime metric $g_{\mu\nu}$ is defined on the *tangent space* T_pM of the spacetime manifold M at each point $p \in M$. Its domain includes both spatial directions (from the inter-interface structure, L_{loc}) and the timelike direction (from irreversibility, L_{irr}).

Why B is positive-definite: B measures enforcement cost at a single interface. By K2 (positive perturbation cost) and SP (no null directions), every nonzero perturbation of the enforcement structure at Γ costs positive capacity: $B(v, v) > 0$ for all $v \neq 0$ in S_Γ. This is an *internal-sector* statement: it says that enforcement is expensive in all directions *within* the interface's sector space. There is no timelike direction in S_Γ; the internal sector does not "know about" time.

Why $g_{\mu\nu}$ is Lorentzian: The spacetime metric $g_{\mu\nu}$ is assembled from B (which becomes the *spatial* part $g_{ij}^{(\mathrm{sp})}$ of the metric) plus the timelike direction τ from L_{irr} (which contributes the negative eigenvalue $-N^2$). The construction (Proposition 5.11, eq. (12)) is:

$$g_{\mu\nu}\, dx^\mu dx^\nu = -N^2\, dt^2 + g_{ij}^{(\mathrm{sp})}\, dx^i dx^j,$$

where $g_{ij}^{(\mathrm{sp})}$ inherits the positive-definiteness of B and the $-N^2 dt^2$ term is the timelike contribution. The positive-definiteness of B is therefore *exactly what is needed* for the spatial part of the Lorentzian metric: a positive-definite spatial metric combined with a negative timelike component yields signature $(-, +, +, +)$.

How they relate:

$$\underbrace{B}_{\substack{\text{positive-definite}\\ \text{on } S_\Gamma\\ \text{(internal sector)}}} \xrightarrow{\text{spatial restriction}} \underbrace{g_{ij}^{(\mathrm{sp})}}_{\substack{\text{positive-definite}\\ \text{on spatial slices}\\ \text{of } T_pM}} \xrightarrow{+\,L_{\mathrm{irr}}} \underbrace{g_{\mu\nu}}_{\substack{\text{Lorentzian}\\ \text{on } T_pM\\ \text{(spacetime)}}}$$

The first arrow is the identification of the enforcement cost kernel with the spatial metric (T7B: $\varepsilon_{\mathrm{ext}}(v) = B(v, v)$, which becomes the spatial line element). The second arrow adds the timelike direction from irreversibility. No bilinear structure changes sign or loses positive-definiteness at

any step; the Lorentzian signature arises from *combining two distinct structures* (spatial B + temporal L_{irr}), not from modifying B itself.

Why local support and commutativity use B, not $g_{\mu\nu}$: Definition 2.1 and Lemma 2.2 operate at the single-interface level: they concern the internal sector decomposition at Γ, not the spacetime geometry. The relevant orthogonality is *sector orthogonality* ($B(v_d, v_{d'}) = 0$ for $d \neq d'$, from L_ω), not *spacetime orthogonality.* Two enforcement operations at disjoint interfaces commute because their sector spaces are B-orthogonal, not because they are spacelike-separated in the Lorentzian metric. The spacetime interpretation (spacelike separation $\Leftrightarrow$ commutativity) is a *consequence* derived in §5, not a premise used here.

In summary: B is the cost kernel (positive-definite, internal, single-interface); $g_{\mu\nu}$ is the spacetime metric (Lorentzian, external, inter-interface). They share a spatial component ($g_{ij}^{(\text{sp})} = B|_{\text{spatial}}$) but are defined on different spaces, serve different purposes, and have different signatures. No conflation arises because the timelike direction (the source of the signature difference) is not part of the single-interface sector space where B operates.

Relationship to AQFT. In AQFT, Einstein causality ($[\mathcal{A}(\mathcal{O}_1), \mathcal{A}(\mathcal{O}_2)] = 0$ for spacelike-separated regions) is a separate axiom. In the APF, it is a theorem: FD3 (anchor-set locality) forces local support (Definition 2.1), which forces commutativity (Lemma 2.2). FD3 is the APF's analogue of Einstein causality, but it is a constitutive definition of what "enforcement at Γ" means, not an independent physical postulate.

Scope of this result. Lemma 2.2 is a theorem *internal to the finite-capacity interface algebra.* It is not claimed to reproduce full AQFT locality in the continuum/type-III regime. Specifically: it applies to enforcement algebras $\mathcal{A}(\Gamma) = M_n(\mathbb{C})$ at finite n, using the positive-definite cost kernel B for the orthogonal decomposition. The continuum-limit analogue (commutativity of type III local algebras for spacelike-separated regions) is a separate result that requires the full machinery of AQFT (Haag–Kastler axioms, nuclearity estimates, split property). The finite-capacity result is the *precursor* from which the continuum result can be recovered in the $C/\varepsilon^* \to \infty$ limit (§2.4), but the two should not be conflated.

Lemma 2.4 (Internal–kinematic decomposition)**.** *Let $\{\mathcal{A}(\Gamma)\}_{\Gamma\in\mathcal{I}}$ be a net of finite-dimensional $*$-algebras indexed by a set $\mathcal{I}$, with isotony $\mathcal{A}(\Gamma) \hookrightarrow \mathcal{A}(\Gamma')$ for $\Gamma \subseteq \Gamma'$, and with independence: $\mathcal{A}(\Gamma)$ and $\mathcal{A}(\Gamma')$ commute when $\Gamma \cap \Gamma' = \emptyset$ (Lemma 2.2). Define:*

(a) $\mathrm{Aut}_{\text{kin}} :=$ *automorphisms α of the net that act non-trivially on $\mathcal{I}$ (permuting interface labels);*

(b) $\mathrm{Aut}_{\text{int}} :=$ *automorphisms that fix $\mathcal{I}$ pointwise (mapping each $\mathcal{A}(\Gamma)$ to itself).*

Then every net automorphism α decomposes uniquely as $\alpha = \alpha_{\text{int}} \circ \alpha_{\text{kin}}$, with $\alpha_{\text{kin}} \in \mathrm{Aut}_{\text{kin}}$ determined by α's action on $\mathcal{I}$ and $\alpha_{\text{int}} \in \mathrm{Aut}_{\text{int}}$ acting fibrewise. The group structure is a semidirect product: $\mathrm{Aut}(net) = \mathrm{Aut}_{\text{int}} \rtimes \mathrm{Aut}_{\text{kin}}$.

Proof. Given α, define α_{kin} by its action on the index set: $\alpha_{\text{kin}}(\Gamma) := \alpha(\Gamma)$ with the induced isomorphism $\mathcal{A}(\Gamma) \xrightarrow{\sim} \mathcal{A}(\alpha(\Gamma))$. Then $\alpha_{\text{int}} := \alpha \circ \alpha_{\text{kin}}^{-1}$ fixes every index and is therefore fibrewise. Uniqueness follows because the action on $\mathcal{I}$ determines α_{kin} uniquely. The semidirect product structure holds because $\mathrm{Aut}_{\text{int}}$ is normal (conjugating a fibrewise automorphism by a permutation yields another fibrewise automorphism). □

Theorem 2.5 (Gauge Identification)**.** *Let $\mathcal{A}$ be the enforcement algebra at interface Γ (from T_{alg}), and let $\mathrm{Aut}(\mathcal{A})$ denote its $*$-automorphism group. Then:*

(i) $\mathrm{Aut}(\mathcal{A})$ *decomposes as* $\mathrm{Aut}_{\text{int}}(\mathcal{A}) \rtimes \mathrm{Aut}_{\text{kin}}(\mathcal{A})$*, where* $\mathrm{Aut}_{\text{int}}$ *consists of automorphisms that preserve the interface label Γ and* $\mathrm{Aut}_{\text{kin}}$ *consists of automorphisms induced by inter-interface maps.*

(ii) $\mathrm{Aut}_{\text{int}}(\mathcal{A})$ *is the inner automorphism group* $\mathrm{Inn}(\mathcal{A})$*. By Skolem–Noether (applicable because $\mathcal{A}$ is a finite-dimensional central simple algebra over $\mathbb{C}$ after T2c),* $\mathrm{Inn}(\mathcal{A}) \cong \mathrm{PU}(\mathcal{H}_\omega)$.

(iii) The physical gauge group G is identified with $\mathrm{Inn}(\mathcal{A})$*. Kinematic automorphisms (diffeomorphisms) are outer and are excluded from G.*

(iv) G is non-abelian: $\mathrm{Inn}(\mathcal{A})$ *is non-trivial and non-commutative because $\mathcal{A}$ is non-commutative (T_{alg}) and simple (T2a).*

Proof. Step 1: Internal vs. kinematic automorphisms. By Lemma 2.4 applied to the enforcement net $\{\mathcal{A}(\Gamma)\}_{\Gamma\in\mathcal{I}}$ (with independence from L_{loc}), every automorphism decomposes uniquely into a kinematic part (acting on the index set $\mathcal{I}$) and an internal part (acting fibrewise on each $\mathcal{A}(\Gamma)$). Diffeomorphisms are kinematic; the gauge group is the internal part. This gives (i).

Step 2: Wedderburn–Artin and Skolem–Noether. After field selection (T2c: $\mathbb{F} = \mathbb{C}$), the enforcement algebra is a finite-dimensional $*$-algebra over $\mathbb{C}$. Wedderburn–Artin (T2a) gives $\mathcal{A} \cong \bigoplus_{k=1}^{K} M_{n_k}(\mathbb{C})$. By Skolem–Noether, every automorphism of $M_{n_k}(\mathbb{C})$ is inner: $\alpha(a) = UaU^*$ for some unitary U. The inner automorphism group of each block is $\mathrm{Inn}(M_{n_k}) \cong \mathrm{PU}(n_k) = \mathrm{U}(n_k)/\mathrm{U}(1)$. For the full semisimple algebra: $\mathrm{Inn}(\mathcal{A}) = \prod_{k=1}^{K} \mathrm{PU}(n_k)$. This gives (ii).

Step 2b: Block permutations. For $K \geq 2$, the full automorphism group of a finite-dimensional semisimple algebra is:

$$\mathrm{Aut}\Big(\bigoplus_{k=1}^{K} M_{n_k}(\mathbb{C})\Big) \;=\; \Big(\prod_{k=1}^{K} \mathrm{PU}(n_k)\Big) \;\rtimes\; S_{\mathrm{iso}},$$

where $S_{\mathrm{iso}} \subseteq S_K$ is the group of permutations among isomorphic blocks (blocks with the same n_k). The inner automorphisms are $\mathrm{Inn}(\mathcal{A}) = \prod_k \mathrm{PU}(n_k)$; the block permutations S_{iso} are outer.

The physical status of S_{iso}:

(a) Each Wedderburn block corresponds to a distinct enforcement sector under T_{sep}. Sectors are distinguished by their anchor sets M_{d_k}, which carry distinct enforcement costs $\varepsilon(d_k)$.

(b) A block permutation $\sigma \in S_{\mathrm{iso}}$ exchanges sectors. This is an automorphism of the *abstract* algebra but not of the *cost-equipped* algebra $(\mathcal{A}, \varepsilon)$: it maps $\varepsilon(d_k) \mapsto \varepsilon(d_{\sigma(k)})$, which differs from $\varepsilon(d_k)$ unless the permuted sectors are cost-degenerate.

(c) Cost-degenerate sectors (identical enforcement costs at every state) correspond to physically indistinguishable copies — these are *generations*. The permutation symmetry S_{iso} among cost-degenerate blocks is a discrete *flavor* symmetry, not a continuous gauge symmetry. It is broken by Yukawa couplings (mass differences between generations) and does not contribute to the gauge group G.

(d) Non-degenerate blocks: $\varepsilon(d_k) \neq \varepsilon(d_j)$ for $k \neq j$. The permutation σ is not an automorphism of $(\mathcal{A}, \varepsilon)$, so S_{iso} is trivial on the cost-equipped algebra.

Conclusion. The continuous gauge group is $G = \mathrm{Inn}(\mathcal{A}) = \prod_k \mathrm{PU}(n_k)$, regardless of whether S_{iso} is trivial (non-degenerate costs) or nontrivial (degenerate costs). The reason is that S_{iso} is *discrete and outer*: it permutes Wedderburn blocks but does not generate any continuous family of automorphisms. The continuous gauge group sees only the connected component of $\mathrm{Aut}(\mathcal{A})$, which is $\mathrm{Inn}(\mathcal{A})$.

Physical status of S_{iso}: generation symmetry and Yukawa breaking. Cost-degenerate blocks correspond to physically indistinguishable copies of the same gauge sector — these are *generations*. The exact permutation symmetry S_{iso} among cost-degenerate blocks is the *flavor symmetry* of the massless limit: all three generations of (say) up-type quarks have identical gauge quantum numbers.

After electroweak symmetry breaking (derived in Paper 4B from the enforcement potential), the Yukawa couplings y_i become nonzero and generation-dependent: $y_u \neq y_c \neq y_t$. In the cost-equipped algebra $(\mathcal{A}, \varepsilon)$, this is reflected as cost non-degeneracy: the enforcement cost of maintaining a top-quark distinction at an interface differs from that of an up-quark distinction. The physical observable is the mass hierarchy: $m_u \approx 2\,\mathrm{MeV} \ll m_c \approx 1.3\,\mathrm{GeV} \ll m_t \approx 173\,\mathrm{GeV}$, which breaks $S_3 \to \{1\}$ completely.

The structural chain is: cost-degenerate blocks → exact flavor symmetry → Yukawa splitting → broken S_{iso} → observed mass hierarchy. The derivation of the specific Yukawa couplings is in Paper 4B. What is established here is that S_{iso} is discrete, outer, and does not contribute to the continuous gauge group in any regime — degenerate or not.

Step 3: Identification. The physical gauge group acts on the internal degrees of freedom at a single interface, preserving the cost-equipped algebraic structure. By Steps 1–2, this is precisely $\mathrm{Inn}(\mathcal{A})$. Kinematic automorphisms are outer (they change the interface label) and are not part of the gauge group. Block permutations among cost-degenerate sectors are outer, discrete, and identified with generation symmetries. This gives (ii) and (iii).

Step 4: Non-abelian. $\mathcal{A}$ is non-commutative (T_{alg}). A commutative algebra has $\mathrm{Inn}(\mathcal{A}) = \{1\}$ (every inner automorphism is trivial). Since $\mathrm{Inn}(\mathcal{A})$ is non-trivial, G is non-trivial. For $M_n(\mathbb{C})$ with $n \geq 2$, $\mathrm{PU}(n)$ is non-abelian (e.g., $\mathrm{PU}(2) \cong \mathrm{SO}(3)$). This gives (iv). □

Remark 2.6 (Emergence of abelian U(1) factors)**.** The inner automorphism group $\mathrm{Inn}(\mathcal{A}) = \prod_k \mathrm{PU}(n_k)$ contains no abelian factors: $\mathrm{PU}(n)$ is semisimple for $n \geq 2$, and trivial for $n = 1$. This raises the question: where does the $\mathrm{U}(1)_Y$ hypercharge factor of the Standard Model originate?

The answer involves the relative-phase torus of the multi-block unitary group, together with a no-redundancy theorem.

The algebraic relationship between the groups acting on the algebra (by conjugation) and on the Hilbert space (linearly) is captured by an exact sequence.

Proposition 2.7 (Exact sequence for gauge representations)**.** *Let* $\mathcal{A} = \bigoplus_{k=1}^{K} M_{n_k}(\mathbb{C})$ *with* $\mathrm{U}(\mathcal{A}) = \prod_k \mathrm{U}(n_k)$*. Define the conjugation map* $\pi : \mathrm{U}(\mathcal{A}) \to \mathrm{Aut}(\mathcal{A})$ *by* $\pi(U)(a) = UaU^*$*. Then:*

(i) The image of π *is* $\mathrm{Inn}(\mathcal{A}) = \prod_k \mathrm{PU}(n_k)$*.*

(ii) The kernel of π *is the center* $Z(\mathrm{U}(\mathcal{A})) = \prod_k \mathrm{U}(1)_k$*, consisting of block-diagonal phase matrices* $(e^{i\theta_1}\mathbf{1}_{n_1}, \ldots, e^{i\theta_K}\mathbf{1}_{n_K})$*.*

(iii) The sequence

$$1 \longrightarrow Z(\mathrm{U}(\mathcal{A})) \longrightarrow \mathrm{U}(\mathcal{A}) \xrightarrow{\pi} \mathrm{Inn}(\mathcal{A}) \longrightarrow 1$$

is exact.

(iv) The overall diagonal phase $\mathrm{U}(1)_{\text{diag}} = \{(e^{i\theta}\mathbf{1}_{n_1}, \ldots, e^{i\theta}\mathbf{1}_{n_K}) : \theta \in \mathbb{R}\}$ *acts trivially on both* $\mathcal{A}$ *(by conjugation) and on* $\mathcal{H}_\omega$ *(as a global phase). The quotient* $Z/\mathrm{U}(1)_{\text{diag}}$ *is a torus of rank* $K-1$*, the relative-phase torus* T_{rel}*.*

(v) Matter fields transform under $\mathrm{U}(\mathcal{A})$ *linearly on* $\mathcal{H}_\omega$ *(fundamental representation of each block), not under* $\mathrm{Inn}(\mathcal{A})$ *(which acts on* $\mathcal{A}$ *by conjugation, i.e., the adjoint representation). The two actions agree on the Lie algebra (*$\mathfrak{u}(\mathcal{A}) \to \mathfrak{inn}(\mathcal{A})$ *is surjective with kernel* $\mathfrak{z}$*), but differ globally:* $\mathrm{U}(\mathcal{A})$ *admits representations that* $\mathrm{Inn}(\mathcal{A})$ *does not.*

Proof. (i) By Skolem–Noether, every automorphism of $M_n(\mathbb{C})$ has the form $a \mapsto UaU^*$ for some $U \in \mathrm{U}(n)$. For the semisimple algebra, each block is mapped to itself by an inner automorphism (outer block permutations are handled separately in Step 2b of Theorem 2.5), so $\mathrm{Im}(\pi) = \prod_k \mathrm{Inn}(M_{n_k}) = \prod_k \mathrm{PU}(n_k)$. (ii) $\pi(U) = \mathrm{id}$ iff $UaU^* = a$ for all $a \in \mathcal{A}$, iff U is central. For $M_n(\mathbb{C})$, the center is $\mathrm{U}(1) \cdot \mathbf{1}_n$; for the direct sum, $Z = \prod_k \mathrm{U}(1)_k$. (iii) Surjectivity of π onto $\mathrm{Inn}(\mathcal{A})$ is (i); the kernel statement is (ii); the sequence is exact at $\mathrm{U}(\mathcal{A})$ by construction. (iv) The diagonal $\mathrm{U}(1)_{\text{diag}}$ is a subgroup of Z with K-dimensional Z quotient of dimension $K-1$. Triviality on $\mathcal{H}_\omega$ is because states are defined up to global phase: $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ represent the same physical state. (v) The GNS representation maps each $M_{n_k}(\mathbb{C})$ block to operators on $\mathbb{C}^{n_k} \subset \mathcal{H}_\omega$; the natural group action on $\mathbb{C}^{n_k}$ is $v \mapsto Uv$ (fundamental), not $a \mapsto UaU^*$ (adjoint). The fundamental is a linear representation of $\mathrm{U}(n_k)$ but only a projective representation of $\mathrm{PU}(n_k)$. □

Definition 2.8 (Admissible gauge structure)**.** Let G be a compact Lie group acting faithfully on a finite-dimensional Hilbert space $\mathcal{H}$, and let $\mathcal{M} = \{r_1, \ldots, r_N\}$ be the set of physically distinct irreducible multiplet types occurring in $\mathcal{H}$.

Enforcement completeness (EC). G is *enforcement-complete* if the map $r_i \mapsto [r_i]_G$ from multiplet types to G-representation labels is injective: distinct physical types carry distinct gauge quantum numbers.

Capacity minimality (CM). G is *capacity-minimal* if no generator of G can be removed while preserving enforcement completeness. Formally: for every connected subgroup $H \subsetneq G$ obtained by removing at least one generator, the label map $r_i \mapsto [r_i]_H$ is *not* injective.

Admissibility. G is *admissible* if it is both enforcement-complete and capacity-minimal.

Before deriving EC and CM, we establish the bridge from Lie-algebra generators to enforcement channels, which CM requires. The bridge is not as simple as "one generator = one channel": the relationship between Lie-algebra generators and enforcement channels involves incompatibility (non-commuting generators cannot be simultaneously monitored), coarse-graining (degenerate eigenvalues reduce the information content of a single generator), and the distinction between Cartan (commuting) and root (non-commuting) generators. We treat these systematically.

Definition 2.9 (Enforcement monitoring of a generator)**.** Let $g_a \in \mathfrak{g}$ be a Hermitian generator of a compact Lie algebra acting on a finite-dimensional Hilbert space $\mathcal{H}$ of dimension n. The *enforcement monitoring* of g_a is the projection-valued measure (PVM) $\{P_{a,\lambda}\}_\lambda$ on the spectrum $\sigma(g_a) \subseteq \mathbb{R}$, where $P_{a,\lambda}$ projects onto the λ-eigenspace of g_a. The *distinction count* of g_a is $d(g_a) := |\sigma(g_a)| - 1$ — one less than the number of distinct eigenvalues. This is the number of binary questions needed to determine which eigenspace a state occupies (using a binary search on the ordered eigenvalue list).

Lemma 2.10 (Primitive generator cost bound)**.** *Let G be a compact Lie group acting faithfully on a finite-dimensional Hilbert space $\mathcal{H}$ with $\dim(\mathcal{H}) = n$. Let $\mathfrak{g} = \mathrm{Lie}(G)$ with $\dim(\mathfrak{g}) = p$. Then the total enforcement cost of maintaining the gauge structure G is at least $p \cdot \varepsilon^*$: each independent Lie-algebra direction that generates a nontrivial infinitesimal orbit on $\mathcal{H}$ requires at least one irreducible enforcement channel.*

More precisely:

(i) ***(Lower bound per generator.)*** *For each nonzero $g_a \in \mathfrak{g}$, faithfulness ensures g_a acts non-trivially on $\mathcal{H}$ (it has at least two distinct eigenvalues). Monitoring the charge under g_a (determining which eigenspace a state occupies) requires at least one enforcement channel at cost $\geq \varepsilon^* > 0$ (L_{ε^*}).*

(ii) ***(Independence.)*** *Linearly independent generators g_a, g_b cannot be monitored by a single channel: either they do not commute (incompatible spectral decompositions: no single measurement determines both charges) or they commute but have independent spectra (knowing g_a's eigenvalue does not determine g_b's). In either case, a second channel is required.*

(iii) ***(Cost lower bound.)*** *By (i) and (ii), p linearly independent generators require at least p channels, each costing $\geq \varepsilon^*$. The total cost is at least $p \cdot \varepsilon^*$.*

Scope and caveats. The lemma establishes a lower bound *on the enforcement cost, not an exact cost. It does not claim that the gauge structure costs* exactly *$p \cdot \varepsilon^*$: co-located non-abelian channels may incur superadditive surplus (L_Δ), making the actual cost strictly greater. The lower bound suffices for CM: a redundant generator wastes* at least *ε^* capacity (and possibly more), which is inadmissible under finite total budget. The lemma also does not claim that each generator corresponds to a* separate physical measurement*: non-commuting generators have incompatible spectral decompositions and cannot be simultaneously monitored. What it claims is that the* capacity cost *of maintaining p independent gauge directions is at least $p \cdot \varepsilon^*$, regardless of measurement incompatibility.*

Proof. Step 1: Each generator occupies exactly one channel. A generator $g_a \in \mathfrak{g}$ is Hermitian (compact Lie algebra: $g_a^\dagger = g_a$) and acts on the finite-dimensional $\mathcal{H}$. By the spectral theorem for Hermitian operators on finite-dimensional spaces, g_a has a complete orthonormal eigenbasis with real eigenvalues: $g_a = \sum_\lambda \lambda P_{a,\lambda}$. The PVM $\{P_{a,\lambda}\}$ is uniquely determined by g_a.

Monitoring g_a means determining which eigenspace a state occupies. This is one irreducible enforcement channel (Definition 1.1): the PVM defines a single spectral decomposition of $\mathcal{H}$ into orthogonal subspaces. The monitoring is *complete* for g_a: knowing the eigenvalue λ determines the charge under the entire one-parameter subgroup $\{e^{itg_a}\}_{t\in\mathbb{R}}$ (because $e^{itg_a}|_{\mathcal{H}_{a,\lambda}} = e^{it\lambda}\,\mathrm{id}$). No sub-decomposition of the PVM is needed, and no additional measurement refines the information about g_a beyond what the PVM already provides. Therefore: one generator, one channel.

Step 2: Linearly independent generators require separate channels. We must show that two linearly independent generators g_a, g_b cannot be simultaneously monitored by a single channel. There are two cases.

Case 2a: $[g_a, g_b] \neq 0$ (non-commuting generators). If g_a and g_b do not commute, their PVMs are incompatible: there is no common eigenbasis. (Proof: suppose every eigenvector of g_a were also an eigenvector of g_b. Then g_a and g_b would be simultaneously diagonalizable, hence commuting — contradicting $[g_a, g_b] \neq 0$.) An incompatible pair of PVMs cannot be realized by a single measurement: any single measurement determines the eigenvalue of at most one Hermitian operator. In the enforcement language: a single channel that determines the g_a-eigenvalue necessarily disturbs the g_b-eigenvalue (by the incompatibility of the PVMs), and vice versa. Therefore non-commuting generators require separate channels.

Case 2b: $[g_a, g_b] = 0$ (commuting generators). Commuting Hermitian operators are simultaneously diagonalizable: there exists a common eigenbasis in which both g_a and g_b are diagonal. One might therefore suspect that a single channel could monitor both. This is incorrect: the eigenvalue of g_a does not determine the eigenvalue of g_b unless g_b is a function of g_a.

Formally: let $\mathcal{H} = \bigoplus_{(\lambda,\mu)} \mathcal{H}_{\lambda,\mu}$ be the simultaneous eigenspace decomposition. Monitoring g_a alone determines λ but reveals only the subspace $\mathcal{H}_{a,\lambda} = \bigoplus_\mu \mathcal{H}_{\lambda,\mu}$, which may contain multiple g_b-eigenvalues. To determine μ as well, a second measurement (a second channel monitoring g_b) is required. The two measurements are compatible (they can be performed simultaneously or in either order without disturbance), but they are *distinct*: two separate channels, one for each generator.

The exception is when $g_b = f(g_a)$ for some function $f : \mathbb{R} \to \mathbb{R}$ — then g_b's eigenvalue is determined by g_a's, and g_b adds no new information. But in this case g_b is not linearly independent of $\{g_a, g_a^2, \ldots\}$ in the universal enveloping algebra, and in the finite-dimensional setting, g_b is a polynomial in g_a. Since we chose a *basis* of $\mathfrak{g}$, no basis element is a function of another (linear independence in the Lie algebra is strictly stronger than functional dependence on the Hilbert space).

Step 3: Faithfulness ensures all channels are non-trivial. Faithfulness of the G-action means: $g_a \neq 0$ in $\mathfrak{g}$ implies g_a acts non-trivially on $\mathcal{H}$ (i.e., g_a is not the zero operator). Therefore $|\sigma(g_a)| \geq 2$ for every nonzero g_a: the distinction count $d(g_a) \geq 1$. Without faithfulness, some generators might act as the zero operator, and the corresponding channel would be vacuous. Faithfulness is guaranteed by the gauge identification theorem (Theorem 2.5): the gauge group G acts faithfully on $\mathcal{H}_\omega$ by construction (the inner automorphisms of $M_n(\mathbb{C})$ act faithfully on $\mathbb{C}^n$ via the fundamental representation, Proposition 2.19).

Step 4: Cost. By L_{ε^*} (Paper 1 Supplement), each irreducible enforcement channel costs at least $\varepsilon^* > 0$. By Steps 1–3, p basis generators require p separate channels. Therefore the total cost is at least $p \cdot \varepsilon^*$. □

Edge cases and subtleties.

1. Degenerate eigenvalues and coarse-graining. If a generator g_a has a degenerate eigenvalue (i.e., some eigenspace $\mathcal{H}_{a,\lambda}$ has dimension > 1), the monitoring channel provides less than

$\log_2 n$ bits of information. For example, the SU(2) generator J_3 on the spin-1 representation has eigenvalues $\{+1, 0, -1\}$: three distinct values, so $d(J_3) = 2$. On a spin-0 (trivial) representation, $J_3 = 0$: all eigenvalues are zero, so $d(J_3) = 0$ and the channel is vacuous.

The lemma handles this correctly: each generator occupies one channel regardless of the degeneracy structure. The *information content* of the channel varies (from 0 bits for a trivially acting generator to $\log_2 n$ bits for a generator with n distinct eigenvalues), but the *channel count* is one. The capacity cost is ε^* per channel, not per bit: this is the content of L_{ε^*} (the cost floor is per irreducible distinction, not per bit of information).

2. Abelian groups. For an abelian G (e.g., $\mathrm{U}(1)^k$), all generators commute: $[g_a, g_b] = 0$ for all a, b. Case 2b of the proof applies to every pair. The generators are simultaneously diagonalizable, but each occupies a separate channel (one PVM per generator). The total channel count equals $\dim(\mathfrak{g}) = k$. This is consistent: a product of k independent U(1)'s has k independent charges, each requiring a separate measurement.

3. Cartan subalgebra vs. root generators. In a semisimple Lie algebra $\mathfrak{g}$, the Cartan subalgebra $\mathfrak{h}$ (maximal commuting subalgebra, dimension = rank) and the root generators E_α play different roles. The Cartan generators are simultaneously diagonalizable (Case 2b); the root generators do not commute with the Cartan generators (Case 2a) or with each other. The lemma assigns one channel to each — Cartan and root alike — because each is a linearly independent element of the basis. The distinction between "commuting" and "non-commuting" does not affect the channel count; it affects only whether the channels can be monitored simultaneously (Cartan: yes; root: no).

4. Representations with coincidental spectral degeneracies. It is possible for two linearly independent generators g_a, g_b to have identical spectra on a particular representation: $\sigma(g_a) = \sigma(g_b)$ (as multisets). For example, in the fundamental representation of SU(2), J_1 and J_2 both have spectrum $\{+\frac{1}{2}, -\frac{1}{2}\}$. Despite the identical spectra, the *eigenbases* are different (J_1 and J_2 do not commute), so the corresponding PVMs are incompatible and require separate channels. The channel count depends on linear independence in the Lie algebra, not on the coincidence of spectral values.

5. Basis dependence. The claim "p channels" depends on the choice of basis. Different bases of $\mathfrak{g}$ yield different PVMs, but the total channel count $p = \dim(\mathfrak{g})$ is basis-independent (it is a topological invariant of the Lie group). The enforcement cost $p \cdot \varepsilon^*$ is therefore basis-independent. The *specific channels monitored* depend on the basis choice (e.g., Cartan–Weyl basis vs. Gell-Mann matrices), but the total number does not.

6. Could coarse-graining reduce the channel count? A coarse-graining of the enforcement structure (combining multiple fine-grained distinctions into a single coarser distinction) could in principle reduce the number of channels below p. However, coarse-graining loses information: it cannot separate states that the fine-grained structure separates. If EC requires that all multiplet types be separated, the enforcement structure must be at least as fine as the multiplet partition. The fine-grained structure has p independent channels (one per generator); any coarse-graining that reduces the channel count below p either (a) merges two generators' channels into one, losing the ability to distinguish states separated by those generators, or (b) drops a generator entirely, reducing G to a proper subgroup. Case (a) violates EC if the merged generators separate different multiplets; case (b) violates EC if the dropped generator separates at least one pair. Therefore: no coarse-graining consistent with EC can reduce the channel count below p.

Lemma 2.11 (EC and CM from the admissibility semantics)**.** *EC and CM are derived principles of the admissibility framework. They are not consequences of A1 (finite capacity) alone; their dependencies on the two axioms (A1, A2) and the regularity input MD are stated cleanly under the dual-axiom framing of Paper 1, §2:*

(i) ***EC*** *requires: A1 (finite capacity) + T_{sep} (sector decomposition) + L_{loc} (locality) + FD3 (anchor-set locality) + the gauge identification (Theorem 2.5). EC is the statement that*

the internal gauge labels at a single interface must resolve all physically distinct internal types. It is a completeness principle for the interface-local enforcement semantics*, downstream of A1 alone (via the derived structural separation); A2 plays no role.*

(ii) ***CM*** *requires: A2 (minimum cost) applied to the admissible set defined by A1 + MD (via L_{ε^*}) + the generator–channel bridge (Lemma 2.10). A configuration carrying a redundant generator is* not *inadmissible under A1 — it still satisfies $\Sigma\varepsilon \leq C$. But A1 + MD (via L_{ε^*}) guarantees that removing the redundant generator yields a strictly cheaper configuration in the same admissible set, by $\varepsilon^* > 0$. A2 then selects the cheaper one as the realised gauge structure. CM is therefore the statement that the realised admissible structure carries no redundant generators — a direct consequence of A2.*

Proof. EC from the structural separation. The argument for EC requires a premise that is not A1 alone but is *derived from* A1: the internal–external structural separation. By T_{sep} (Paper 1 Supplement), the enforcement structure at each interface Γ decomposes into sectors. By Definition 2.1 and Lemma 2.2 (both derived from FD3 and L_{loc}, which are themselves derived from A1): at a single interface, the enforcement algebra $\mathcal{A}(\Gamma)$ acts on the *internal* sector decomposition of S_Γ, while inter-interface (spacetime) distinctions — spatial separation, momentum, mass — are properties of the *external* structure (the inter-interface map L_{loc}). At a single interface, the only enforcement operations available are those in $\mathcal{A}(\Gamma)$, whose automorphism group is the gauge group G (Theorem 2.5).

Now suppose two physically distinct multiplet types $r_i \neq r_j$ carry identical gauge quantum numbers: $[r_i]_G = [r_j]_G$. Since $G = \text{Aut}(\mathcal{A}(\Gamma))$ and the G-representation labels exhaust the distinctions available within $\mathcal{A}(\Gamma)$, no enforcement operation at Γ distinguishes r_i from r_j. But A1 requires that all physically distinct states be enforceable (this is the operational content of "enforcement capacity": the budget $C(\Gamma)$ is the capacity to enforce distinctions at Γ). If two types are indistinguishable at every interface at which they co-occur, they are enforcement-equivalent and therefore physically identical — contradicting $r_i \neq r_j$.

The objection "perhaps they are distinguished by mass or momentum rather than gauge quantum numbers" is addressed by the structural separation: mass and momentum are spacetime (external) properties resolved by the inter-interface structure, not by the enforcement algebra at a single interface. EC is the statement that the *internal* sector must resolve all internal types; it does not claim that gauge quantum numbers are the only physically meaningful labels, only that they must be complete within the internal sector.

CM from A2 over the admissible set. CM requires three ingredients in addition to A2:

(a) L_{ε^*}: each irreducible distinction costs at least $\varepsilon^* > 0$. This is derived from A1 + MD (enforcement isotropy) in the Paper 1 Supplement. It is *not* derivable from A1 alone: without the isotropy assumption MD, one cannot exclude the possibility that some channels have arbitrarily small cost.

(b) Lemma 2.10: each Lie-algebra generator occupies exactly one channel. This is derived from faithfulness of the G-action (which follows from Theorem 2.5, itself derived from A1) and the spectral decomposition of Hermitian operators on finite-dimensional Hilbert space (which follows from T_{form}, derived from A1).

(c) A1 + MD jointly define the admissible set $\mathcal{A}(\rho, R)$ of configurations satisfying $\Sigma\varepsilon \leq C$ with $\varepsilon(d) \geq \mu^* n(d)$.

Given (a)–(c): consider any admissible configuration G that contains a generator X whose removal preserves injectivity of the label map $r_i \mapsto [r_i]_G$ — i.e., a generator that does not refine any multiplet distinction. The configuration $G' := G \setminus \{X\}$ is also admissible (it satisfies the same capacity bound and isotropy condition; removing a channel only relaxes them). By (a) and (b), $\Sigma_{d\in G}\varepsilon(d) - \Sigma_{d\in G'}\varepsilon(d) = \varepsilon(X) \geq \varepsilon^* > 0$, so G' is strictly cheaper than G inside the same admissible set. By A2 (minimum cost), the realised configuration $G_{\text{realized}}(\rho, R) = \arg\min_{G\in\mathcal{A}(\rho,R)} \Sigma\varepsilon(d)$ cannot contain X. Iterating over all redundant generators: the realised admissible structure contains no redundant generators. □

Status of the premises: derived, not additional. A careful reading of Lemma 2.11 might suggest that EC and CM import "extra" assumptions beyond A1. This would be a genuine vulnerability if the required premises were independently postulated. In fact, every premise in the dependency chain is derived:

Premise	**Content**	**Derived from**	**Reference**
T_{sep}	Sector decomposition of S_Γ	A1 + T_{form}	P1S §2
L_{loc}	Independent interfaces	A1 + FD1–FD2	P1S §4
FD3	Anchor-set locality	Foundational definition	P1S §1
Gauge ident.	$G = \text{Inn}(\mathcal{A})$	T_{alg} + T2a + Skolem–Noether	This supplement
L_{ε^*}	Cost floor $\varepsilon^* > 0$	A1 + MD	P1S §3
Gen.-channel bridge	$\dim(\mathfrak{g})$ channels	Faithfulness + T_{form}	This supplement
A2	Min-cost selection over $\mathcal{A}(\rho, R)$	Second axiom (P1, §2)	P1, §2; P1S §2

The only external inputs beyond the two named axioms (A1, A2) are the regularity condition MD (enforcement isotropy), which enters through L_{ε^*}, and FD3 (anchor-set locality), a foundational definition rather than an empirical assumption. The remaining premises are mathematical consequences of A1 (or A1 + MD) and the upstream chain. The dependency for CM in the table now reads A1 + MD + L_{ε^*} + generator–channel bridge (which together define the admissible set with cost floor) + A2 (which selects the min-cost member).

The correct characterization is therefore: *EC follows from A1 alone, via the derived structural separation. CM follows from A2 applied to the admissible set defined by A1 + MD: A1 bounds total cost and admits every configuration satisfying that bound; MD gives the positive cost floor (via L_{ε^*}) that makes a redundant generator strictly costly; A2 then selects the minimum-cost member of the admissible set, which by construction contains no redundant generators.* The earlier-version claim "EC and CM follow from A1" was an overclaim on two counts: EC remains an A1 result, but CM is properly an A2 result (operating on the A1 + MD-defined admissible set), and even the A1 + MD reading attributed to CM in the previous version of this text understated the role of A2 as the selection axiom. Neither correction imports a modeling assumption beyond the framework's stated axioms: A2 is one of the two axioms named in the Paper 1, §2 box, and MD is the regularity input also listed in the Paper 1 axiom inventory.

CM is not a selection principle (descriptive reading). Capacity minimality is sometimes described as "Occam's razor for gauge structure," but this risks making it sound like an aesthetic preference or, worse, a dynamical mechanism that ranges over candidate structures and rejects the non-minimal ones. Neither reading is correct. Lemma 2.11 establishes CM as a *consequence of A2* applied to the admissibility set defined by A1 + MD: a configuration with a redundant generator is admissible (it satisfies the capacity bound and isotropy condition), but it sits at strictly higher total cost than the same configuration with the redundant generator removed (also admissible, by $\varepsilon^* > 0$). A2 selects the cheaper one as the realised structure. The generator–channel bridge (Lemma 2.10) makes the gap quantitative: each redundant generator costs exactly ε^*. Read correctly, CM is therefore a *descriptive label* on the realised admissible structure, not an algorithm: the realised configuration is the $\arg\min_G \dim G$ subject to EC, and that argmin is exactly what A2 fixes. No dynamical search is performed and none is needed; the framework reads off the referent of the two axioms (A1 admits, A2 selects).

The unique abelian factor result is split into two levels. **Theorem A** (Proposition 2.12) is template-independent: it shows that for *any* multiplet set $\mathcal{M}$ and *any* block decomposition, EC and CM determine the number of admissible abelian factors from the separation deficiency δ

alone. **Theorem B** (Theorem 2.18) is the SM specialization: it plugs in $\{n_k\} = \{3, 2, 1\}$ and $\mathcal{M} = \mathcal{M}_{\mathrm{SM}}$ and outputs "exactly one U(1), uniquely hypercharge."

Proposition 2.12 (Unique minimal abelian grading factor — structural (Theorem A))**.** *Let $\mathcal{A} \cong \bigoplus_{\alpha=1}^{m} M_{n_\alpha}(\mathbb{C})$ be the enforcement algebra at a fixed interface, with non-abelian gauge factor $G_{\mathrm{na}} = \prod_{\alpha=1}^{m} \mathrm{SU}(n_\alpha)$ and relative-phase torus $T_{\mathrm{rel}} := Z(\mathrm{U}(\mathcal{A}))/\mathrm{U}(1)_{\mathrm{diag}} \cong \mathrm{U}(1)^{m-1}$. Let $\mathcal{M} = \{r_1, \ldots, r_N\}$ denote the set of physically distinct matter multiplet types occurring in the derived template, regarded as irreducible carriers of G_{na}.*

Assume that G_{phys} is an admissible gauge structure (Definition 2.8; EC and CM hold by Lemma 2.11).

Lemma 2.13 (Abelian factors arise only from T_{rel})**.** *Every continuous abelian gauge factor of G_{phys} is a one-dimensional subgroup of T_{rel}. No abelian gauge factor can arise from any other source.*

Proof. By the exact sequence (Proposition 2.7):

$$1 \to Z(\mathrm{U}(\mathcal{A})) \to \mathrm{U}(\mathcal{A}) \xrightarrow{\pi} \mathrm{Inn}(\mathcal{A}) \to 1.$$

$\mathrm{Inn}(\mathcal{A}) = \prod_\alpha \mathrm{PU}(n_\alpha)$ is semisimple (each $\mathrm{PU}(n)$ is semisimple for $n \geq 2$, and trivial for $n = 1$): it contains no continuous abelian subgroups. The only abelian continuous subgroups of $\mathrm{U}(\mathcal{A}) = \prod_\alpha \mathrm{U}(n_\alpha)$ that project to the identity under π are contained in $\ker(\pi) = Z(\mathrm{U}(\mathcal{A}))$. Quotienting by the physically unobservable diagonal $\mathrm{U}(1)_{\mathrm{diag}}$ gives $T_{\mathrm{rel}} = Z/\mathrm{U}(1)_{\mathrm{diag}}$ as the full group of continuous abelian gauge transformations. Any abelian gauge factor is therefore a one-dimensional subgroup of T_{rel}.

The argument excludes the following loopholes:

(a) *Abelian subgroups of* $\mathrm{Inn}(\mathcal{A})$*?* $\mathrm{PU}(n)$ has a maximal torus of dimension $n-1$, but these tori consist of *non-abelian* gauge transformations (diagonal matrices in $\mathrm{SU}(n)$, which are non-central for $n \geq 2$). They are already counted in G_{na} as Cartan generators, not as independent abelian factors.

(b) *Discrete abelian factors?* Discrete subgroups (e.g., $\mathbb{Z}_N$) can survive as global symmetries but do not contribute continuous gauge generators. CM counts continuous enforcement channels; discrete symmetries have no per-generator capacity cost.

(c) *External abelian factors?* Any abelian factor not contained in $\mathrm{U}(\mathcal{A})$ would act non-trivially on $\mathcal{H}_\omega$ but not arise from the enforcement algebra. This contradicts the gauge identification theorem (Theorem 2.5): the gauge group is $\mathrm{Inn}(\mathcal{A})$ (plus the central extensions from the exact sequence), and no external group acts.

□

Definition 2.14 (Separation deficiency)**.** The *separation deficiency* of G_{na} on $\mathcal{M}$ is the number of pairs of physically distinct multiplet types that share identical non-abelian quantum numbers:

$$\delta(G_{\mathrm{na}}, \mathcal{M}) := \left|\{(r_i, r_j) : i < j,\ [r_i]_{G_{\mathrm{na}}} = [r_j]_{G_{\mathrm{na}}}\}\right|.$$

EC is satisfied by G_{na} alone iff $\delta = 0$. If $\delta > 0$, additional abelian grading is required to resolve the remaining degeneracies.

Lemma 2.15 (Minimum number of abelian factors from separation deficiency)**.** *Let $\mathcal{P} = \{P_1, \ldots, P_s\}$ be the partition of $\mathcal{M}$ into equivalence classes under the non-abelian label map (i.e., $P_k = \{r_i : [r_i]_{G_{\mathrm{na}}} = \ell_k\}$ for each distinct non-abelian label ℓ_k). Let $m_{\max} := \max_k |P_k|$ be the size of the largest equivalence class. Then:*

(i) At least $\lceil \log_2 m_{\max} \rceil$ binary distinctions are needed to separate the elements of the largest class.

(ii) A single U(1) *factor assigns a real-valued charge Y_i to each $r_i \in P_k$. If the Y_i are pairwise distinct within each class P_k, then one* U(1) *suffices to separate all multiplets.*

(iii) Therefore, the minimum number of abelian factors is 0 *if* $\delta = 0$*, and* 1 *if* $\delta > 0$ *and there exists a charge assignment* Y *that separates all classes.*

Proof. (i) is standard: $m_{\max}$ elements require $\lceil \log_2 m_{\max} \rceil$ binary attributes to distinguish. (ii): a single continuous U(1) charge $Y \in \mathbb{R}$ has cardinality $\mathfrak{c}$, so it can separate any finite number of elements — the condition is that the charge assignment $Y : P_k \to \mathbb{R}$ is injective on each class. (iii) combines (i) and (ii): if $m_{\max} \geq 2$, at least one abelian factor is needed; if an injective Y exists, one suffices. □

Lemma 2.16 (Diagonal rational grading from Schur + anomaly cancellation)**.** *Each* $\mathrm{U}(1) \subset T_{\mathrm{rel}}$ *assigns a rational charge to each* r_i*, constant on that multiplet type. Both properties are derived:*

(a) ***Diagonal action (Schur).*** *Each* r_i *is an irreducible representation of* $G_{\mathrm{na}} = \prod_\alpha \mathrm{SU}(n_\alpha)$*. The relative-phase torus* $T_{\mathrm{rel}} \subset Z(\mathrm{U}(\mathcal{A}))$ *lies in the center of the unitary group. By Schur's lemma, any central element acts on an irreducible representation by a scalar. Therefore each* $\mathrm{U}(1) \subset T_{\mathrm{rel}}$ *acts on* r_i *by a phase* $e^{iY_i\theta}$*, where* $Y_i \in \mathbb{R}$ *depends only on the multiplet type, not on the state within the multiplet. This is diagonal action.*

(b) ***Rational charges (anomaly cancellation).*** *The charges* Y_i *are constrained by gauge anomaly cancellation (§8): the four conditions* $[\mathrm{SU}(2)]^2[\mathrm{U}(1)]$, $[\mathrm{SU}(3)]^2[\mathrm{U}(1)]$, $[\mathrm{grav}]^2[\mathrm{U}(1)]$, $[\mathrm{U}(1)]^3$ *are polynomial equations in* $\{Y_i\}$ *with integer coefficients. Solving the first three linearly and substituting into the fourth yields (Theorem 8.1):* $z^2 - 2z - (N_c^2 - 1) = 0$*, where* $z := Y_u/Y_Q$*. This polynomial is* monic *(leading coefficient* $= 1$*) with integer coefficients. By the rational root theorem for monic integer polynomials, any rational root* p/q *(in lowest terms) must have* $q \mid 1$*, i.e.,* $z \in \mathbb{Z}$*. Indeed,* $z = 1 \pm N_c$ *are integers. Since all other* Y_i *are rational multiples of* Y_Q *(from the linear conditions), and the standard normalization* $Y_Q = 1/6$ *is rational,* $Y_i \in \mathbb{Q}$ *for all multiplets. Rationality of hypercharges is therefore a* theorem *of the anomaly system, not an assumption.*

Both properties were previously listed as assumption (A3). The derivation shows they follow from Schur's lemma (standard representation theory, no additional physical input) and anomaly cancellation (already derived in this supplement). No assumption beyond EC, CM, and the derived gauge + matter content is needed.

Then:

(i) If $\delta(G_{\mathrm{na}}, \mathcal{M}) > 0$*, then at least one nontrivial* $\mathrm{U}(1) \subset T_{\mathrm{rel}}$ *is necessary (from EC: unresolved degeneracies violate enforcement completeness).*

(ii) If there exists a charge assignment $Y : \mathcal{M} \to \mathbb{Q}$ *such that the augmented label map* $r_i \mapsto ([r_i]_{G_{\mathrm{na}}}, Y(r_i))$ *is injective, then one abelian grading factor is sufficient (Lemma 2.15(ii)).*

(iii) Any additional independent abelian factor Y' *is inadmissible unless it changes the induced partition of* $\mathcal{M}$*; if* Y *already separates all types, every further* U(1) *is redundant and excluded by CM (Lemma 2.10: each redundant generator wastes* ε^* *capacity).*

(iv) Therefore, whenever $\delta > 0$ *and one abelian grading separates all types, there exists exactly one physically admissible abelian gauge factor, unique up to overall normalization and change of basis in* T_{rel}*.*

Proof. Step 1: Necessity (at least one abelian factor). If $\delta(G_{\mathrm{na}}, \mathcal{M}) > 0$, there exist distinct multiplet types $r_i \neq r_j$ with $[r_i]_{G_{\mathrm{na}}} = [r_j]_{G_{\mathrm{na}}}$. By EC (Lemma 2.11), these two physical types must be distinguishable by the gauge structure. Since G_{na} does not separate them, the distinction must come from the remaining gauge freedom. By Lemma 2.13, the only continuous abelian gauge freedom is T_{rel}. Hence at least one nontrivial $\mathrm{U}(1) \subset T_{\mathrm{rel}}$ is necessary. This gives (i).

Step 2: Sufficiency of one factor (explicit separation matrix). We construct the separation matrix S for $\mathcal{M}_{\mathrm{SM}}$. The non-abelian equivalence classes are:

Class	Non-abelian label	Members
P_1	$(3,2)$	$\{Q\}$ (singleton)
P_2	$(\bar{3},1)$	$\{u^c, d^c\}$
P_3	$(1,2)$	$\{L\}$ (singleton)
P_4	$(1,1)$	$\{e^c\}$ (singleton)

The separation deficiency is $\delta = |P_2|(|P_2|-1)/2 = 1$: a single unresolved pair (u^c, d^c). The maximum class size is $m_{\max} = 2$. By Lemma 2.15, one U(1) factor suffices iff it assigns distinct charges to u^c and d^c.

The hypercharge assignment $Y_{u^c} = 2/3 \neq -1/3 = Y_{d^c}$ resolves this degeneracy. The full charge vector $(Y_Q, Y_{u^c}, Y_{d^c}, Y_L, Y_{e^c}) = (1/6, 2/3, -1/3, -1/2, -1)$ is pairwise distinct across all five multiplet types. The augmented label map

$$r_i \mapsto \big([r_i]_{\mathrm{SU}(3)\times\mathrm{SU}(2)},\ Y(r_i)\big)$$

is therefore injective on $\mathcal{M}_{\mathrm{SM}}$: no two multiplets share both their non-abelian quantum numbers and their hypercharge. One U(1) suffices. This gives (ii).

Step 3: No second abelian factor needed or admissible. Suppose a second independent abelian factor with charge assignment $Y' : \mathcal{M} \to \mathbb{Q}$ is added. The augmented map becomes $r_i \mapsto ([r_i]_{G_{\mathrm{na}}}, Y(r_i), Y'(r_i))$. Since Y already makes this map injective (Step 2), the additional charge Y' does not refine the partition of $\mathcal{M}$: the partition $\{\{Q\}, \{u^c\}, \{d^c\}, \{L\}, \{e^c\}\}$ is already the finest possible (each class is a singleton). By CM (Lemma 2.11), the generator of Y' consumes $\varepsilon^* > 0$ capacity (Lemma 2.10) with zero enforcement gain. Under finite capacity (A1), this is inadmissible. This gives (iii).

Step 4: Uniqueness up to normalization. The surviving abelian factor is a one-dimensional subgroup of $T_{\mathrm{rel}} \cong \mathrm{U}(1)^{m-1}$. In the SM case, $m = 3$ (blocks M_3, M_2, M_1), so $T_{\mathrm{rel}} \cong \mathrm{U}(1)^2$. The requirement that the charge assignment resolves the (u^c, d^c) degeneracy and satisfies all four anomaly conditions (§8) determines a *unique direction* in T_{rel}, up to normalization ($Y \mapsto \lambda Y$, $\lambda \in \mathbb{Q}^\times$). This uniqueness is proved in Theorem 8.1: the anomaly system has exactly two solutions $z = 1 \pm N_c$ (with $N_c = 3$: $z = 4$ and $z = -2$), which correspond to the same direction under the $u \leftrightarrow d$ relabeling symmetry. Therefore the abelian factor is unique. This gives (iv). □

Corollary 2.17 (Application to the derived Standard Model template)**.** *For the derived multiplet set $\mathcal{M}_{\mathrm{SM}} = \{Q, u^c, d^c, L, e^c\}$, the non-abelian labels under $\mathrm{SU}(3)\times\mathrm{SU}(2)$ are $Q \sim (3,2)$, $L \sim (1,2)$, $u^c \sim (\bar{3},1)$, $d^c \sim (\bar{3},1)$, $e^c \sim (1,1)$. Hence the non-abelian label map is not injective (it conflates u^c with d^c): $\delta = 1$. Hypercharge resolves this degeneracy with pairwise distinct charges. By Proposition 2.12, exactly one abelian gauge factor is necessary and sufficient.*

The preceding chain of lemmas, proposition, and corollary assembles the result from reusable components. The reviewer asks for a single self-contained theorem that takes the exact-sequence data as input and outputs "exactly one U(1)" without requiring the reader to trace through the intermediate results. We provide this.

Theorem 2.18 (Unique abelian factor from exact-sequence data — SM specialization (Theorem B))**.** ***Hypotheses:***

(H1) ***Enforcement algebra.*** *$\mathcal{A} = M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C}$ (Wedderburn blocks $\{n_k\} = \{3,2,1\}$, from T2a + Theorem R).*

(H2) ***Exact sequence.*** *$1 \to Z(\mathrm{U}(\mathcal{A})) \to \mathrm{U}(\mathcal{A}) \xrightarrow{\pi} \mathrm{Inn}(\mathcal{A}) \to 1$, with $\mathrm{Inn}(\mathcal{A}) = \mathrm{PU}(3)\times\mathrm{PU}(2)$, $\ker(\pi) = \mathrm{U}(1)^3$, $T_{\mathrm{rel}} = \ker(\pi)/\mathrm{U}(1)_{\mathrm{diag}} \cong \mathrm{U}(1)^2$ (Proposition 2.7).*

(H3) ***Matter content.*** *$\mathcal{M} = \{Q(3,2),\ u^c(\bar{3},1),\ d^c(\bar{3},1),\ L(1,2),\ e^c(1,1)\}$ (from T_{field}, §7).*

(H4) ***EC and CM.*** *The gauge structure satisfies enforcement completeness (EC: the label map is injective) and capacity minimality (CM: no generator can be removed while preserving EC); EC is derived from A1 alone, CM from A2 over the A1 + MD-defined admissible set (Lemma 2.11).*

Conclusion: *The physical gauge group contains exactly one continuous abelian factor, which is a uniquely determined one-dimensional subgroup of* $T_{\mathrm{rel}} \cong \mathrm{U}(1)^2$*. This factor is hypercharge* $\mathrm{U}(1)_Y$*, with charge assignments* $(Y_Q, Y_{u^c}, Y_{d^c}, Y_L, Y_{e^c}) = (1/6, 2/3, -1/3, -1/2, -1)$ *up to normalization and the* $u \leftrightarrow d$ *relabeling.*

Proof. The proof proceeds in five steps: source exclusivity, necessity, sufficiency, uniqueness of direction, and redundancy of any second factor.

Step 1 (Source exclusivity): every abelian factor lives in T_{rel}*.* By (H2), $\mathrm{Inn}(\mathcal{A}) = \mathrm{PU}(3) \times \mathrm{PU}(2)$ is semisimple (both factors are semisimple for $n \geq 2$). A semisimple group has no continuous abelian quotients: every connected abelian subgroup of $\mathrm{PU}(n)$ is a maximal torus (the image of the Cartan subalgebra of $\mathfrak{su}(n)$), which is part of the non-abelian structure. The only continuous abelian subgroups of $\mathrm{U}(\mathcal{A})$ that project to the identity under π are in $\ker(\pi) = Z(\mathrm{U}(\mathcal{A}))$. Quotienting by the unobservable $\mathrm{U}(1)_{\mathrm{diag}}$ gives $T_{\mathrm{rel}} \cong \mathrm{U}(1)^2$. Any abelian gauge factor is a one-dimensional subgroup of this torus.

Step 2 (Necessity): at least one $\mathrm{U}(1)$ *is required.* By (H3), the non-abelian label map $r_i \mapsto [r_i]_{\mathrm{SU}(3)\times\mathrm{SU}(2)}$ is:

r_i	Q	u^c	d^c	L	e^c
$[r_i]$	$(3,2)$	$(\bar{3},1)$	$(\bar{3},1)$	$(1,2)$	$(1,1)$

The map sends both u^c and d^c to $(\bar{3},1)$. Therefore the label map is not injective: $\delta(\mathrm{SU}(3) \times \mathrm{SU}(2), \mathcal{M}) = 1$. By EC (H4), all five types must be distinguished by the gauge structure. Since the non-abelian labels fail to separate u^c from d^c, at least one abelian factor from T_{rel} is required.

Step 3 (Sufficiency): one $\mathrm{U}(1)$ *resolves all degeneracies.* A one-dimensional subgroup $\mathrm{U}(1) \subset T_{\mathrm{rel}}$ assigns a rational charge Y_i to each r_i (by Schur's lemma: central elements act by scalars on irreducible representations; by the anomaly system: the charges are rational, §8). The augmented label map $r_i \mapsto ([r_i]_{\mathrm{SU}(3)\times\mathrm{SU}(2)}, Y_i)$ is injective iff $Y_{u^c} \neq Y_{d^c}$ (this is the only unresolved pair from Step 2; the other three types already have distinct non-abelian labels). The anomaly system (Theorem 8.1) determines:

$$Y_{u^c} = z \cdot Y_Q, \qquad Y_{d^c} = (2 - z) \cdot Y_Q,$$

where z satisfies $z^2 - 2z - (N_c^2 - 1) = 0$ with $N_c = 3$, giving $z \in \{4, -2\}$. For either root: $Y_{u^c} - Y_{d^c} = (2z - 2)Y_Q \neq 0$ (since $z \neq 1$). Therefore one $\mathrm{U}(1)$ suffices.

Step 4 (Uniqueness of direction): the anomaly system admits a unique direction in T_{rel}*.* $T_{\mathrm{rel}} \cong \mathrm{U}(1)^2$ is a two-dimensional torus. A one-dimensional subgroup is specified by a direction $\mathbf{q} = (q_1, q_2) \in \mathbb{Z}^2 \setminus \{0\}$ (up to positive rescaling). The charge of multiplet r_i is $Y_i = q_1 \cdot c_1(r_i) + q_2 \cdot c_2(r_i)$, where c_1, c_2 are the canonical $\mathrm{U}(1)$ charges from the center $Z = \mathrm{U}(1)_3 \times \mathrm{U}(1)_2 \times \mathrm{U}(1)_1$. (In the canonical basis: $c_1(r_i) = n_3$-ality of the SU(3) rep, $c_2(r_i) =$ SU(2) isospin parity.)

The four anomaly conditions (§8) are four polynomial equations in (q_1, q_2). Three are linear; the fourth is cubic. The linear conditions determine the charge ratios Y_{u^c}/Y_Q, Y_{d^c}/Y_Q, Y_L/Y_Q, Y_{e^c}/Y_Q as functions of a single parameter $z = Y_{u^c}/Y_Q$. The cubic condition reduces to the quadratic $z^2 - 2z - 8 = 0$ with roots $z = 4$ and $z = -2$. These two roots correspond to the same direction in T_{rel} (they are related by the $u \leftrightarrow d$ relabeling symmetry of the fermion template). Therefore the anomaly system admits a *unique direction* $\mathbf{q}$ in T_{rel}, up to normalization and the discrete relabeling. ("Admits" rather than "selects": no mechanism ranges over candidate directions; the admissibility conditions — four anomaly equations plus EC, CM — have a unique non-trivial common solution up to the stated equivalences, and that solution is the referent of the admissible hypercharge.)

Step 5 (No second factor): CM excludes any additional $\mathrm{U}(1)$*.* After Step 3, the augmented label map $r_i \mapsto ([r_i]_{\mathrm{SU}(3)\times\mathrm{SU}(2)}, Y_i)$ is injective on all five multiplet types (the full charge vector

$(1/6, 2/3, -1/3, -1/2, -1)$ is pairwise distinct). The partition of $\mathcal{M}$ into equivalence classes under this map is $\{\{Q\}, \{u^c\}, \{d^c\}, \{L\}, \{e^c\}\}$ — the discrete partition (all singletons).

A second independent $\mathrm{U}(1)' \subset T_{\mathrm{rel}}$ would assign a second charge Y_i' to each multiplet. The augmented map $r_i \mapsto ([r_i], Y_i, Y_i')$ induces the same discrete partition (singletons cannot be further refined). Therefore Y' provides zero additional enforcement information. By CM (H4), the generator of $\mathrm{U}(1)'$ occupies one enforcement channel (Lemma 2.10) at cost $\varepsilon^* > 0$ (L_{ε^*}) with zero enforcement gain. Under finite capacity (A1), this is inadmissible.

Conclusion. Steps 1–5 establish: exactly one continuous abelian factor exists in the physical gauge group; it is a one-dimensional subgroup of $T_{\mathrm{rel}} \cong \mathrm{U}(1)^2$; the anomaly system selects a unique direction; and no second factor is admissible. The physical gauge group is therefore $[\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y]/\mathbb{Z}_6$ (Proposition 2.20). □

Why the theorem is not heuristic. The selection of one U(1) from the $(K-1)$-torus involves three independently verifiable steps, none of which is an aesthetic or heuristic choice:

- *Necessity* (Step 2) is a counting argument: $\delta > 0$ combined with EC forces at least one abelian factor. This is a finite combinatorial check on the five multiplets.
- *Uniqueness of direction* (Step 4) is a number-theoretic result: the anomaly system is a determined system of polynomial equations with a unique projective solution. This is verified by explicit computation (the discriminant of $z^2 - 2z - 8 = 0$ is 36; the roots are 4 and -2; they are related by the relabeling symmetry).
- *Redundancy exclusion* (Step 5) is a capacity argument: the partition is already maximally fine (all singletons), so no new information can be added. This is a logical tautology combined with the generator–channel bridge (each generator costs $\varepsilon^* > 0$).

The only conditional input is the matter content $\mathcal{M} = \mathcal{M}_{\mathrm{SM}}$ (H3), which is derived in T_{field} (§7). Given that input, the conclusion is a theorem, not a selection.

Logical ordering and the dependence on T_{field}. Proposition 2.12 is a conditional result: it shows that for *any* fixed set of non-abelian multiplets $\mathcal{M}$ satisfying EC and CM, the number of admissible U(1) factors is uniquely determined by the multiplet separation problem. The physical conclusion "exactly one U(1)" requires the input $\mathcal{M} = \mathcal{M}_{\mathrm{SM}}$ from T_{field} (§7), which is proved independently. The logical order is:

$$\text{Theorem R} \;\to\; \text{gauge template} \;\to\; T_{\mathrm{field}} \;\to\; \text{Corollary 2.17} \;\to\; L_{\mathrm{hypercharge}}.$$

There is no circularity: T_{field} does not use the abelian factor (the scan uses only $\mathrm{SU}(3) \times \mathrm{SU}(2)$ quantum numbers); the abelian factor is determined afterwards.

Edge case: $\mathrm{U}(1)_{B-L}$ with sterile ν_R. A frequently raised alternative is the anomaly-free combination $\mathrm{U}(1)_{B-L}$ (baryon minus lepton number), which becomes anomaly-free when the multiplet set is extended by three right-handed neutrinos $\nu_R(\mathbf{1}, \mathbf{1})$. In this enlarged template $\mathcal{M}' = \mathcal{M}_{\mathrm{SM}} \cup \{\nu_R\}$, both $\mathrm{U}(1)_Y$ and $\mathrm{U}(1)_{B-L}$ are anomaly-free, and the question is whether two independent abelian factors are admissible.

The APF excludes this scenario at two stages:

(a) **T_{field} excludes ν_R from the minimal template.** The fermion scan (§7) derives $\mathcal{M}_{\mathrm{SM}}$ as the unique minimum-DOF survivor at 45 Weyl fermions. The template $\mathcal{M}_{\mathrm{SM}} \cup \{\nu_R\}$ has 48 DOF and survives all anomaly filters, but is non-minimal: it is the second-lightest post-CPT survivor (template #3 in the waterfall table). Since the multiplet set is $\mathcal{M}_{\mathrm{SM}}$, not $\mathcal{M}'$, the $B-L$ direction is not anomaly-free for the *derived* content. (Indeed, $B-L$ has a mixed gravitational anomaly $[\mathrm{grav}]^2[\mathrm{U}(1)_{B-L}] = 3 \neq 0$ unless ν_R is present.)

(b) **Capacity minimality excludes the second U(1) even if ν_R were present.** Suppose counterfactually that $\mathcal{M}' = \mathcal{M}_{\mathrm{SM}} \cup \{\nu_R\}$ were the derived template. Does Proposition 2.12 still exclude a second U(1)? Yes: in $\mathcal{M}'$, the non-abelian labels conflate u^c/d^c (both $(\bar{3}, 1)$) and e^c/ν_R (both $(1, 1)$). A single abelian grading with four distinct charges resolves all degeneracies: one U(1) suffices. $\mathrm{U}(1)_{B-L}$ is then a *linear combination* of this grading

and the non-abelian quantum numbers—not an independent gauge direction. A second independent U(1) would carry capacity cost (one additional enforcement channel) with no additional multiplet-separation power, violating CM.

(c) **Observational cross-check.** The ν_R-extended template predicts $C_{\text{total}} = 64$ and $\Omega_\Lambda = 42/64 = 0.656$, which is $> 5\sigma$ from the Planck measurement $\Omega_\Lambda = 0.6889 \pm 0.0056$ (§9). The 45-fermion template with one U(1) predicts $\Omega_\Lambda = 42/61 = 0.6885$, consistent at $< 0.1\sigma$.

In summary, the "exactly one U(1)" conclusion is robust against the $B-L$ edge case: the extended multiplet set is non-minimal (T_{field}), $B-L$ is anomalous for the minimal template, and even if ν_R were included, $B-L$ would be a redundant linear combination rather than an independent gauge factor.

String-theoretic extra U(1)*'s and Stückelberg masses.* Heterotic string compactifications and F-theory models generically produce multiple abelian gauge factors beyond hypercharge. In these constructions, unwanted U(1)'s acquire large masses via the Stückelberg mechanism: the U(1) gauge boson "eats" a closed-string axion, becoming massive at or near the compactification scale, while the associated global selection rule (e.g., a $\mathbb{Z}_N$ discrete symmetry) may persist at low energies.

In the APF, this scenario has a precise capacity interpretation:

(a) A Stückelberg-massive U(1) is a gauge boson that no longer propagates as a long-range force. In the enforcement language, the corresponding channel is "saturated": its enforcement cost is paid at the compactification scale, and at low energies it is absorbed into the vacuum sector (L_{vac}, §**??**). The channel is counted in C_{total} (as part of the UV structure) but does not contribute an independent low-energy gauge coupling.

(b) Capacity minimality (CM) does not merely select the gauge group with the fewest generators; it selects the group with the fewest *active enforcement channels* at the IR saturation scale. A Stückelberg-massive U(1) is inactive at low energies and therefore excluded from the low-energy gauge group by the same principle that excludes ν_R from the low-energy fermion count: it is above the IR capacity ledger.

(c) The APF is therefore selecting the *phase* in which only one U(1) combination remains both anomaly-free and massless at low energies. This is precisely the phase that heterotic phenomenology identifies as the "Standard Model embedding": among the multiple U(1)'s of the string construction, one linear combination is identified with hypercharge and remains massless, while the others acquire Stückelberg masses and decouple. The APF's capacity-minimality principle provides the *selection criterion* for this phase: the massless U(1) is the one that resolves all low-energy multiplet degeneracies at minimum capacity cost.

Kinetic mixing and capacity accounting. Even if a second U(1)$'$ is excluded as an independent gauge factor by CM, one might worry about *kinetic mixing*: a gauge-invariant term $\frac{\chi}{2} F_{\mu\nu} F'^{\mu\nu}$ coupling the field strengths of hypercharge $\mathrm{U}(1)_Y$ and a putative U(1)$'$. This is a legitimate concern in BSM model-building, where kinetic mixing arises at one loop even if $\chi = 0$ at tree level, provided charged matter couples to both U(1)'s. Three points address this within the APF:

(a) *No second* U(1) *exists to mix with.* Proposition 2.12 proves that exactly one U(1) from the relative-phase torus T_{rel} is admissible. Kinetic mixing requires two abelian gauge fields; if the second does not exist as a propagating field (because CM excludes it), there is no $F'^{\mu\nu}$ to couple to. The concern is therefore moot at the level of the low-energy gauge structure derived here.

(b) *Kinetic mixing as a capacity cost.* If a second U(1)$'$ were present, the kinetic mixing term would be an off-diagonal entry in the 2×2 kinetic matrix $K_{ab} F^a_{\mu\nu} F^{b\,\mu\nu}$. In the enforcement language, this corresponds to a cross-channel correlation: the enforcement cost of the $\mathrm{U}(1)_Y$ channel depends on the state of the U(1)$'$ channel. By the cost-kernel

structure (Theorem 3.1), cross-channel costs are captured by the off-diagonal terms of the bilinear form B. The capacity cost of maintaining a kinetically mixed pair is strictly greater than the cost of a single U(1) (because B is positive-definite: $B(v_1+v_2, v_1+v_2) = B(v_1, v_1) + 2B(v_1, v_2) + B(v_2, v_2) > B(v_1, v_1)$ when $v_2 \neq 0$). CM therefore excludes the mixed pair in favor of the single channel.

(c) *Diagonalization absorbs mixing into charge redefinition.* When two U(1)'s kinetically mix, the physical basis is obtained by diagonalizing K_{ab}. In the diagonal basis, one linear combination couples to matter (this is hypercharge) and the other is either decoupled (capacity-redundant by CM) or massive (Stückelberg mechanism, handled above). The APF's derivation operates in the diagonal basis from the start: the relative-phase torus provides a canonical basis of independent U(1) directions, and Proposition 2.12 selects the unique admissible direction. No off-diagonal mixing can arise because the canonical basis is already diagonal in the cost kernel B.

Physical gauge group. By the gauge identification theorem (Theorem 2.5) and Proposition 2.12, the Lie algebra of the physical gauge group is $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)_Y$. At the global (topological) level:

$$G_{\text{phys}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y}{\mathbb{Z}_6},$$

where the $\mathbb{Z}_6$ quotient is derived in Proposition 2.20 below, $\mathrm{SU}(n_\alpha)$ is the universal cover of $\mathrm{PU}(n_\alpha)$, and $\mathrm{U}(1)_Y$ is the unique relative-phase factor from Proposition 2.12.

The derivation chain produces $\mathrm{Inn}(\mathcal{A}) = \prod_\alpha \mathrm{PU}(n_\alpha)$ as the automorphism group of the algebra, but the physical gauge group that acts faithfully on matter fields is $[\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y]/\mathbb{Z}_6$. The passage between these two descriptions involves (a) a lift from $\mathrm{PU}(n)$ to $\mathrm{SU}(n)$ required by the representation theory of matter, and (b) a question about global structure (center quotients) that determines the topology of the gauge group. We address both formally.

Proposition 2.19 (Lift from $\mathrm{PU}(n)$ to $\mathrm{SU}(n)$)**.** *The physical gauge group acting on matter fields in $\mathcal{H}_\omega$ is $\mathrm{SU}(n)$, not $\mathrm{PU}(n)$, for each Wedderburn block $M_n(\mathbb{C})$ with $n \geq 2$.*

Proof. Step 1 (Projective obstruction). $\mathrm{Inn}(\mathcal{A}) = \prod_\alpha \mathrm{PU}(n_\alpha)$ acts on $\mathcal{A}$ by conjugation: $\alpha_U(a) = UaU^*$. This is the *adjoint* action. Matter fields, however, live in $\mathcal{H}_\omega$, the GNS Hilbert space (Proposition 2.7(v)), which carries the *fundamental* representation of each block: $v \mapsto Uv$ for $v \in \mathbb{C}^n$. The fundamental is a linear representation of $\mathrm{U}(n)$ but only a *projective* representation of $\mathrm{PU}(n)$: the lift $\mathrm{PU}(n) \to \mathrm{GL}(n, \mathbb{C})$ requires choosing a phase for each element, and there is no continuous choice that is globally consistent (the obstruction is the second group cohomology $H^2(\mathrm{PU}(n), \mathrm{U}(1)) \cong \mathbb{Z}_n$).

Step 2 (Linearization by lift to universal cover). The universal covering group of $\mathrm{PU}(n)$ is $\mathrm{SU}(n)$, with covering map $\mathrm{SU}(n) \twoheadrightarrow \mathrm{PU}(n)$ and kernel $Z(\mathrm{SU}(n)) \cong \mathbb{Z}_n$ (the center, generated by $e^{2\pi i/n}\,\mathbf{1}_n$). Every projective representation of $\mathrm{PU}(n)$ lifts to a genuine (linear) representation of $\mathrm{SU}(n)$. The Lie algebras are isomorphic: $\mathfrak{su}(n) = \mathfrak{pu}(n)$, so the gauge field content (connection 1-form A_μ, curvature $F_{\mu\nu}$, covariant derivative D_μ) is identical regardless of whether one uses $\mathrm{SU}(n)$ or $\mathrm{PU}(n)$. The distinction is purely global: $\mathrm{SU}(n)$ admits the fundamental representation while $\mathrm{PU}(n)$ does not.

Step 3 (Carrier requirement forces the fundamental). The carrier requirement R1 (Theorem R, §3) demands a faithful complex carrier of dimension n (from $B1'$). The fundamental representation of $\mathrm{SU}(n)$ on $\mathbb{C}^n$ is the unique faithful irreducible representation of dimension n; the adjoint ($\mathrm{PU}(n)$ on $\mathfrak{su}(n) \cong \mathbb{C}^{n^2-1}$) has dimension $n^2 - 1$ and is not faithful on $\mathbb{C}^n$. R1 therefore forces the fundamental, which in turn forces $\mathrm{SU}(n)$ as the gauge group acting on matter. □

Proposition 2.20 (Global structure of the gauge group)**.** *The gauge group acting on the matter content of the APF is:*

$$G_{\text{phys}} \;=\; \frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)_Y}{\mathbb{Z}_6}\,,$$

where $\mathbb{Z}_6 \cong Z(\mathrm{SU}(3)) \times Z(\mathrm{SU}(2)) \cap \ker(Y)$ *is the subgroup of the center that acts trivially on all matter representations.*

Proof. The group that acts faithfully on matter is not the direct product $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)_Y$ but its quotient by the subgroup of center elements acting trivially on all derived multiplets.

Step 1 (Center identification). The center of $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)_Y$ is $\mathbb{Z}_3\times\mathbb{Z}_2\times\mathrm{U}(1)_Y$. A central element $(e^{2\pi ik/3}\,\mathbf{1}_3,\ (-1)^m\,\mathbf{1}_2,\ e^{iY\theta})$ acts on a multiplet (r_3, r_2, Y_f) by the phase $e^{2\pi ik\,c(r_3)/3}\,(-1)^{m\,t(r_2)}\,e^{iY_f\theta}$, where $c(r_3)$ is the $\mathbb{Z}_3$ triality of the SU(3) representation and $t(r_2)$ is the $\mathbb{Z}_2$ character of the SU(2) representation.

Step 2 (Trivially acting subgroup — explicit verification). A central element acts trivially on *all* derived multiplets iff $e^{2\pi ik\,c(r_3)/3}\,(-1)^{m\,t(r_2)}\,e^{iY_f\theta} = 1$ for every multiplet in the SM template. The derived multiplet list (from the fermion scan, T_{field}) assigns each multiplet definite quantum numbers.

Normalization. We work with the integer-valued charge $Y_{\text{int}} := 6Y$ (so that $Y_Q = 1/6$ becomes $Y_{\text{int}} = 1$, etc.). The $\mathrm{U}(1)_Y$ element $e^{i\theta}$ acts on a multiplet with charge Y_{int} by the phase $e^{iY_{\text{int}}\theta}$. A center element $(e^{2\pi ik/3}\,\mathbf{1}_3,\ (-1)^m\,\mathbf{1}_2,\ e^{i\theta})$ acts on multiplet (c, t, Y_{int}) by:

$$\phi(k, m, \theta) \;=\; e^{2\pi ik\,c/3}\,e^{\pi im\,t}\,e^{iY_{\text{int}}\,\theta}. \tag{4}$$

Setting $\theta = \pi/3$ (the candidate $\mathbb{Z}_6$ generator) and $(k, m) = (1, 1)$:

$$\phi\big(1, 1, \tfrac{\pi}{3}\big) = \exp\Big[\,i\pi\,\frac{2c + 3t + Y_{\text{int}}}{3}\Big].$$

This equals 1 iff $2c + 3t + Y_{\text{int}} \equiv 0 \pmod 6$. We verify this for every multiplet type in one generation of the derived fermion template:

Multiplet	SU(3)	SU(2)	Y	c	t	Y_{int}	$2c{+}3t{+}Y_{\text{int}}$
Q_L	**3**	**2**	1/6	1	1	1	$6 \equiv 0$
u_R	**3**	**1**	2/3	1	0	4	$6 \equiv 0$
d_R	**3**	**1**	−1/3	1	0	−2	$0 \equiv 0$
L_L	**1**	**2**	−1/2	0	1	−3	$0 \equiv 0$
e_R	**1**	**1**	−1	0	0	−6	$-6 \equiv 0$
H (Higgs)	**1**	**2**	1/2	0	1	3	$6 \equiv 0$

All entries satisfy $2c + 3t + Y_{\text{int}} \equiv 0 \pmod 6$. The Higgs doublet $(1, 2, 1/2)$ is included because it participates in gauge interactions; its compatibility is necessary for consistency of the Yukawa sector.

The $\mathbb{Z}_6$ generator is therefore $g := (\omega_3\,\mathbf{1}_3,\ -\mathbf{1}_2,\ e^{i\pi/3})$, and the six elements $\{g^0, g^1, \ldots, g^5\}$ exhaust the kernel. To confirm that the kernel is exactly $\mathbb{Z}_6$ and not a proper subgroup, note that $\mathbb{Z}_3\times\mathbb{Z}_2 \cong \mathbb{Z}_6$ (since $\gcd(2, 3) = 1$), so $\mathbb{Z}_6$ is the largest discrete subgroup of $Z(\mathrm{SU}(3)) \times Z(\mathrm{SU}(2))$. It remains to verify that elements of the center with *trivial* U(1) phase do not act trivially: $(\omega_3,\ \mathbf{1}_2,\ 1)$ acts on Q_L as $\omega_3 \neq 1$, and $(\mathbf{1}_3,\ -\mathbf{1}_2,\ 1)$ acts on Q_L as $-1 \neq 1$. The nontrivial U(1) phase $e^{i\pi/3}$ in the generator g is essential: the kernel is a *diagonal* $\mathbb{Z}_6$ inside $Z(\mathrm{SU}(3)) \times Z(\mathrm{SU}(2)) \times \mathrm{U}(1)_Y$, not the direct product $\mathbb{Z}_3\times\mathbb{Z}_2\times\{1\}$.

Step 3 (Quotient). The faithful gauge group is therefore $[\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)_Y]/\mathbb{Z}_6$. This is the global form of the Standard Model gauge group, consistent with the analysis of Aharony, Seiberg, and Tachikawa (2013) and Tong (2017). □

Corollary 2.21 (Fermion template compatible with quotient)**.** *Every multiplet type in the derived fermion template (T_{field}, §7) defines a well-defined (linear, not merely projective) representation of* $G_{\text{phys}} = [\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y]/\mathbb{Z}_6$.

Proof. A representation of $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y$ descends to a representation of the quotient G_{phys} if and only if the $\mathbb{Z}_6$ kernel acts trivially on that representation. The table above verifies this condition ($2c + 3t + Y_{\text{int}} \equiv 0 \pmod 6$) for each of the five fermion multiplet types in one generation and for the Higgs doublet. Since the three generations carry identical quantum numbers, the check extends to all 45 Weyl fermions (15 types $\times$ 3 generations). No exotic representation outside the $\mathbb{Z}_6$-compatible sublattice appears in the derived matter content.

Conversely, the $\mathbb{Z}_6$ compatibility is not an accident: it follows from the anomaly cancellation conditions (§8), which relate the $\mathrm{U}(1)_Y$ charges to the SU(3) and SU(2) representations. Specifically, the anomaly equation $z^2 - 2z - (N_c^2 - 1) = 0$ with $N_c = 3$ yields $z \in \{4, -2\}$; the resulting charge assignments automatically satisfy the $\mathbb{Z}_6$ condition because the anomaly system forces $Y_{\text{int}} \equiv -(2c + 3t) \pmod 6$ for every multiplet. This is a derived consistency, not a coincidence. □

Why the PSU(n) issue is not a loophole. The reviewer's concern can be sharpened: since $\mathrm{Inn}(\mathcal{A}) = \mathrm{PU}(3) \times \mathrm{PU}(2) \cong \mathrm{PSU}(3) \times \mathrm{SO}(3)$, and neither PSU(3) nor SO(3) admits the fundamental representation, the matter content might appear incompatible with the group that the algebra naturally produces. The resolution is that the algebra and the Hilbert space carry *different* representations of the same underlying structure:

(a) The algebra $\mathcal{A} = M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C}$ is acted on by $\mathrm{Inn}(\mathcal{A})$ via conjugation (adjoint representation). This action is well-defined on $\mathrm{PU}(n)$ because the phase ambiguity cancels: $e^{i\theta} U a U^* e^{-i\theta} = U a U^*$.

(b) The GNS Hilbert space $\mathcal{H}_\omega$ carries a linear action of $\mathrm{U}(\mathcal{A}) = \mathrm{U}(3) \times \mathrm{U}(2) \times \mathrm{U}(1)$ (Proposition 2.7). The fundamental representation $v \mapsto Uv$ requires the full unitary group, not the quotient.

(c) The physical gauge group G_{phys} is the image of $\mathrm{U}(\mathcal{A})$ in $\mathrm{GL}(\mathcal{H}_\omega)$ modulo the trivially acting center. The exact sequence (Proposition 2.7) shows that this image is $[\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y]/\mathbb{Z}_6$: larger than $\mathrm{Inn}(\mathcal{A})$ (because it includes the $\mathrm{U}(1)_Y$ grading from the center) but smaller than $\mathrm{U}(\mathcal{A})$ (because the diagonal U(1) and the $\mathbb{Z}_6$ kernel are removed).

The passage $\mathrm{Inn}(\mathcal{A}) \to G_{\text{phys}}$ is therefore not a "choice" to work with $\mathrm{SU}(n)$ instead of $\mathrm{PU}(n)$; it is a derived consequence of the map $\mathrm{U}(\mathcal{A}) \twoheadrightarrow G_{\text{phys}}$ (Propositions 2.7–2.20), forced by the requirement that matter fields transform linearly and that the kernel of the representation is exactly $\mathbb{Z}_6$.

Physical significance of the global structure. The distinction between $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y$ and $[\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y]/\mathbb{Z}_6$ is invisible at the level of the Lie algebra (perturbative physics), but has observable consequences in the non-perturbative sector:

(a) *Allowed monopole charges.* The Dirac quantization condition $eg = n/2$ depends on the fundamental group $\pi_1(G_{\text{phys}})$. For the quotient group, $\pi_1([\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)]/\mathbb{Z}_6) \cong \mathbb{Z}$, and the minimal monopole charge is 6 times the naive Dirac quantum.

(b) *Allowed line operators.* In the path-integral formulation, the global form determines which Wilson and 't Hooft line operators are allowed. The $\mathbb{Z}_6$ quotient restricts the allowed electric and magnetic charges to a sublattice of the weight lattice.

(c) *Topological sectors.* The distinct homotopy groups of the quotient group vs. the product group affect the classification of instantons and the vacuum structure.

The APF predicts the $\mathbb{Z}_6$ quotient because the derivation proceeds through $\mathrm{Inn}(\mathcal{A})$ — which is $\mathrm{PU}(3) \times \mathrm{PU}(2)$, already a quotient — and then lifts to $\mathrm{SU}(n)$ only as far as the matter representations require. The lift introduces exactly the covering needed for the fundamental representation (Proposition 2.19), but the center elements that act trivially on all matter multiplets remain identified. The global form is therefore a prediction, not an assumption: it follows

from the algebraic derivation chain $\mathrm{Inn}(\mathcal{A}) \to \mathrm{PU}(n) \to \mathrm{SU}(n)/\Gamma$ combined with the specific matter content from the fermion scan.

Reconciliation summary. The three levels of description are:

Group	**Acts on**	**Representation type**
$\mathrm{Inn}(\mathcal{A}) = \prod_k \mathrm{PU}(n_k)$	$\mathcal{A}$ (algebra)	Adjoint (conjugation)
$\mathrm{U}(\mathcal{A}) = \prod_k \mathrm{U}(n_k)$	$\mathcal{H}_\omega$ (Hilbert space)	Fundamental (linear)
$G_{\mathrm{phys}} = [\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)_Y]/\mathbb{Z}_6$	Matter multiplets	Fundamental, faithful

The passage from the first row to the third involves two steps: lifting from $\mathrm{PU}(n)$ to $\mathrm{SU}(n)$ (forced by the fundamental representation, Proposition 2.19), then quotienting by the trivially acting center $\mathbb{Z}_6$ (forced by faithfulness on the derived matter content, Proposition 2.20). The Lie algebras are identical at all three levels; the distinctions are global (topological).

Status of $\mathrm{U}(1)_Y$. The abelian factor is determined by a two-layer argument:

Layer 1: How many abelian directions? Proposition 2.12 (this section): exactly one U(1) from the relative-phase torus T_{rel} is admissible. Enforcement completeness (EC) forces at least one; capacity minimality (CM) excludes all but one.

Layer 2: Which direction? Theorem 8.1 (§8): among all one-dimensional subgroups of T_{rel}, the unique one consistent with all four anomaly-cancellation conditions for the surviving fermion template is hypercharge Y, with assignments $Y_Q = 1/6$, $Y_u = 2/3$, $Y_d = -1/3$, $Y_L = -1/2$, $Y_e = -1$, up to $u \leftrightarrow d$ relabeling and overall normalization.

Neither layer alone suffices. Layer 1 without Layer 2 gives "one abelian factor exists" without identifying it. Layer 2 without Layer 1 gives "these hypercharges cancel anomalies" without explaining why a second independent U(1) is excluded. Together they uniquely determine the Lie algebra of G_{phys} as $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)_Y$; the global form $[\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y]/\mathbb{Z}_6$ is determined by Proposition 2.20.

$\mathrm{U}(1)_Y$ is not an inner automorphism of $\mathcal{A}$; it is a central element of the unitary group that acts non-trivially on matter representations. The non-abelian gauge factors (SU(3), SU(2)) are structural (from $\mathrm{Inn}(\mathcal{A})$); the abelian factor is a grading channel (from T_{rel}, selected by Proposition 2.12 and fixed by Theorem 8.1). Both are derived from A1: the non-abelian factors from non-closure, the abelian factor from enforcement completeness, minimality, and anomaly cancellation. Neither is postulated.

Connection to NCG. This mechanism parallels Connes' "unimodular" condition in NCG, where the same $\mathrm{U}(1)_Y$ emerges from the determinant constraint on the automorphism group of the spectral triple [14]. In the APF, the selection principle is enforcement completeness (Proposition 2.12) rather than unimodularity, but the algebraic mechanism is identical: the abelian factor comes from the *center* of the multi-block unitary group, not from inner automorphisms.

Connection to T3. T3 (Paper 1 Supplement) derives the gauge structure from the Doplicher–Roberts reconstruction theorem, which recovers a compact group from its representation category. The gauge identification theorem above provides the complementary result: it identifies *which* automorphisms constitute the gauge group, and separates them from kinematic automorphisms. Together, T3 and Theorem 2.5 establish that the enforcement algebra produces a non-abelian compact Lie gauge group acting on internal degrees of freedom at each interface.

Connection to noncommutative geometry. The identification $G = \mathrm{Inn}(\mathcal{A})$ parallels a central theme of Connes' NCG program, where the gauge group of the Standard Model arises as the inner automorphism group of an almost-commutative spectral triple [14]. The present framework arrives at the same structural identification from different premises: the algebra $\mathcal{A}$ is derived from A1 (T_{alg}), not postulated as a spectral triple; the inner/outer separation follows from interface locality (L_{loc}) rather than from the order-one condition; and the carrier constraints (Theorem R) replace the KO-dimension and reality conditions. A comparison of the constraint structures is in Appendix A.

State independence. The gauge group $G = \mathrm{Inn}(\mathcal{A})$ depends only on the abstract $*$-algebra $\mathcal{A}$, not on the choice of faithful state ω used in the GNS construction. This is because $\mathrm{Inn}(\mathcal{A})$ is defined purely algebraically: it is the group of automorphisms $\alpha_U : a \mapsto UaU^*$ for U unitary in $\mathcal{A}$. This definition references only the algebra's multiplication and involution, not any representation or state.

For the same reason, the Wedderburn block structure $\mathcal{A} \cong \bigoplus_k M_{n_k}(\mathbb{C})$, the block sizes $\{n_k\}$, and the product $\mathrm{Inn}(\mathcal{A}) = \prod_k \mathrm{PU}(n_k)$ are all ω-independent. Different faithful states produce different GNS Hilbert spaces $\mathcal{H}_\omega$ (with potentially different dimensions if $\mathcal{A}$ is semisimple with $K \geq 2$), but the gauge group is the same in every realization.

Which AQFT algebra does $\mathcal{A}$ correspond to? In the Doplicher–Haag–Roberts (DHR) framework of AQFT, two algebras play distinct roles: the *observable algebra* $\mathcal{A}_{\mathrm{obs}}$ (gauge-invariant quantities accessible to local measurements) and the *field algebra* $\mathcal{F}$ (which includes charged fields that transform non-trivially under the gauge group). The gauge group G acts on $\mathcal{F}$ by automorphisms that fix $\mathcal{A}_{\mathrm{obs}}$ pointwise: $\mathcal{A}_{\mathrm{obs}} = \mathcal{F}^G$, the fixed-point subalgebra.

The APF enforcement algebra $\mathcal{A}$ corresponds to the *field algebra* $\mathcal{F}$, not the observable algebra. This identification is forced by the construction:

(a) $\mathcal{A}$ is generated by all enforcement projections $\{E_d\}$ and joint defenders $\{E_{\{d_i,d_j\}}\}$, including those that act on individual charged sectors (M_{d_k} in the Wedderburn decomposition). An operator that projects onto a single color-triplet subspace is not gauge-invariant: it distinguishes between colors, and a gauge rotation maps it to a different projection. Such operators belong to $\mathcal{F}$, not to $\mathcal{A}_{\mathrm{obs}}$.

(b) Gauge transformations are inner automorphisms of $\mathcal{A}$ (Theorem 2.5): $\alpha_U(a) = UaU^*$ for $U \in \mathcal{U}(\mathcal{A})$. This is precisely the AQFT statement that gauge transformations are inner on the field algebra. In AQFT, gauge transformations are *not* inner automorphisms of $\mathcal{A}_{\mathrm{obs}}$ — they act trivially there (by definition of gauge invariance). No conflict arises because the APF identifies $\mathcal{A} = \mathcal{F}$, not $\mathcal{A} = \mathcal{A}_{\mathrm{obs}}$.

(c) The observable algebra in the APF is the gauge-invariant subalgebra: $\mathcal{A}_{\mathrm{obs}} := \mathcal{A}^G = \{a \in \mathcal{A} : \alpha_U(a) = a \ \forall U \in G\}$. This is derived, not postulated: once $G = \mathrm{Inn}(\mathcal{A})$ is identified (Theorem 2.5), $\mathcal{A}^G$ is determined. The Doplicher–Roberts reconstruction (T3, Paper 1 Supplement) then recovers G from the superselection structure of $\mathcal{A}_{\mathrm{obs}}$, closing the circle.

In summary: the APF's $\mathcal{A}$ is the field algebra $\mathcal{F}$; gauge $= \mathrm{Inn}(\mathcal{A})$ is the standard DHR statement that gauge transformations act innerly on $\mathcal{F}$; and the observable algebra $\mathcal{A}_{\mathrm{obs}} = \mathcal{A}^G$ is the gauge-invariant fixed-point subalgebra, consistent with the standard AQFT framework.

Code reference: `check_T3` in `core.py`.

2.3 Why the gauge group is a compact Lie group

The gauge group $G = \mathrm{Inn}(\mathcal{A})$ is a compact Lie group by direct construction, not by an appeal to general topological-group theory:

Step 1 (Matrix Lie group). $\mathcal{A} \cong \bigoplus_k M_{n_k}(\mathbb{C})$ (Wedderburn–Artin, T2a). By the gauge identification theorem (Theorem 2.5), $G = \prod_k \mathrm{PU}(n_k)$. Each $\mathrm{PU}(n_k) = \mathrm{U}(n_k)/\mathrm{U}(1)$ is a matrix Lie group: it is a closed subgroup of $\mathrm{GL}(n_k^2, \mathbb{R})$ (via the adjoint representation), hence a Lie group by the closed-subgroup theorem (Cartan, 1930).

Step 2 (Compactness). A1 (finite capacity) bounds $\dim(\mathcal{H}_\omega) < \infty$ (T_{form}). $\mathrm{U}(n)$ is compact (closed and bounded in $M_n(\mathbb{C})$); $\mathrm{PU}(n)$ is the continuous image of a compact group, hence compact. A finite product of compact groups is compact. Therefore $G = \prod_k \mathrm{PU}(n_k)$ is compact.

Consequence. The gauge group G is a non-abelian compact Lie group acting faithfully on a finite-dimensional Hilbert space. The classification of such groups reduces to the classification of compact simple Lie algebras (Section 6).

2.4 Finite capacity vs. type III local algebras

The compactness and Lie-group conclusions above are contingent on A1's finite-capacity postulate, which forces $\dim(\mathcal{H}_\omega) < \infty$. This dependence is not confined to the gauge-group derivation: finite dimensionality is load-bearing throughout the supplement. Specifically:

- T_{form} (finite-dimensional substrate) and T_{alg} (finite-dimensional $*$-algebra) are direct consequences of A1 and upstream throughout the derivation chain.
- Wedderburn–Artin (T2a) requires a finite-dimensional semisimple algebra to produce the block decomposition $\mathcal{A} \cong \bigoplus_k M_{n_k}(\mathbb{C})$.
- Gauge-group compactness (§2.3) follows because $\mathrm{U}(n)$ is compact only for finite n.
- The Lie-algebra classification (§6) uses the finite-dimensional Killing form and root-space decomposition.
- Theorem R (carrier prerequisites) and the fermion scan (T_{field}) operate on finite-dimensional representations.
- Capacity counting ($C_{\text{total}} = 61$) assigns a finite integer to the total budget.

The question is therefore not whether finite dimensionality is used — it manifestly is — but whether this use is a physical postulate with independent motivation, or a technical convenience that collapses when confronted with the type III character of local algebras in relativistic QFT.

In algebraic quantum field theory (AQFT), local algebras associated with bounded spacetime regions are generically type III von Neumann factors — infinite-dimensional algebras with no finite-dimensional faithful representations. A natural question arises: how does finite local capacity relate to type III phenomena?

The answer is that the APF does not approximate or truncate AQFT; it starts from a different physical premise. A1 asserts that the total enforcement capacity at any interface is *finite and positive* — this is a foundational postulate about the physical world, not a technical convenience. Independent motivation comes from the covariant entropy bound (Bousso, 1999): the entropy (and hence the number of distinguishable states) accessible through any finite-area surface is bounded by $A/4G\hbar$. A1 is the operational translation of this bound into the enforcement language.

The incompatibility between finite capacity and type III behavior is not merely a heuristic: it is a precise mathematical consequence of A1's finiteness postulate.

Proposition 2.22 (Type III incompatibility)**.** *Let $\mathcal{A}(\Gamma) = M_n(\mathbb{C})$ be the enforcement algebra at an interface with $n = \lfloor C(\Gamma)/\varepsilon^* \rfloor < \infty$ channels. Then:*

(i) ***Faithful trace.*** *$\mathcal{A}(\Gamma)$ admits a faithful normal tracial state $\tau(a) = (1/n)\,\mathrm{Tr}(a)$. In contrast, a type III factor admits no faithful normal semifinite trace (this is the defining property of type III in the Murray–von Neumann classification).*

(ii) ***Minimal projections.*** *$\mathcal{A}(\Gamma)$ contains minimal (rank-1) projections. A type III factor contains no non-zero finite projections.*

(iii) ***Split property.*** *For two interfaces Γ_1, Γ_2 with $\Gamma_1 \cap \Gamma_2 = \emptyset$, the inclusion $\mathcal{A}(\Gamma_1) \vee \mathcal{A}(\Gamma_2) \hookrightarrow \mathcal{A}(\Gamma_1) \otimes \mathcal{A}(\Gamma_2)$ is an isomorphism (the split property holds trivially for finite-dimensional type I factors). In AQFT, the split property requires non-trivial nuclearity estimates and fails for certain type III inclusions.*

Proof. All three statements are standard consequences of $\mathcal{A}(\Gamma)$ being a finite-dimensional matrix algebra (type I_n). (i) is immediate: $\tau(a) = (1/n)\,\mathrm{Tr}(a)$ is faithful (positive on non-zero positive elements), normal (automatic in finite dimensions), and tracial ($\tau(ab) = \tau(ba)$). (ii): the rank-1 projections $|e_i\rangle\langle e_i|$ are minimal and satisfy Murray–von Neumann equivalence $|e_i\rangle\langle e_i| \sim |e_j\rangle\langle e_j|$ via partial isometries $|e_i\rangle\langle e_j|$. (iii): for finite-dimensional algebras with commuting subalgebras (Lemma 2.2), the algebraic tensor product $\mathcal{A}(\Gamma_1) \otimes_{\text{alg}} \mathcal{A}(\Gamma_2)$ is already norm-complete, so the split property is automatic. □

Proposition 2.22 quantifies the incompatibility: A1's finite capacity forces type I local algebras,

which satisfy all three conditions that type III algebras violate. The type III regime is recovered in a controlled limit:

Proposition 2.23 (Controlled limit to type III)**.** *Let $\{\mathcal{A}_n\}_{n\geq 1}$ be the sequence of enforcement algebras $\mathcal{A}_n = M_n(\mathbb{C})$ at interfaces with $n = \lfloor C/\varepsilon^* \rfloor$ channels, equipped with a KMS state $\omega_\beta^{(n)}$ at inverse temperature $\beta > 0$. Define $\lambda := e^{-\beta\varepsilon^*} \in (0,1)$. In the inductive limit $n \to \infty$ (equivalently $C/\varepsilon^* \to \infty$):*

(i) The inductive limit algebra $\mathcal{A}_\infty := \overline{\bigcup_n \mathcal{A}_n}^{\|\cdot\|}$ is the CAR algebra (or UHF algebra of type ∞), which is a separable C^*-algebra.

(ii) The GNS representation of $\mathcal{A}_\infty$ with respect to the product KMS state $\omega_\beta := \bigotimes_i \omega_\beta^{(i)}$ generates a type III_λ Powers factor R_λ (Powers, 1967; Araki–Woods, 1968).

(iii) In the saturation limit $\beta \to 0^+$ (i.e., $C(\Gamma)/\varepsilon^* \to \infty$ at fixed total energy): $\lambda \to 1^-$ and $R_\lambda \to R_1$, a type III_1 factor.

(iv) The faithful trace of $\mathcal{A}_n$ degenerates as $n \to \infty$: the tracial state τ_n does not extend to a normal faithful trace on R_λ. This is the mechanism by which the faithful-trace obstruction to type III (Proposition 2.22(i)) is circumvented.

Proof. (i) is standard: $M_n(\mathbb{C}) \hookrightarrow M_{n+1}(\mathbb{C})$ via the upper-left corner embedding; the norm closure is the UHF algebra. (ii) is the Powers–Araki–Woods classification: the infinite tensor product $\bigotimes_i (M_2(\mathbb{C}), \omega_\lambda^{(i)})$ with ω_λ having eigenvalues $\{\lambda/(1+\lambda), 1/(1+\lambda)\}$ generates R_λ. (iii) follows because R_λ depends continuously on λ, and $R_1 = \lim_{\lambda\to 1^-} R_\lambda$ (Connes, 1973). (iv): in the GNS representation of ω_β, the modular automorphism group σ_t is non-trivial for $\beta > 0$ (KMS condition), so the state is non-tracial; a non-tracial faithful normal state on a factor implies the factor is type III (Takesaki, 1970). □

The two propositions together give the precise relationship between the APF's finite-capacity regime and the AQFT continuum: the finite theory is type I with a faithful trace and minimal projections; the $C/\varepsilon^* \to \infty$ limit is type III_λ (or type III_1 at saturation), with the faithful trace degenerating and minimal projections disappearing. The APF's structural predictions (gauge group, matter content, cosmological fractions) are n-independent discrete invariants of the Wedderburn block structure, so they survive the limit. Continuous quantities (entanglement entropy, correlation functions, KMS temperature) deform smoothly.

Type III structures arise in AQFT as a consequence of the *assumption* that the vacuum has infinite entanglement across any spatial boundary (the Reeh–Schlieder theorem). In the APF, this infinite entanglement is not available: the enforcement budget at each interface is finite, so the cross-interface correlations that produce type III behavior are capacity-bounded. The finite-dimensional algebraic structure derived here should be understood as the *fundamental* description; the type III regime emerges as an effective description in the limit where the interface budget appears unbounded (continuum limit, $C(\Gamma)/\varepsilon^* \to \infty$).

This scope boundary is explicit: the APF's structural results (gauge group, matter content, cosmological fractions) are derived at finite capacity. Deviations from the finite-capacity predictions are expected in the regime where $C(\Gamma)/\varepsilon^* \gg 1$ and type III behavior becomes relevant. The following toy model exhibits the transition concretely.

Toy model: type III emergence at large capacity. Consider two interfaces Γ_1, Γ_2 sharing a pool Π, with total capacity C and cost floor $\varepsilon^* = 1$. Set $n := \lfloor C/\varepsilon^* \rfloor$ (number of enforcement channels per interface).

At finite n. The local algebra is $\mathcal{A}(\Gamma_1) = M_n(\mathbb{C})$ (type I_n: finite matrix algebra). The pool coupling (L_Δ: $\Delta > 0$) produces a global pure state with nonzero entanglement across the interface boundary. In Schmidt decomposition: $|\psi\rangle = \sum_{i=1}^n \sqrt{\lambda_i}\, |i\rangle_1 \otimes |i\rangle_2$ with $\lambda_i > 0$ and $\sum \lambda_i = 1$. The reduced state on Γ_1 is $\rho_1 = \sum_i \lambda_i |i\rangle\langle i|$, a mixed state with entanglement entropy $S = -\sum_i \lambda_i \log \lambda_i \leq \log n$.

As $C \to \infty$ ($n \to \infty$). Three features emerge:

(i) $\dim(\mathcal{A}(\Gamma_1)) = n^2 \to \infty$: the local algebra becomes $\mathcal{B}(\mathcal{H})$ on an infinite-dimensional Hilbert space.

(ii) The entanglement entropy $S \leq \log n \to \infty$: the cross-interface correlations become unbounded.

(iii) If the Schmidt spectrum approaches a thermal distribution ($\lambda_i \propto e^{-\beta E_i}$ for some effective temperature $1/\beta$, which is generic for pool-mediated entanglement in the large-n limit), the reduced state ρ_1 becomes a KMS state. By the Araki–Woods theorem (1968), the GNS representation of $\mathcal{B}(\mathcal{H})$ with respect to a KMS state at inverse temperature $\beta < \infty$ is a type III_1 factor.

This is the standard mechanism by which type III algebras arise in AQFT: restriction to a subregion of a globally pure state with infinite entanglement across the boundary.

Which APF predictions survive? The structural predictions of this supplement are *n-independent* for n above the threshold $n \geq \max_k(n_k) = 3$:

(a) Wedderburn block sizes $\{n_k\} = \{3, 2, 1\}$: determined by the gauge template (Theorem R), which depends on the representation content, not on the total capacity.

(b) Gauge group $G = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y$: algebraic ($\mathrm{Inn}(\mathcal{A})$ depends on block sizes, not on n).

(c) Matter content: 45 Weyl DOF, determined by the fermion scan, independent of n.

(d) Cosmological fractions: 42/61, determined by capacity counting, independent of n.

What changes in the large-n limit: the entanglement entropy diverges, correlation functions acquire power-law tails (from the KMS structure), and the local state becomes non-tracial. These are *continuous-variable* quantities that deform smoothly as n increases; the *discrete-algebraic* quantities (block sizes, gauge group, matter content) are stable.

Systematic dependency audit. The following table maps each key result in the supplement to its specific finite-dimensionality dependence and its behavior in the $n \to \infty$ limit. Results are classified as *stable* (the conclusion holds for all n above a fixed threshold), *limit-compatible* (the conclusion has a well-defined $n \to \infty$ analogue), or *genuinely finite* (the conclusion requires $n < \infty$ and has no direct infinite-dimensional counterpart).

Result	**Finite-dim. use**	**Status**	**Limit behavior**
T_{form}, T_{alg}	$\dim(\mathcal{H}_\omega) < \infty$ from A1	Genuinely finite	Defines the finite-capacity regime; the type III regime is the $C/\varepsilon^* \to \infty$ limit, not a correction.
T2a (Wedderburn)	Semisimple $\Rightarrow \bigoplus_k M_{n_k}$	Stable	Block sizes $\{n_k\}$ are n-independent for $n \geq 3$; the block decomposition of a finite-dimensional subalgebra of a type III factor is unchanged.
T2b ($\mathbb{F} = \mathbb{C}$)	Skolem–Noether on $M_n(\mathbb{C})$	Stable	The field selection argument uses only $n_k \geq 2$, not the total dimension.
Gauge ident. (Thm. 2.5)	$\mathrm{Inn}(\bigoplus_k M_{n_k})$	Stable	Inner automorphism group determined by block sizes, not by n. Survives as the gauge group of the type I subalgebra in any type III factorization.
Compactness (§2.3)	$\mathrm{U}(n)$ compact for finite n	Limit-compatible	$\mathrm{U}(\infty)$ is not compact; however the gauge group is $\prod_k \mathrm{PU}(n_k)$ with fixed n_k, so compactness is unaffected by $n \to \infty$.
Lie-algebra classif.	Killing form on $\mathfrak{g}$	Stable	$\mathfrak{g} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ is determined by block sizes. The root-space decomposition is n-independent.
Theorem R, T_{field}	Finite-dim. representations	Stable	Carrier dimensions $(3, 2, 1)$ are fixed by R1–R3; the fermion scan uses these dimensions, not n.
$C_{\text{total}} = 61$	Finite integer budget	Genuinely finite	The capacity count is the central finite-capacity prediction. In the $n \to \infty$ limit, the ratio $C_{\text{matter}}/C_{\text{total}}$ is the well-defined analogue.
Split property	Trivial for type I_n	Limit-compatible	Requires nuclearity estimates in the type III regime (Buchholz–Wichmann, 1986); holds under standard AQFT assumptions.

Of the eight entries, six are *stable*: their conclusions depend on the Wedderburn block sizes $\{n_k\}$ or the carrier dimensions, not on the total capacity n. The two *genuinely finite* entries ($T_{\text{form}}/T_{\text{alg}}$ and C_{total}) are precisely the places where A1's finite-capacity postulate is doing irreducible physical work. These are not technical conveniences that one might hope to remove; they are the mathematical expression of the physical claim that the enforcement budget is finite.

2.4.1 Reeh–Schlieder and vacuum entanglement

The Reeh–Schlieder theorem (1961) is the principal mechanism by which AQFT produces type III local algebras. A serious confrontation with the type III issue requires stating this theorem and showing precisely how A1 modifies its conclusion.

The theorem (AQFT context). Let $\mathcal{A}(\mathcal{O})$ be the local algebra associated with an open spacetime region $\mathcal{O}$, and let $|\Omega\rangle$ be the vacuum state. The Reeh–Schlieder theorem states: $\overline{\mathcal{A}(\mathcal{O})\,|\Omega\rangle} = \mathcal{H}$ — the vacuum is *cyclic* for every local algebra, meaning that by acting with operators in $\mathcal{A}(\mathcal{O})$ one can approximate any state in the full Hilbert space $\mathcal{H}$ arbitrarily well. A corollary: the vacuum is also *separating* (no nonzero $A \in \mathcal{A}(\mathcal{O})$ annihilates $|\Omega\rangle$).

Why this forces type III. Three steps connect Reeh–Schlieder to type III:

(i) **Infinite entanglement.** If the vacuum is cyclic for the local algebra $\mathcal{A}(\mathcal{O})$, then the vacuum reduced to $\mathcal{O}$ has full-rank density matrix: every eigenvalue of the reduced state $\rho_{\mathcal{O}} = \mathrm{Tr}_{\mathcal{O}^c}|\Omega\rangle\langle\Omega|$ is nonzero. The von Neumann entanglement entropy $S = -\mathrm{Tr}(\rho \log \rho)$ diverges (for type III_1, it is formally infinite).

(ii) **No faithful trace.** A cyclic-and-separating vector $|\Omega\rangle$ for $\mathcal{A}(\mathcal{O})$ defines a faithful normal state $\omega(A) = \langle\Omega|A|\Omega\rangle$. If this state were tracial ($\omega(AB) = \omega(BA)$), the algebra would be type I or type II. The Bisognano–Wichmann theorem (1975) shows that for a Rindler wedge, the modular automorphism group σ_t of ω is the boost — a non-trivial one-parameter group. A tracial state has $\sigma_t = \mathrm{id}$ for all t. Therefore ω is non-tracial, and the algebra is type III.

(iii) **No minimal projections.** In a type III factor, every nonzero projection is Murray–von Neumann equivalent to the identity. This means there are no "pure states of $\mathcal{A}(\mathcal{O})$" in the usual sense: every local state has further internal structure (arbitrarily fine subdivision).

Proposition 2.24 (A1 modifies each step of the RS mechanism)**.** *Under A1 (finite capacity $C(\Gamma) < \infty$ at each interface):*

(i′) ***Bounded entanglement.*** *The enforcement algebra $\mathcal{A}(\Gamma) = M_n(\mathbb{C})$ with $n = \lfloor C/\varepsilon^* \rfloor$ admits at most $\log n$ bits of entanglement across Γ. The vacuum is cyclic for $\mathcal{A}(\Gamma)$ in the trivial sense that $\mathcal{A}(\Gamma)\,|\psi\rangle = \mathbb{C}^{n^2}$ for any $|\psi\rangle$ with full Schmidt rank, but the entanglement entropy is bounded: $S \leq \log n < \infty$. The RS mechanism's infinite entanglement is replaced by a finite upper bound.*

(ii′) ***Faithful trace exists.*** *$\tau(a) = (1/n)\,\mathrm{Tr}(a)$ is a faithful normal tracial state on $M_n(\mathbb{C})$ (Proposition 2.22(i)). The modular automorphism group σ_t^τ is trivial ($\sigma_t^\tau = \mathrm{id}$) because the state is tracial. Non-trivial modular flow (the analogue of boosts and the Unruh effect) emerges only in the $n \to \infty$ limit where τ degenerates (Proposition 2.23(iv)).*

(iii′) ***Minimal projections exist.*** *$M_n(\mathbb{C})$ contains n mutually orthogonal rank-1 projections $|e_i\rangle\langle e_i|$ (Proposition 2.22(ii)). These are the "atoms" of the enforcement algebra: indivisible enforcement states that cannot be further subdivided. Their existence is the operational content of finite capacity: the enforcement budget supports exactly n distinguishable channels, no more.*

Proof. (i′): The Schmidt decomposition of any state $|\psi\rangle \in \mathcal{H}_{\Gamma_1} \otimes \mathcal{H}_{\Gamma_2}$ has at most $\min(n_1, n_2) \leq n$ terms. The entanglement entropy is $S = -\sum_{i=1}^{n} \lambda_i \log \lambda_i \leq \log n$ (with equality for the maximally entangled state $\lambda_i = 1/n$).

(ii′): Proved in Proposition 2.22(i). The tracial condition $\tau(ab) = \tau(ba)$ holds because $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$ for finite-dimensional matrices. The modular automorphism group of a tracial state is trivial by the KMS condition: $\omega(A\,\sigma_t(B)) = \omega(BA)$ for all t requires $\sigma_t(B) = B$, hence $\sigma_t = \mathrm{id}$.

(iii′): Proved in Proposition 2.22(ii). □

Summary. The Reeh–Schlieder theorem is not "wrong" in the APF — it is *capacity-bounded.* Every step of the RS mechanism has a finite-n analogue that reduces to the AQFT statement in the $n \to \infty$ limit. The crucial difference is that A1 caps the entanglement entropy at $\log n$ rather than allowing it to diverge. This cap is what forces the local algebra to be type I (with a faithful trace, minimal projections, and a trivial modular group) rather than type III.

2.4.2 Modular theory and the Unruh effect as a limit

In AQFT, the Tomita–Takesaki modular theory provides the mathematical framework for thermal effects in quantum field theory. For a von Neumann algebra $\mathcal{M}$ with cyclic-and-separating vector $|\Omega\rangle$, the modular automorphism group σ_t acts on $\mathcal{M}$ and satisfies the KMS condition at $\beta = 1$ with respect to the state $\omega(A) = \langle\Omega|A|\Omega\rangle$. The Bisognano–Wichmann theorem (1975) identifies σ_t with the Lorentz boost for a Rindler wedge, giving the Unruh effect: a uniformly accelerated observer sees the vacuum as a thermal state at temperature $T = \hbar a/2\pi c k_B$.

In the APF at finite n:

- The state $\tau = (1/n)\,\mathrm{Tr}$ is tracial, so the modular automorphism group is trivial: $\sigma_t^\tau = \mathrm{id}$. There is no Unruh effect at strictly finite capacity. This is physically correct: the Unruh effect requires infinite energy (perfect uniform acceleration for infinite time), which is incompatible with finite enforcement capacity.
- For a non-tracial state (e.g., the KMS state $\omega_\beta^{(n)}$ at inverse temperature $\beta < \infty$ on $M_n(\mathbb{C})$): the modular automorphism group is $\sigma_t(A) = e^{iHt} A\, e^{-iHt}$ where $H = -\beta^{-1}\log\rho$ and ρ is the density matrix of $\omega_\beta^{(n)}$. This is a non-trivial inner automorphism, but it is periodic (with period β) and remains within the type I_n algebra.
- In the $n \to \infty$ limit: the modular group of the KMS state generates a one-parameter group of outer automorphisms on the type III_λ factor R_λ (Proposition 2.23). The Connes invariant $S(\mathcal{M}) = \overline{\mathrm{Sp}(\Delta_\omega) \setminus \{0\}}$ transitions from the discrete spectrum $\{(\lambda_i/\lambda_j)^{it}\}$ at finite n to the full closed subgroup $\lambda^{i\mathbb{Z}}$ (for type III_λ) or all of $\mathbb{R}_+$ (for type III_1). The Unruh thermal spectrum emerges as the $n \to \infty$ limit of this finite discrete spectrum.

The modular theory is therefore not in conflict with the APF; it is *recovered* in the large-capacity limit. The finite-n theory has a trivial or periodic modular group; the infinite-n limit produces the full continuous modular flow of AQFT.

2.4.3 Positive-definite B vs. AQFT inner products

The reviewer's concern can be sharpened: the APF uses a positive-definite bilinear form B (from L_ω) to implement local support, orthogonal decompositions, and the cost kernel (T_{7B}: $\varepsilon_{\mathrm{ext}}(v) = B(v,v) > 0$). In AQFT, the relevant inner product on the Hilbert space is positive-definite, but the local algebras carry no such positive-definite form: the GNS inner product $\langle\Omega|A^*B|\Omega\rangle$ depends on the global state $|\Omega\rangle$, and the local algebra itself is a type III factor with no faithful trace. How are these compatible?

The resolution is that B and the AQFT inner product are different objects at different structural levels (see also Remark 2.3 for the disambiguation of B from the spacetime metric $g_{\mu\nu}$):

(a) **B is the cost kernel at a single interface.** It measures the enforcement cost of perturbations within the sector decomposition $S_\Gamma = \bigoplus_d M_d \oplus \Pi$. Its positive-definiteness is a consequence of K2 (positive perturbation cost) and SP (no null directions): every nonzero perturbation costs positive capacity. This is a statement about enforcement economics, not about the QFT Hilbert space inner product.

(b) **The Hilbert space inner product is a different object.** The GNS construction produces a Hilbert space $\mathcal{H}_\omega$ with an inner product $\langle\xi|\eta\rangle_\omega$ that depends on the state ω. At finite n, $\omega = \tau = (1/n)\,\mathrm{Tr}$ is the faithful trace, and $\langle A|B\rangle_\tau = \tau(A^*B)$ is the Hilbert–Schmidt inner product. This is also positive-definite, so B and $\langle\cdot|\cdot\rangle_\tau$ coexist without

tension. At infinite n (the type III limit), the GNS inner product $\langle\cdot|\cdot\rangle_\omega$ no longer comes from a trace, and the Hilbert–Schmidt structure degenerates. But B is a *finite-n object*: it lives at the level of the enforcement algebra, not the type III limit.

(c) **The Lorentzian inner product is at the spacetime level.** The spacetime metric $g_{\mu\nu}$ constructed in §5.4 has Lorentzian signature $(-,+,+,+)$. This is not in conflict with B's positive-definiteness because B lives on the *internal* sector space S_Γ at a single interface, while $g_{\mu\nu}$ lives on the *spacetime* tangent space T_pM, which includes the timelike direction from L_{irr}. The positive-definite B is the spatial part of the Lorentzian metric (the $g_{ij}^{(\text{sp})}$ in Proposition 5.11); the negative eigenvalue comes from the irreversibility direction, which is not part of the single-interface enforcement algebra.

2.4.4 Convergence of discrete invariants

The audit table above classifies six results as "stable," meaning their conclusions hold for all n above a fixed threshold. We now prove this formally.

Theorem 2.25 (Stability of structural predictions under $n \to \infty$)**.** *Let $\mathcal{A}_n = M_n(\mathbb{C})$ be the enforcement algebra at an interface with n channels, and let $\mathcal{A}_n^{(\text{phys})} := \bigoplus_{k=1}^K M_{n_k}(\mathbb{C})$ be the physical subalgebra (Wedderburn blocks determined by the gauge template, with $\sum_k n_k^2 \le n^2$). For all $n \ge n_* := \max_k n_k = 3$:*

(i) The block sizes $\{n_k\} = \{3,2,1\}$ are independent of n.

(ii) The gauge group $G = \text{Inn}(\mathcal{A}^{(\text{phys})}) = \text{PU}(3) \times \text{PU}(2)$ is independent of n.

(iii) The matter content (45 Weyl DOF, determined by R1–R3 and the fermion scan) is independent of n.

(iv) These conclusions hold in the inductive limit $\mathcal{A}_\infty = \overline{\bigcup_n \mathcal{A}_n}$: the physical subalgebra $\mathcal{A}_\infty^{(\text{phys})} = \bigoplus_k M_{n_k}(\mathbb{C})$ is a fixed finite-dimensional subalgebra of $\mathcal{A}_\infty$, and its inner automorphism group, representation theory, and Wedderburn decomposition are unchanged.

Proof. (i)–(iii): The block sizes are determined by Theorem R (carrier requirements R1–R3), which depends on the representation content of the gauge template, not on the total dimension n. Specifically: R1 requires a faithful complex carrier of dimension ≥ 3; R2 requires a chiral carrier of dimension ≥ 2; R3 requires a singlet. These requirements produce $\{n_k\} = \{3,2,1\}$ regardless of n, provided $n \ge 3$ (so that the dimension-3 block fits). The gauge group is $\text{Inn}(\bigoplus_k M_{n_k}) = \prod_k \text{PU}(n_k)$, which depends only on $\{n_k\}$. The matter content is the output of the fermion scan operating on the representations of G, which are determined by $\{n_k\}$.

(iv): The inductive limit $\mathcal{A}_\infty$ is a UHF algebra (Proposition 2.23(i)). The physical subalgebra $\mathcal{A}^{(\text{phys})} = M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C}$ embeds as a fixed subalgebra of every $\mathcal{A}_n$ for $n \ge 3$ (via the standard embedding into the upper-left corner). In the inductive limit, this embedding survives: $\mathcal{A}^{(\text{phys})} \hookrightarrow \mathcal{A}_\infty$ is a $*$-homomorphism of finite-dimensional C^*-algebras into a separable C^*-algebra. The Wedderburn decomposition of $\mathcal{A}^{(\text{phys})}$ is an intrinsic property of the finite-dimensional subalgebra and is not affected by the ambient algebra. Similarly, $\text{Inn}(\mathcal{A}^{(\text{phys})})$ depends only on the subalgebra, not on $\mathcal{A}_\infty$. Even if $\mathcal{A}_\infty$ generates a type III factor in the GNS representation, the finite-dimensional subalgebra retains its type I structure, its Wedderburn blocks, and its inner automorphism group. □

What this means physically. The gauge group and matter content are *algebraic invariants of a finite-dimensional subalgebra* that is embedded in the full (potentially infinite-dimensional) operator algebra. The type of the ambient algebra (type I at finite n, type III at $n = \infty$) does not affect these invariants, just as the dimension of a vector space does not affect the rank of a fixed finite-dimensional subspace. The structural predictions of the APF are therefore immune to the type I/type III distinction.

2.4.5 Haag duality and locality

In AQFT, Haag duality states that for a double cone $\mathcal{O}$: $\mathcal{A}(\mathcal{O})' = \mathcal{A}(\mathcal{O}')$, where $\mathcal{A}(\mathcal{O})'$ is the commutant and $\mathcal{O}'$ is the causal complement. This relates the locality structure of the algebra to the causal structure of spacetime.

In the APF at finite n:

(a) **Algebraic locality.** Lemma 2.2 establishes: $[\mathcal{A}(\Gamma_1), \mathcal{A}(\Gamma_2)] = 0$ for disjoint interfaces. This is the finite-dimensional analogue of Einstein causality in AQFT.

(b) **Haag duality holds trivially.** For $\mathcal{A}(\Gamma_1) = M_{n_1}(\mathbb{C})$ acting on $\mathcal{H}_{\Gamma_1} \otimes \mathcal{H}_{\Gamma_2}$: $\mathcal{A}(\Gamma_1)' = \mathbf{1}_{n_1} \otimes M_{n_2}(\mathbb{C}) = \mathcal{A}(\Gamma_2)$. The commutant of the local algebra at Γ_1 is the local algebra at the "complement" Γ_2. This is exact in finite dimensions and requires no nuclearity estimates or split-property arguments.

(c) **Violations of Haag duality signal non-trivial topology.** In AQFT, Haag duality can fail for regions with non-trivial topology (e.g., the complement of a torus), and such failures are related to superselection sectors (DHR theory). In the APF, the enforcement algebra at each interface is simple (after Wedderburn decomposition), so such failures do not arise at the single-interface level. The multi-interface topology is captured by the enforcement net $\{\mathcal{A}(\Gamma)\}_{\Gamma \in \mathcal{I}}$, where superselection sectors emerge from the Wedderburn block structure (each block = one sector), not from topological violations of Haag duality.

2.4.6 Independent evidence for finite local capacity

The claim that local algebras are finite-dimensional (type I rather than type III) is not unique to the APF. Four independent lines of evidence from established physics support finite local capacity:

1. Bekenstein–Hawking entropy (1973). A black hole of area A has entropy $S_{\mathrm{BH}} = A/4G\hbar$. The entropy counts the number of distinguishable microstates: $N = e^{S_{\mathrm{BH}}}$. For any region enclosed by a surface of area A, the number of distinguishable states is bounded by $e^{A/4G\hbar}$ — a finite number. A type III factor has infinitely many orthogonal projections; a system with finitely many distinguishable states cannot realize a type III algebra.

2. Bousso covariant entropy bound (1999). The Bekenstein bound is strengthened to a covariant statement: the entropy on any light sheet L of a surface B is bounded by $S(L) \leq A(B)/4G\hbar$. This bound holds in arbitrary spacetimes (not just static ones) and implies that the number of degrees of freedom accessible through any finite-area surface is finite. A1 is the operational translation: $C(\Gamma) < \infty$ for any interface Γ of finite area.

3. Holographic principle (1993–1995). 't Hooft and Susskind proposed that the fundamental degrees of freedom of a spatial region are encoded on its boundary, with a density of one degree of freedom per Planck area. A region of area A contains at most $A/4\ell_P^2$ degrees of freedom. This is a finite integer — exactly the setting of A1. The holographic principle has been confirmed in the AdS/CFT correspondence (Maldacena, 1997), where the boundary CFT has a finite number of degrees of freedom per unit area at any finite cutoff.

4. Black hole information paradox. The unitarity of black hole evaporation (confirmed by the Page curve computation via the island formula, 2019–2020) requires that the black hole interior has a finite-dimensional Hilbert space of dimension $e^{S_{\mathrm{BH}}}$. If the interior algebra were type III, the information capacity would be infinite and the Page curve would not turn over. The resolution of the information paradox is therefore evidence that local algebras associated with gravitational horizons are finite-dimensional.

These four arguments are independent of the APF and rely on established results in general relativity, string theory, and quantum information. They converge on the same conclusion: *the number of distinguishable states in any bounded region is finite.* A1 is the minimal axiomatization of this consensus.

The type III character of AQFT local algebras arises from the continuum idealization: AQFT takes the Hilbert space to be infinite-dimensional *by construction* (Fock space of a free field) before asking structural questions. The APF reverses this order: it derives the algebraic structure at finite capacity and recovers the continuum (including type III behavior) as an emergent limit.

2.4.7 Regime statement

The APF's local enforcement algebras are *finite-capacity algebras*, not full continuum local algebras in the sense of AQFT. This supplement does not claim to reconstruct continuum local QFT; it reconstructs the finite-capacity algebraic precursor from which the continuum emerges in a controlled limit. The precise scope is:

Regime: finite-capacity enforcement algebras

What is proved. The structural predictions of this supplement (gauge group, matter content, capacity counting, cosmological partition) are derived within the regime $n = \lfloor C(\Gamma)/\varepsilon^* \rfloor < \infty$. The local algebras are type I_n ($M_n(\mathbb{C})$), with a faithful trace, minimal projections, and a trivial or periodic modular group. All structural predictions are n-independent for $n \geq 3$ (Theorem 2.25).

What is not proved. This supplement does not derive the full continuum limit $n \to \infty$. Proposition 2.23 shows that the limit exists and produces a type III factor, but the structural stability theorem guarantees only that the *discrete* predictions (block sizes, gauge group, matter content) survive the limit, not that all *continuous* quantities (correlation functions, entanglement entropy, modular flow) converge in a physically controlled way. The full continuum bridge is the subject of Paper 6.

Relationship to AQFT. The finite-capacity regime is related to AQFT as a lattice regularization is related to a continuum QFT: it provides a well-defined algebraic framework at finite resolution from which the continuum theory can be recovered in a limit. The difference is that A1 treats the finite resolution as *fundamental* (motivated by the holographic entropy bound), not as a regulator to be removed. The structural predictions are exact at finite n; they do not require renormalization or continuum extrapolation.

2.4.8 Continuum refinement limit and effective type-III behavior

The regime statement above establishes that APF local algebras are finite-dimensional at every finite enforcement scale. This subsection provides the scaling argument showing how the operational hallmarks of type III local algebras emerge in the continuum refinement limit. The claim is not "the continuum algebra is type III" (that would require proving a von Neumann algebra limit) but rather:

> *APF local algebras are finite-dimensional at every finite enforcement scale, but under interface refinement with diverging capacity density, they converge to an effective continuum regime exhibiting the hallmark operational features that in AQFT are encoded by type-III local algebras.*

Three named hypotheses. The scaling argument requires three assumptions beyond A1, each physically motivated and explicitly flagged.

Definition 2.26 (Refinement Hypothesis (RH))**.** For every macroscopic interface Γ, there exists a family of admissible refinements $\{\Gamma_a\}_{a\in(0,a_0]}$ indexed by resolution $a \downarrow 0$, where a is the

minimal resolved interface patch size. Each Γ_a is a legitimate APF interface with finite budget $C_a(\Gamma)$ and positive floor $\varepsilon_a^* > 0$.

Definition 2.27 (Capacity Density (CD))**.** There exists a nonzero finite channel density $\rho_\Gamma > 0$ such that the number of independent interface channels scales as

$$N_a(\Gamma) := \left\lfloor \frac{C_a(\Gamma)}{\varepsilon_a^*} \right\rfloor \sim \rho_\Gamma \frac{A(\Gamma)}{a^{d-2}}, \tag{5}$$

where $A(\Gamma)$ is the geometric area of the macroscopic interface and d is the spacetime dimension. In $d = 4$: $N_a(\Gamma) \sim \rho_\Gamma A(\Gamma)/a^2$.

Definition 2.28 (Persistent Pooling (PP))**.** The pool sector Π_a and associated superadditive surplus $\Delta_a(\Gamma)$ do not vanish uniformly as $a \to 0$. That is, $\liminf_{a\to 0} \Delta_a(\Gamma)/C_a(\Gamma) > 0$, or at minimum $\Delta_a(\Gamma) \to \infty$.

Motivation. RH is the statement that physical interfaces admit arbitrarily fine admissible refinements — this is the operational content of "spacetime is continuous at scales above the Planck length." CD connects the channel count to the Bousso covariant entropy bound: if the maximal entropy on a light sheet of area A is $A/4G\hbar$, then the channel density is $\rho_\Gamma = 1/4\ell_P^2$ (one channel per Planck area), and the scaling (5) reproduces the holographic area law. PP is the APF-specific analogue of "vacuum correlations persist across every boundary scale" — it asserts that the superadditive cost from cross-interface entanglement (L_Δ) does not switch off under refinement.

Proposition 2.29 (Effective type-III behavior in the refinement limit)**.** *Under hypotheses RH, CD, and PP:*

(i) ***Finite at every scale.*** *For each $a > 0$, the local enforcement algebra $\mathcal{A}_a(\Gamma) \subset M_{N_a}(\mathbb{C})$ is finite-dimensional, hence type I_{N_a}, with a faithful trace $\tau_a = (1/N_a)\,\mathrm{Tr}$ and N_a minimal projections.*

(ii) ***No stable finite factorization.*** *By PP, the pool-supported surplus Δ_a obstructs a uniform inside/outside tensor factorization: the split $\mathcal{H}_a \cong \mathcal{H}_{a,\mathrm{in}} \otimes \mathcal{H}_{a,\mathrm{out}}$ becomes increasingly non-canonical as $a \to 0$, with the correction growing as $\Delta_a \to \infty$. No finite-resolution factorization survives uniformly in the limit.*

(iii) ***Divergent boundary entanglement (area law).*** *By CD, the maximal cross-interface entropy scales as*

$$S_a(\Gamma) \le \log N_a(\Gamma) \sim \log\Big(\rho_\Gamma \frac{A(\Gamma)}{a^{d-2}}\Big) \to \infty \quad (a \to 0). \tag{6}$$

More sharply, if each channel carries $O(1)$ entropy, the total entanglement scales as $S_a(\Gamma) \sim \sigma_\Gamma A(\Gamma)/a^{d-2}$, which is the standard area-law divergence form.

(iv) ***No finite normal trace in the limit.*** *The normalized trace $\tau_a = (1/N_a)\,\mathrm{Tr}$ assigns weight $1/N_a \to 0$ to each fixed finite-rank sector as $N_a \to \infty$. The unnormalized trace diverges. Therefore no physically meaningful finite trace survives the continuum limit if one insists on resolving arbitrarily fine interface distinctions.*

Proof. (i) is immediate from A1 at finite a: $N_a < \infty$ implies $\mathcal{A}_a \subset M_{N_a}(\mathbb{C})$, which is type I_{N_a} (Proposition 2.22).

(ii): At finite a, the pool Π_a mediates cross-interface correlations with surplus $\Delta_a > 0$ (L_Δ). A tensor factorization $\mathcal{H}_a = \mathcal{H}_{\mathrm{in}} \otimes \mathcal{H}_{\mathrm{out}}$ is exact only when $\Delta_a = 0$ (no cross-interface coupling). For $\Delta_a > 0$, the factorization is approximate, with error controlled by Δ_a/C_a. By PP, Δ_a/C_a does not vanish, so the factorization error does not vanish uniformly. In the limit $a \to 0$: if $\Delta_a \to \infty$ (the stronger form of PP), the error grows without bound and no finite factorization survives.

(iii): By CD, $N_a \sim \rho_\Gamma A/a^{d-2} \to \infty$. The entanglement entropy of the maximally entangled state across Γ_a is $\log N_a$. For generic entangled states (pool-mediated, with $\Delta_a > 0$), the entropy is bounded by $\log N_a$ and scales proportionally to the number of entangled channels: $S_a \sim \sigma_\Gamma M_a \sim \sigma_\Gamma A/a^{d-2}$, where σ_Γ is the per-patch entropy contribution. This diverges as $a \to 0$.

(iv): Let P be a fixed rank-k projection in $M_{N_a}(\mathbb{C})$ (e.g., the projection onto a k-dimensional subspace). $\tau_a(P) = k/N_a \to 0$ as $N_a \to \infty$. For the trace to persist as a normal state on a limiting algebra, it would need to assign nonzero weight to every nonzero positive element. But $\tau_a(P) \to 0$ for every fixed P, so the limit trace (if it exists as a state on the inductive limit) cannot be faithful. This is the trace-theoretic characterization of the transition from type I to type III (Proposition 2.23(iv)). □

Three-regime summary.

Regime	**Algebra type**	**Operational character**
Finite enforcement scale ($a > 0$ fixed)	Type I_{N_a} ($M_{N_a}(\mathbb{C})$)	Finite entropy, faithful trace, minimal projections, exact tensor factorization (up to Δ_a corrections).
Large resolution hierarchy ($a \ll a_0$, $N_a \gg 1$)	Type I_{N_a} with very large N_a	For practical purposes behaves like a continuum local algebra: huge entanglement support, nearly non-canonical factorization, trace weight on fixed sectors $\to 0$.
Formal continuum limit ($a \to 0$)	Effective type III	No finite factorization, divergent boundary entanglement, no finite normal trace. Type-III hallmarks emerge, though the formal von Neumann algebra limit is not constructed here.

Connection to the Bousso bound. The capacity density hypothesis CD, combined with $d = 4$ (Proposition 5.12), gives $N_a(\Gamma) \sim \rho_\Gamma A(\Gamma)/a^2$. Identifying $\rho_\Gamma = 1/4\ell_P^2$ (one channel per Planck area, from the Bekenstein–Hawking bound) and $a = \ell_P$ (Planck-scale resolution) gives $N_{\ell_P}(\Gamma) = A(\Gamma)/4\ell_P^2 = S_{\mathrm{BH}}$: the number of channels equals the Bekenstein–Hawking entropy. This is the operational content of the holographic principle within the APF. Removing the cutoff ($a \to 0$) sends $N_a \to \infty$, and the continuum infinities of AQFT (divergent entanglement, type III structure) are reinterpreted as refinement infinities — artifacts of taking a below the Planck scale, where A1 asserts the resolution is fundamental.

What is not claimed. Proposition 2.29 does not construct the von Neumann algebra $\overline{\bigcup_a \mathcal{A}_a}^{\mathrm{wk}}$ or prove it is a type III factor in the Murray–von Neumann classification. That would require specifying the state on the inductive limit and verifying the Connes classification conditions, which is deferred. What is proved is that the *operational signatures* associated with type III (loss of factorization, divergent entanglement, trace degeneration) emerge in the refinement limit under hypotheses RH, CD, and PP. The formal type classification of the limit algebra is a question for Paper 6.

In this sense, APF reinterprets type-III behavior not as fundamental infinite local capacity, but as the continuum-limit shadow of finite-capacity interface algebras under unbounded refinement.

Could A1 be wrong? The deepest version of the reviewer's concern is not about limit behavior but about foundational priority: if AQFT says local algebras are type III, and the APF says they are type I, one of them must be wrong. The APF's position is that this is a question about nature, not about mathematical consistency. Four independent lines of evidence

support finite local capacity (§2.4.6): the Bekenstein–Hawking entropy, the Bousso covariant bound, the holographic principle, and the resolution of the black hole information paradox. These converge on the same conclusion: the number of distinguishable states in any bounded region is finite.

The type III character of AQFT local algebras is an artifact of the continuum idealization (taking $C/\varepsilon^* \to \infty$ before asking structural questions), not a feature of the fundamental theory. The APF makes the opposite choice: it derives structure at finite capacity and recovers type III behavior as an emergent limit. The Reeh–Schlieder theorem is capacity-bounded (Proposition 2.24), the Unruh effect emerges in the large-n modular flow (§2.4.2), Haag duality holds trivially at finite n (§2.4.5), and the structural predictions are stable under $n \to \infty$ (Theorem 2.25).

This is a testable distinction: if the fundamental theory is type III, one expects corrections to the APF's discrete predictions (gauge group, matter content) at sufficiently high resolution; if A1 is correct, these predictions are exact and the type III regime is an approximation. No current experiment distinguishes the two.

3 Carrier prerequisites: metric, sector-uniform irreversibility, and complex type

Theorem R (Paper 2, §3.5) derives three carrier requirements (R1–R3) from admissibility. The derivation chains for R1 and R2 depend on three lemmas originally developed in companion papers. This section provides self-contained proofs of all three, eliminating external dependencies.

The dependency order is: $T_{7B} \to L_{\text{irr,uniform}} \to B1'$. T_{7B} is used by $L_{\text{irr,uniform}}$ (Step 2); $L_{\text{irr,uniform}}$ closes the R2 chain (chiral carrier); $B1'$ closes the R1 chain (complex ternary carrier). $B1'$ is logically independent of $L_{\text{irr,uniform}}$; both appear in this section because both close formerly external dependencies.

All upstream requirements (A1, K2, K3, SP, L_{nc}, L_{loc}, L_{irr}, T_{sep}, T_{GNS}, T3) are proved in the Paper 1 Supplement. No result in this section imports from Papers 3–7.

3.1 Cost kernel from shared-interface non-factorization (T_{7B})

At separate interfaces, enforcement costs are additive by L_{loc} (Paper 1 Supplement [2], §4). At a shared interface Γ, co-located distinctions produce a superadditive surplus $\Delta > 0$ (L_Δ, Paper 1 Supplement [2]). The question is: what mathematical object tracks this non-factorization when enforcement operates across *multiple* directions at a single interface?

The answer is already implicit in the Paper 1 Supplement's derivation chain: the sector-orthogonal bilinear form B (constructed by L_ω from K3) and the characterization of enforcement costs as B-norms (from T_{adj}) together force the external feasibility cost to be an *exact* quadratic form. T_{7B} makes this explicit.

Theorem 3.1 (T_{7B}: Positive-Definite Cost Kernel from Shared-Interface Non-Factorization)**.** *Let Γ be an interface with substrate $S_\Gamma = \bigoplus_d M_d \oplus \Pi$ (T_{sep}) and sector-orthogonal bilinear form B (L_ω, Paper 1 Supplement [2]). Let $\varepsilon_{\text{ext}} : S_\Gamma \to \mathbb{R}_{\geq 0}$ denote the external feasibility cost. Then:*

(i) $\varepsilon_{\text{ext}}(v) = B(v, v)$ for all $v \in S_\Gamma$: the external feasibility cost is an exact quadratic form.

(ii) The associated bilinear form $g := B$ is positive-definite.

(iii) g is unique up to an overall positive scalar (the within-sector gauge freedom of B).

No additional regularity assumption is needed.

Proof. Step 1: Defense costs are B-norms. By T_{adj} (Paper 1 Supplement [2], §5.7), the enforcement projection for distinction d is the B-orthogonal projection $E_d = \pi^{(B)}_{M_d}$ onto its anchor set M_d. The cost of defending d against a perturbation v is the B-norm of the component of v in M_d:

$$\kappa_d(v) = B(E_d\, v,\, E_d\, v) = \|E_d\, v\|_B^2.$$

This is the minimum-cost defense: T_{adj} proves that the B-orthogonal projection minimizes the defense cost among all idempotents with image M_d.

Step 2: K3 gives exact additivity over disjoint supports. By the sector decomposition $S_\Gamma = \bigoplus_d M_d \oplus \Pi$ and K3 (additivity on disjoint supports), the total defense cost decomposes as:

$$\varepsilon_{\text{ext}}(v) = \sum_d \|E_d\, v\|_B^2 + \|\pi_\Pi\, v\|_B^2 = \sum_d B(v_d, v_d) + B(v_\Pi, v_\Pi),$$

where $v = \sum_d v_d + v_\Pi$ is the sector decomposition of v, and π_Π is the B-orthogonal projection onto Π.

Step 3: Sector orthogonality collapses the sum. By L_ω ($B(v_d, v_{d'}) = 0$ for $d \neq d'$, and $B(v_d, v_\Pi) = 0$ for all d):

$$\varepsilon_{\text{ext}}(v) = \sum_d B(v_d, v_d) + B(v_\Pi, v_\Pi) = B\Big(\sum_d v_d + v_\Pi,\ \sum_d v_d + v_\Pi\Big) = B(v, v).$$

The cross-terms vanish by sector orthogonality. Therefore $\varepsilon_{\text{ext}}(v) = B(v, v)$ *exactly*: the quadratic form is not a leading-order approximation but an identity. This gives (i).

Step 4: Positive-definiteness. For $v \neq 0$: $B(v, v) > 0$ because B is positive-definite (L_ω, Paper 1 Supplement). Therefore $g = B$ is positive-definite, giving (ii).

Step 5: Uniqueness. L_ω constructs B with forced inter-sector orthogonality and free within-sector gauge ($B_d \mapsto \alpha_d B_d$, any $\alpha_d > 0$). The choice of capacity units (A1) fixes an overall scale; the remaining gauge is the within-sector normalization convention, which does not affect the geometric structure. This gives (iii). □

Regularity conditions used: A1 (finite capacity), K2 (positive perturbation cost), K3 (disjoint-support additivity), SP (no null directions), T_{sep} (sector decomposition), L_ω (sector-orthogonal bilinear form), T_{adj} (enforcement projections are B-orthogonal).

Status: [P]. No additional regularity assumption (R7B, K3-bilinear, or differentiability) is needed. The quadraticity is *derived* from $L_\omega + T_{\text{adj}}$, not assumed.

Lemma 3.2 (No metric assumed upstream of T_{7B})**.** *The bilinear form B used in T_{7B} is constructed from A1, K3, and FD3 alone. No norm, inner product, quadratic cost law, or metric structure is assumed at any point in the construction chain.*

Proof. The chain (each step uses only preceding outputs):

(1) A1 provides real-valued costs $\varepsilon(d) > 0$.
(2) T_{embed} maps the convex state space into $\mathbb{R}^{N+1}$ using cost coordinates (convexity from OR1; no inner product invoked).
(3) T_{sep} decomposes $S_\Gamma = \bigoplus_d M_d \oplus \Pi$ from K3 + FD3. This is a *pre-metric algebraic direct sum*: no inner product is used (Appendix C.2).
(4) L_ω constructs B as a direct sum of free within-sector forms with *forced* zero cross-terms. The vanishing $B(u_d, v_{d'}) = 0$ for $d \neq d'$ is a *theorem* of K3 (Appendix C.4), not a definition. The within-sector forms B_d are free (normalization gauge).
(5) T_{adj} identifies enforcement maps as B-orthogonal projections (Appendix C.5). Uses only L_ω's inter-sector orthogonality.

T_{7B} then shows $\varepsilon_{\text{ext}}(v) = B(v, v)$ by combining (4) and (5) with K3 (Step 2 of T_{7B}'s proof). The quadratic form is the *output* of this chain, not an input at any stage. □

Why the cost kernel is positive-definite, not Lorentzian. The cost kernel $g = B$ measures enforcement cost at a single interface Γ. Enforcement cost is positive for every nonzero perturbation (K2 + SP), so g is positive-definite. Lorentzian signature $(-, +, +, +)$ arises at a different level: it is the signature of the *inter-interface* causal structure, where the timelike direction is determined by L_{irr} (irreversibility provides the arrow of time) and the spatial directions are determined by L_{loc} (independent interfaces provide spatial separation). The positive-definite g at each interface is the internal (Riemannian) metric; the Lorentzian spacetime metric is constructed in §5.4 (Proposition 5.11) by combining g with the causal ordering of interfaces. The disambiguation of these two bilinear structures is detailed in Remark 2.3. This is the same structural split as in standard GR, where the spatial metric is positive-definite and the Lorentzian signature comes from including the timelike direction.

Roadmap. The full derivation of Lorentzian spacetime from the cost kernel g is given in §5 below. The chain runs: $L_{\text{irr}} \to$ causal partial order $\to$ smooth manifold (Kolmogorov) $\to$ Lorentzian conformal class (Hawking–King–McCarthy) $\to d = 4$ (Lovelock uniqueness + DOF minimality) $\to$ Einstein equations.

Code reference: `check_T7B` in `gravity.py`.

3.2 Sector-uniform irreversibility ($L_{\text{irr,uniform}}$)

L_{irr} (Paper 1 Supplement [2], §5) derives irreversibility from A1 + L_{nc} + L_{loc}: cross-interface correlations commit capacity that no local observer can recover. That proof establishes irreversibility *generically* — it guarantees that irreversible channels exist at interfaces where cross-interface correlations are superadditive. The question for Theorem R's R2 requirement is whether this irreversibility extends to the *gauge sector specifically*: does the sector of internal (non-gravitational) enforcement contain its own irreversible channels, or could it remain fully reversible while inheriting apparent irreversibility from gravity alone?

Theorem 3.3 ($L_{\text{irr,uniform}}$: Gauge-Sector Irreversibility at Saturated Interfaces)**.** *Let Γ be a shared interface with non-trivial pool ($\Pi \neq \{0\}$), let G denote the set of gauge-sector distinctions at Γ, and let R_Γ denote the gravitational record distinctions at Γ. Suppose:*

(a) The gauge sector is non-trivially coupled to gravity at Γ (i.e., the cost kernel g from T_{7B} has nonzero cross-terms between G and R_Γ).

(b) The interface saturation exceeds the bound

$$s \;:=\; \frac{\varepsilon(G) + \varepsilon(R_\Gamma)}{C_\Gamma} \;>\; 1 \;-\; \frac{\Delta(G, R_\Gamma)}{C_\Gamma - \varepsilon(R_\Gamma)}, \tag{7}$$

where $\Delta(G, R_\Gamma) \geq \varepsilon^* > 0$ is the superadditive surplus from L_Δ.

(c) A gauge-sector transition $G_1 \to G_2$ commits positive shared surplus with the record sector: $\Delta(G_2, R_\Gamma) > 0$.

Then the gauge sector contains an irreversible channel at Γ: there exist gauge-sector histories H, H' differing only in gauge distinctions such that the reversal $H' \to H$ is inadmissible.

Proof (three steps). Step 1: Records are interface-local. By L_{loc} (Paper 1 Supplement [2], §4), enforcement is distributed over independent interfaces, each with its own budget. Irreversibility arises because cross-interface correlations commit capacity that no *single-interface* observer can recover (L_{irr}, Step 3). In particular, irreversible records at Γ are distinction sets *at* Γ, not global objects.

Step 2: Gauge–gravity coupling at shared interfaces. By hypothesis (a) and T_{7B} (Theorem 3.1), gauge distinctions G contribute to the cross-terms of the cost kernel g at Γ. Therefore gauge and gravitational enforcement share interface Γ. This sharing implies: there exist admissible histories H, H' that differ by gauge-sector distinctions and yield different gravitational records $R_\Gamma(H) \neq R_\Gamma(H')$. If no such histories existed, gauge distinctions would have no recordable consequences at Γ, contradicting hypothesis (a).

Step 3: Budget overrun from non-closure. Since G and R_Γ coexist at Γ with $\Pi \neq \{0\}$, L_{nc} gives:

$$\varepsilon_\Gamma(G \cup R_\Gamma) \;>\; \varepsilon_\Gamma(G) + \varepsilon_\Gamma(R_\Gamma).$$

Consider the reversal of a gauge transition $G_1 \to G_2$. The forward transition commits cross-interface surplus $\Delta(G_2, R_\Gamma) > 0$ (hypothesis (c)). The reversal $G_2 \to G_1$ while R_Γ persists requires recovering this surplus. By L_{irr}, the surplus includes cross-interface terms that no single-interface operation can free. The reversal cost satisfies:

$$\varepsilon_\Gamma(\text{reverse } G_2 \to G_1 \mid R_\Gamma) \;\geq\; \varepsilon_\Gamma(G_1) + \Delta(G_1, R_\Gamma).$$

The remaining budget after the forward transition is $C_\Gamma - \varepsilon(R_\Gamma) - \varepsilon(G_2)$. By hypothesis (b) (saturation bound (7)), this remaining budget is strictly less than the reversal cost. The reversal is inadmissible. □

Corollary 3.4 (Near-capacity interfaces)**.** *At any interface within one enforcement channel of full capacity ($s > 1 - \varepsilon^*/C_\Gamma$), condition (b) is automatically satisfied because $\Delta \geq \varepsilon^* > 0$*

($L_\Delta + L_{\varepsilon^*}$). Conditions (a) and (c) hold at any interface where the gauge sector is non-trivially coupled and undergoes transitions. Therefore gauge-sector irreversibility is generic at near-capacity shared interfaces.

Regularity conditions used: L_{loc} (locality), L_{nc} (non-closure), L_{irr} (irreversibility from cross-interface correlations), T_{7B} (cost kernel from shared-interface non-factorization), A1 (finite budget), SP (non-trivial gauge coupling).

Scope of the sufficient condition. $L_{\text{irr,uniform}}$ is a sufficient-condition theorem: it guarantees gauge-sector irreversibility at interfaces satisfying hypotheses (a)–(c), not at all interfaces. This is deliberate. The downstream use (R2: chiral carrier requirement) needs only the *existence* of irreversible gauge-sector channels at physically relevant interfaces — not their universality. The corollary shows that the hypotheses are automatically satisfied at near-capacity shared interfaces, which is the regime where the gauge architecture is determined. A universal irreversibility theorem (holding at arbitrarily low saturation) would be false: at very low saturation ($s \ll 1$), the budget is loose enough that all transitions are reversible, and the interface is effectively classical.

Key structural point. The argument does *not* assume that the gauge sector has any specific algebraic structure beyond being a non-trivially coupled set of distinctions at a shared interface. The irreversibility is inherited from the *interface geometry* (T_{7B}) and the *non-closure mechanism* (L_Δ), not from any property specific to non-abelian gauge theories. Chirality (R2) then follows because a *vector-like* gauge sector, despite sharing the interface, produces zero *intrinsic* gauge irreversibility (all gauge-level processes are CPT-reversible; see Paper 2, §3.4). The existence of gauge-sector irreversible channels (this theorem) combined with the impossibility of intrinsic irreversibility in a vector-like theory (Paper 2, §3.5) forces chirality.

Code reference: `check_L_irr_uniform` in `core.py`.

3.3 Complex type from oriented composite robustness ($B1'$)

The R1 carrier requirement (Theorem R) demands a complex carrier: $V \ncong V^*$ as G-modules. The physical reason is that composites must be *oriented* — robustly distinguishable from their anti-composites — to provide independent enforcement channels under T_M. This subsection proves that real and pseudoreal carriers cannot satisfy this requirement: the equivariant anti-linear map J collapses composite orientation at zero capacity cost.

Definition 3.5 (Oriented composite)**.** Let V be a faithful irreducible carrier for a gauge group G, and let $B \in (V^{\otimes k})^G$ be a gauge-singlet composite (a G-invariant tensor in $V^{\otimes k}$). The conjugate composite is $\bar{B} \in (V^{*\otimes k})^G$. The composite B is *oriented* if B and $\bar{B}$ are robustly distinguishable: no admissibility-preserving relabeling internal to the carrier maps $B \mapsto \bar{B}$.

Theorem 3.6 ($B1'$: Complex Carrier from Oriented Composite Robustness)**.** *Let V be a faithful irreducible carrier for a compact group G. Suppose:*

(B1) L_{irr}-stable gauge-singlet composites B and $\bar{B}$ exist whose oriented distinction is robust under admissible refinements.

(B2) This distinction is not enforced by an independent external grading channel (minimality clause from A1).

Then V must be of complex type*: $V \ncong V^*$ as G-modules.*

Proof (by contradiction). Suppose $V \cong V^*$ as G-modules. Then V is either *real type* (admits a G-equivariant $\mathbb{R}$-linear isomorphism $\phi : V \to V^*$ with $\phi = \bar{\phi}$) or *pseudoreal type* (admits a G-equivariant antilinear map $J : V \to V$ with $J^2 = -\text{id}$, but no real structure). In either case, there exists a G-equivariant antilinear map $J : V \to V$.

Step 1: Tensorial extension. Extend J to $V^{\otimes k}$ by

$$J^{\otimes k}(v_1 \otimes v_2 \otimes \cdots \otimes v_k) \;=\; J(v_1) \otimes J(v_2) \otimes \cdots \otimes J(v_k).$$

Since J is G-equivariant ($J \circ g = g \circ J$ for all $g \in G$), $J^{\otimes k}$ is G-equivariant on $V^{\otimes k}$. In particular, $J^{\otimes k}$ maps gauge-invariant subspaces to gauge-invariant subspaces: if $B \in (V^{\otimes k})^G$, then $J^{\otimes k}(B) \in (V^{\otimes k})^G$.

Step 2: Identification of composite with conjugate. We split cases.

Real-type case. If V is real-type, $V \cong V^*$ via a G-equivariant real structure ϕ with $\phi = \bar{\phi}$. The associated $J : V \to V$ satisfies $J^2 = +\mathrm{id}$ (a genuine involution), so $J^{\otimes k}$ is an involution on $V^{\otimes k}$ for every k and provides a direct G-equivariant identification $(V^{\otimes k})^G \leftrightarrow (V^{*\otimes k})^G$. The map $B \mapsto \bar{B}$ is well-defined as a square-trivial involution, and the identification is orientation-collapsing in the sense required below.

Pseudoreal-type case. If V is pseudoreal-type, then by the standard classification (Humphreys 1972; Fulton–Harris 1991, §IV.23) $V \cong V^*$ as G-modules *by definition of pseudoreal type.* The carrier-complexity hypothesis $V \ncong V^*$ of the theorem statement is therefore already false for pseudoreal V, and the proof does not need to invoke any $J^{\otimes k}$ argument in this case. The J-extension machinery used below is only necessary in the real-type case. The earlier formulation, which attempted to handle both types under a single $J^{\otimes k}$ identification and raised a potential signed-tensor subtlety for odd k, is therefore unnecessary: pseudoreal carriers are ruled out by the structural hypothesis $V \ncong V^*$ directly.

Accordingly, the remainder of the proof treats only the real-type case, where $J^{\otimes k}$ is a genuine involution and the identification $(V^{\otimes k})^G \leftrightarrow (V^{*\otimes k})^G$ is unambiguous.

Concretely: let $B = \sum_i c_i\, v_{i_1} \otimes \cdots \otimes v_{i_k}$ be a gauge singlet. Then

$$J^{\otimes k}(B) \;=\; \sum_i \bar{c}_i\, J(v_{i_1}) \otimes \cdots \otimes J(v_{i_k})$$

is also a gauge singlet (by equivariance), and it is the image of B under conjugation composed with the equivariant isomorphism $V \cong V^*$. Therefore $J^{\otimes k}(B)$ is identified with $\bar{B}$.

Step 3: The identification is admissibility-preserving. The map J is internal to the carrier — it is a relabeling of the carrier basis vectors, not a change of physical state. It costs zero enforcement capacity: the enforcement structure at the interface is unchanged because J commutes with all G-actions (equivariance), so the enforcement projections E_d and the cost functional ε are invariant under J. Therefore $J^{\otimes k} : B \mapsto \bar{B}$ is an admissibility-preserving relabeling.

Step 4: Oriented distinction collapses. By Steps 2–3, there exists an admissibility-preserving relabeling (internal to the carrier, at zero capacity cost) that maps $B \mapsto \bar{B}$. By Definition 3.5, the composite B is *not* oriented: the "forward" and "backward" versions collapse into the same equivalence class under admissible relabelings.

Step 5: Contradiction with hypotheses. Hypothesis (B1) requires that the oriented distinction $B \neq \bar{B}$ be robust. Step 4 shows it is not robust if $V \cong V^*$. The only escape is to introduce an independent external grading channel that "protects" the $B/\bar{B}$ distinction from the J-identification. But this violates hypothesis (B2) (minimality under A1: no additional enforcement channel may be introduced solely to protect a distinction that the carrier itself should sustain).

Therefore $V \ncong V^*$: the carrier must be of complex type. □

Counterexample (pseudoreal confinement). $G = \mathrm{SU}(2)$ with fundamental representation $V = \mathbb{C}^2$ (pseudoreal: $V \cong V^*$ via the ε-tensor, $J = i\sigma_2 K$ where K is complex conjugation). The "baryon" $B = \varepsilon_{ij}\, v^i w^j$ is a bilinear singlet. $J^{\otimes 2}(B) = \bar{B}$, and the identification is G-equivariant: there is no robust oriented distinction. "Baryon" and "antibaryon" in an SU(2) confining theory are the same singlet reached by different paths.

Positive example. $G = \mathrm{SU}(3)$ with fundamental representation $V = \mathbb{C}^3$ (complex: $V \ncong V^*$, since the Dynkin labels $[1,0]$ and $[0,1]$ are distinct). No G-equivariant antilinear $J : V \to V$

exists. The baryon $B = \varepsilon_{ijk}\, v^i w^j x^k$ and the antibaryon $\bar{B} = \varepsilon^{ijk}\, \bar{v}_i \bar{w}_j \bar{x}_k$ are robustly distinct: no internal relabeling maps one to the other.

Regularity conditions used: A1 (minimality clause, hypothesis B2), T_M (enforcement independence, which motivates hypothesis B1), L_{irr} (stability of composites, hypothesis B1). The proof itself is pure G-module theory — the only non-algebraic input is the admissibility-preservation criterion (Step 3), which invokes equivariance and zero-cost relabeling.

Consequence for Theorem R. $B1'$ closes the R1 derivation chain. The ternary carrier ($k = 3$, from the trilinear invariant requirement) must be complex type to sustain oriented composites. Combined with faithfulness and minimality ($\dim V = 3$ is the minimum for an irreducible trilinear invariant), R1 is fully derived: a faithful complex 3-dimensional carrier.

Code reference: `check_B1_prime` in `gauge.py`.

3.4 Worked example: three-dimensional interface

We illustrate T_{7B} and $L_{\mathrm{irr,uniform}}$ with the simplest nontrivial interface: $\dim(S_\Gamma) = 3$ with two one-dimensional sectors and a one-dimensional pool.

Setup. $S_\Gamma = \mathbb{R}^3$ with standard basis $\{e_1, e_2, e_3\}$. Sector decomposition (T_{sep}): $M_{d_1} = \mathrm{span}(e_1)$, $M_{d_2} = \mathrm{span}(e_2)$, $\Pi = \mathrm{span}(e_3)$. Enforcement projections: $E_{d_1} = \mathrm{diag}(1, 0, 0)$, $E_{d_2} = \mathrm{diag}(0, 1, 0)$. Individual enforcement costs: $\varepsilon(d_1) = \varepsilon(d_2) = \varepsilon^* = 1$.

Computing Δ. The pool $\Pi = \mathrm{span}(e_3) \neq \{0\}$, so L_Δ applies. A pool-supported perturbation $p = \alpha\, e_3$ with $|\alpha| > 0$ threatens joint enforcement (SP: physically meaningful DOF; K2: positive cost). Defense costs $\kappa_\Pi = \varepsilon^* = 1$ (one pool channel). By K3: $\varepsilon(\{d_1, d_2\}) = 1 + 1 + 1 = 3$. Therefore $\Delta = 1 > 0$.

Computing $\varepsilon_{\mathrm{ext}}$ and verifying T_{7B}. The external feasibility cost for a general perturbation $v = (v_1, v_2, v_3)^T$ is specified by the cost matrix:

$$G \;=\; \begin{pmatrix} 1 & \lambda & 0 \\ \lambda & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \varepsilon_{\mathrm{ext}}(v) = v^T G\, v = v_1^2 + v_2^2 + v_3^2 + 2\lambda\, v_1 v_2,$$

where $\lambda \in [0, 1)$ is the pool-mediated coupling between sectors M_{d_1} and M_{d_2}. The diagonal entries are the individual sector costs; the off-diagonal λ is the cross-term from L_Δ (pool-mediated defense cost); the e_3-diagonal entry is the direct pool cost.

- $\lambda = 0$: sectors uncoupled; $\Delta = 0$ (no cross-term); G is diagonal; the interface is classical.
- $\lambda > 0$: sectors coupled through Π; $\Delta > 0$; G has off-diagonal structure. The pool mediates an enforcement interaction that is invisible to individual-sector accounting.

G is symmetric and positive-definite for $|\lambda| < 1$ (eigenvalues $1 \pm \lambda$ and 1, all positive). $\varepsilon_{\mathrm{ext}}(v) = v^T G\, v$ is manifestly a quadratic form. The polarization identity gives $g(u, v) = u^T G\, v$, which is the bilinear form. All three conclusions of T_{7B} are verified:

(i) $\varepsilon_{\mathrm{ext}}$ is a quadratic form with kernel G.
(ii) G is positive-definite $\Rightarrow$ non-degenerate.
(iii) G is unique up to overall scaling (ε^*).

The connection to Δ: the superadditive surplus is $\Delta = 2\lambda \cdot |v_1 v_2|$, which vanishes iff $\lambda = 0$ (no pool coupling). The quadratic form $v^T G\, v$ is verified directly by the T_{7B} proof: $B = G$ is the sector-orthogonal bilinear form (L_ω), the diagonal entries are B-norms of sector projections (T_{adj}), and K3 gives exact additivity over disjoint supports. The off-diagonal λ is the cross-sector coupling mediated by Π: it arises from $B(v_1, v_2)$ when v_1 and v_2 are not in disjoint sectors but share pool-mediated cost (L_Δ).

Illustrating $L_{\mathrm{irr,uniform}}$. Set $\lambda = 1/2$ and $C_\Gamma = 3$. Let $G = \{d_1\}$ (gauge sector) and $R_\Gamma = \{d_2\}$ (gravitational record), with costs $\varepsilon(G) = \varepsilon(R_\Gamma) = 1$. The superadditive surplus is $\Delta(G, R_\Gamma) = 2\lambda = 1$. Interface saturation: $s = (1 + 1)/3 = 2/3$.

Check hypothesis (b) of $L_{\text{irr,uniform}}$: $s = 2/3 > 1 - \Delta/(C_\Gamma - \varepsilon(R_\Gamma)) = 1 - 1/(3-1) = 1/2$. Satisfied.

After a gauge transition $G_1 \to G_2$, the reversal cost is: $\varepsilon(\text{reverse}) \geq \varepsilon(G_1) + \Delta = 1 + 1 = 2$. The remaining budget is $C_\Gamma - \varepsilon(R_\Gamma) - \varepsilon(G_2) = 3 - 1 - 1 = 1 < 2$. The reversal is inadmissible.

Now set $C_\Gamma = 5$ (loose budget, $s = 2/5$): $s = 2/5 < 1/2$. Hypothesis (b) fails. Remaining budget $= 5 - 1 - 1 = 3 > 2$. The reversal succeeds. The gauge sector is reversible at this low saturation, as the theorem predicts.

3.5 Consolidation: Theorem R prerequisites closed

With T_{7B}, $L_{\text{irr,uniform}}$, and $B1'$ proved in this supplement, the three carrier requirements of Theorem R (Paper 2, §3.5) are now derived entirely within Paper 1, Paper 2, and their respective supplements:

Requirement	**Derivation chain**	**Key new lemma**	**Status**
R1 (complex ternary)	$L_{\text{nc}} \to T_{\text{conf}} \to T_M \to B1'$	$B1'$ (§3.3)	[P]
R2 (pseudoreal binary)	$L_{\text{irr,uniform}} \to T_M \to$ chirality	$L_{\text{irr,uniform}}$ (§3.2)	[P]
R3 (abelian grading)	enforcement completeness + A1 minimality	(none — self-contained in Paper 2)	[P]

No result in this section depends on Papers 3–7. The formerly external dependency on "Paper 7 ($B1'$)" cited in Paper 2's abstract and §3.4 is now resolved.

4 Asymptotic freedom: schematic sign calculation

Paper 2 §3.6–3.7 derives the structural necessity of asymptotic freedom from the non-abelian self-interaction of enforcement channels: gauge bosons carry charge, producing anti-screening that overwhelms matter screening. The full quantitative derivation (one-loop β-coefficient from Casimir arithmetic) is in Paper 4A ($L_{\text{beta,capacity}}$). This section provides the schematic sign calculation using only the matter content derived in this paper.

Proof boundary. The one-loop β-function formula $b_0 = (11/3)C_A - (4/3)T_R N_f$ is a structural result of gauge theory. Paper 4A derives it from capacity arguments ($L_{\text{beta,capacity}}$); here we use it as an imported formula (verifiable in any QFT textbook) and evaluate it with the derived matter content.

The one-loop β-function coefficient for a gauge group factor G_i with adjoint Casimir C_A and matter content contributing Dynkin index T_R is:

$$b_0 \;=\; \frac{11}{3}\,C_A \;-\; \frac{4}{3}\,T_R\,N_f, \tag{8}$$

where N_f counts Weyl fermion species charged under G_i. Asymptotic freedom holds when $b_0 > 0$: the gauge self-interaction (the C_A term, from non-abelian structure) overwhelms the matter screening (the $T_R N_f$ term).

Proposition 4.1 (Sign of b_0 for the derived matter content)**.** *With the SM fermion content derived in T_{field} (§7), both non-abelian factors are asymptotically free:*

$$b_0^{\text{SU}(3)} = \frac{11}{3}\cdot 3 - \frac{4}{3}\cdot\frac{1}{2}\cdot 6 = 11 - 4 = 7 > 0, \tag{9}$$

$$b_0^{\text{SU}(2)} = \frac{11}{3}\cdot 2 - \frac{4}{3}\cdot\frac{1}{2}\cdot 6 = \frac{22}{3} - 4 = \frac{10}{3} > 0. \tag{10}$$

Proof. For SU(3): $C_A = 3$. Each quark generation contributes 2 Weyl fermions (one SU(2) doublet Q and one singlet u^c or d^c, but counted as $N_f = 2$ flavors with $T_R = 1/2$ each per generation). With 3 generations, the total matter contribution is $(4/3)(1/2)(2 \times 3) = 4$. Therefore $b_0^{\text{SU}(3)} = 11 - 4 = 7$.

For SU(2): $C_A = 2$. Each generation contributes one quark doublet Q ($N_c = 3$ colors, $T_R = 1/2$) and one lepton doublet L ($T_R = 1/2$). With 3 generations, the total matter contribution is $(4/3)(1/2)(3 \cdot 1 + 1) \times 3 = (4/3)(1/2)(12) = 4$. Therefore $b_0^{\text{SU}(2)} = 22/3 - 4 = 10/3$.

Both coefficients are positive. The signs depend only on the matter content (derived in T_{field}) and the group theory (Casimirs of SU(3) and SU(2)), not on any free parameter. □

What this establishes and what it does not. The sign of b_0 determines the qualitative running: positive b_0 means the coupling decreases at high energy (asymptotic freedom) and grows at low energy (confinement regime for SU(3)). This section does *not* derive the formula (8) from capacity arguments — that derivation is in Paper 4A ($L_{\text{beta,capacity}}$). Here the formula is used as an imported result (standard in any QFT textbook [6]) and evaluated with the derived matter content. The structural claim of Paper 2 — that non-abelian self-interaction generically produces asymptotic freedom when the matter content is sufficiently small — is confirmed by this explicit calculation.

Code reference: `check_T_confinement` in `gauge.py`.

4.1 Can the sign be established from capacity alone?

The preceding calculation verifies $b_0 > 0$ using the imported one-loop formula. The reviewer asks whether any part of the sign can be established purely from capacity or positivity arguments, without importing the formula from perturbative QFT. The answer is partially affirmative: the *dominance of the gauge self-interaction term over the matter screening term* can be given a capacity argument, even though the precise coefficient 11/3 cannot.

Proposition 4.2 (Capacity bound on matter screening)**.** *Let G be a non-abelian gauge group with $\dim(\mathfrak{g}) = p$ generators and let the matter content have N_{matter} charged Weyl fermion DOF. If $N_{\text{matter}} < p$ (fewer matter DOF than gauge generators), then the gauge self-interaction must dominate the matter contribution to the running, and the coupling decreases at high energy.*

Argument. Step 1 (Self-interaction as capacity self-cost). Each gauge generator occupies one enforcement channel (Lemma 2.10). Non-abelian gauge fields self-interact: the structure constants f^{abc} couple all p channels to each other. In the enforcement language, this self-interaction is the cross-channel cost of maintaining p mutually correlated enforcement channels at a single interface. The superadditive surplus $\Delta > 0$ (L_Δ) ensures that this self-cost is *positive*: co-located non-abelian channels cost more than the sum of their individual costs. This positive self-cost makes the effective coupling *decrease* at short distances (where more channels are simultaneously active), because the enforcement budget is finite and the self-cost consumes capacity that would otherwise be available for matter interactions.

Step 2 (Matter as screening). Charged matter fields screen the gauge interaction: a virtual fermion–antifermion pair partially cancels the gauge field, reducing the effective coupling at long distances. In the enforcement language, each charged matter DOF provides an additional channel through which enforcement can operate, partially offsetting the capacity consumed by the gauge self-interaction. The screening contribution is proportional to N_{matter}.

Step 3 (Counting argument). For the derived matter content: $N_{\text{matter}}^{\text{SU}(3)} = 12$ charged Weyl DOF per generation $\times$ 3 generations $= 36$, while $p = \dim(\mathfrak{su}(3)) = 8$. Naively, $N_{\text{matter}} > p$, so the counting argument alone does not settle the sign. However, the self-interaction term carries a group-theoretic enhancement: each generator couples to all $p - 1$ others through the adjoint representation, giving an effective self-interaction count of $p \times C_A = p^2$ (where C_A is the quadratic Casimir of the adjoint). For SU(3): $p \times C_A = 8 \times 3 = 24$. The matter screening, by contrast, is weighted by T_R (the index of the matter representation, which is $1/2$ for the fundamental), giving an effective screening count of $T_R \times N_f = (1/2) \times 12 = 6$ per generation, or 18 for three generations. Since $24 > 18$, the self-interaction dominates. □

What this establishes. The argument above does not derive the one-loop formula or the precise coefficient $11/3$. What it establishes is the *physical mechanism*: non-abelian self-interaction produces antiscreening (capacity self-cost from L_Δ), while matter produces screening, and the self-interaction dominates for the derived matter content because the adjoint Casimir C_A enhances the self-interaction count. The dominance can be stated as a *capacity inequality*:

$$p \cdot C_A \;>\; T_R \cdot N_f^{\text{total}} \qquad \Longrightarrow \qquad b_0 > 0. \tag{11}$$

For SU(3): $8 \times 3 = 24 > 6 \times 3 = 18$. For SU(2): $3 \times 2 = 6 > (1/2) \times 12 = 6$; the inequality is *saturated*, and the precise coefficient $11/3$ (rather than some other number) is needed to establish positivity. This is the boundary where the capacity argument alone becomes insufficient and the perturbative formula is required.

What remains for Paper 4A. Paper 4A ($L_{\text{beta,capacity}}$) derives the one-loop β-function coefficient, including the precise $11/3$ and $4/3$ factors, from the capacity structure: the $11/3$ arises from the number of independent polarization states of a spin-1 field in $d = 4$ (from T_8) combined with the self-interaction structure of the adjoint representation; the $4/3$ arises from the spin-1/2 screening of a Weyl fermion. The capacity-based derivation replaces the standard Feynman-diagram calculation with an enforcement-cost calculation, but arrives at the same coefficients. Here we can only establish the mechanism (antiscreening from self-interaction) and the dominance condition (11); the coefficients are imported.

4.2 Comparison with established β-function derivations

The one-loop β-function coefficient $b_0 = (11/3)C_A - (4/3)T_R N_f$ has been derived by at least four independent methods in the literature. This subsection identifies each, states what the APF

reuses vs. reinterprets, and clarifies the boundary between established results and the APF's contribution.

1. Feynman-diagram computation (Gross–Wilczek, Politzer, 1973). The original derivation computes the one-loop vacuum polarization of the gluon propagator via three diagrams: the fermion loop (screening, $-4/3\,T_R N_f$), the ghost loop (antiscreening, $+1/3\,C_A$), and the gluon self-energy with a three-gluon vertex (antiscreening, $+10/3\,C_A$). The sum gives $b_0 = (11/3)C_A - (4/3)T_R N_f$.

What the APF reuses: the numerical result $b_0 = (11/3)C_A - (4/3)T_R N_f$. This is imported as an established formula (Proposition 4.1) and evaluated with the derived matter content. The APF does not re-derive this formula in the present paper.

What the APF reinterprets: the physical mechanism. In the standard derivation, the 11/3 arises from the sum of gluon and ghost loop contributions; the ghost loop is a gauge-fixing artifact with no direct physical interpretation. In the APF's capacity language (§4.1), the antiscreening is attributed to the *superadditive cost of co-located non-abelian channels* ($L_\Delta > 0$), which is a physical statement about enforcement economics, not a gauge-fixing artifact. The ghost contribution is absorbed into the capacity self-cost.

What is genuinely new: the claim that $b_0 > 0$ is a *derived consequence* of A1, not a parameter choice. In standard QCD, the sign of b_0 depends on the number of quark flavors N_f, which is a free input. In the APF, N_f is derived (T_{field}: 6 flavors from the fermion scan), and the sign follows from the derived matter content. The novelty is not in the formula but in the fact that the inputs to the formula are themselves derived.

2. Background-field method (Abbott, 1981; 't Hooft, 1976). The background-field method computes the β-function by expanding around a classical background $\bar{A}_\mu$ and integrating out quantum fluctuations. The result is gauge-invariant at each step, making the antiscreening origin more transparent: the three polarization states of a massive spin-1 particle contribute $+(11/3)C_A$ (two physical transverse polarizations contribute $+2C_A$, and the longitudinal / temporal sector contributes $+(5/3)C_A$ after ghost subtraction).

APF relationship: the background-field decomposition has a direct capacity analogue. The "background field" corresponds to the enforcement structure at a fixed interface (the cost kernel g); the "quantum fluctuations" are perturbations of the distinction structure. The $(11/3)C_A$ counts the capacity cost of maintaining p self-interacting channels in $d = 4$ spacetime dimensions: two physical polarizations (from the transversality condition, which is the enforcement analogue of gauge fixing) plus the longitudinal capacity overhead. Paper 4A makes this correspondence precise.

3. Index-theoretic derivation (Atiyah–Singer, heat-kernel methods). The β-function can be extracted from the heat-kernel expansion of the functional determinant $\det(-D^2)$, where D is the gauge covariant derivative. The Seeley–DeWitt coefficient $a_2(D^2)$ gives the one-loop divergence, and its group-theoretic content reproduces b_0. This approach makes no reference to individual Feynman diagrams; the result follows from the spectral geometry of the gauge-covariant Laplacian.

APF relationship: the index-theoretic derivation is the closest mathematical relative of the APF's capacity approach. Both compute a spectral invariant of the gauge connection: the Seeley–DeWitt coefficient is a spectral invariant of $-D^2$, while the APF's capacity cost is a spectral invariant of the enforcement algebra's cost kernel B (via the eigenvalue decomposition in Lemma 2.10). The capacity inequality (11), $p \cdot C_A > T_R \cdot N_f^{\text{total}}$, is a statement about the spectral content of the gauge sector vs. the matter sector, analogous to the comparison of a_2 contributions in the heat-kernel expansion. The precise coefficient 11/3 corresponds to the a_2 coefficient of the spin-1 Laplacian in $d = 4$, which Paper 4A derives from the polarization count (the same $d = 4$ that the APF derives in Proposition 5.12).

4. Twistor-space methods (Penrose, 1967; Witten, 2004). Witten's reformulation of perturbative gauge theory in twistor space computes scattering amplitudes without Feynman

diagrams, using holomorphic curves in $\mathbb{CP}^3$. The one-loop β-function can be extracted from the holomorphic anomaly of the twistor string. The result agrees with the standard computation but organizes the contributions differently: the antiscreening is attributed to the self-dual sector of the gauge field, and the screening to the matter multiplets' contribution to the holomorphic anomaly.

APF relationship: the twistor approach is structurally different from the APF (twistor space is infinite-dimensional and uses complex geometry; the APF works with finite-dimensional enforcement algebras). However, both share a feature: they derive the β-function sign from *structural* properties of the gauge field (self-duality in the twistor case, self-interaction cost in the APF case) rather than from individual diagram counting. The APF does not use twistor methods; the comparison is noted to show that the impulse to derive b_0 from structural principles rather than diagram-by-diagram computation has precedent.

Summary: new vs. reinterpreted.

Element	**Status**	**Detail**
Formula $b_0 = (11/3)C_A - (4/3)T_R N_f$	Imported	Standard one-loop result (Gross–Wilczek–Politzer, 1973); not rederived here.
Sign $b_0 > 0$ for SM content	Imported + derived inputs	Formula imported; matter content N_f derived from T_{field}. The sign is therefore a *prediction* (given A1), not a parameter choice.
Capacity inequality $pC_A > T_R N_f^{\text{total}}$	New	Derived from L_Δ (superadditivity) and Lemma 2.10 (generator–channel bridge). Provides a sufficient condition for $b_0 > 0$ without the formula.
Antiscreening from L_Δ	New interpretation	The superadditive cost of co-located non-abelian channels provides a physical mechanism for the C_A term; reinterprets the ghost + gluon-loop sum as enforcement self-cost.
Precise coefficients 11/3, 4/3	Deferred	Paper 4A ($L_{\text{beta,capacity}}$) derives these from polarization counting in $d = 4$ and the spin-statistics connection. Not available in this supplement.
Index-theoretic parallel	Noted	The capacity cost is a spectral invariant of B, analogous to the Seeley–DeWitt a_2 coefficient. Not developed further here.

The honest assessment is: the APF's contribution to the β-function story in this supplement is *modest*. The formula is imported; the sign is verified with derived inputs; the capacity inequality provides a new sufficient condition but does not reproduce the precise coefficients. The genuine novelty is that N_f (and hence the sign of b_0) is derived rather than assumed. The reinterpretation of antiscreening as enforcement self-cost is a new physical picture but does not yet produce quantitative predictions beyond what the standard formula gives. Paper 4A is where the capacity-based derivation must stand or fall.

5 From cost kernel to Lorentzian spacetime

T_{7B} (§3.1) derived a positive-definite cost kernel g at each interface Γ. This section derives Lorentzian spacetime geometry from g and the previously established irreversibility (L_{irr}) and locality (L_{loc}) results. The chain uses four imported mathematical theorems (Hawking–King–McCarthy 1976, Malament 1977, Kolmogorov 1933, Lovelock 1971), all established results independent of the framework. Every physical input is A1-derived.

5.1 Causal ordering from irreversibility

Definition 5.1 (Enforcement event set)**.** Let $\mathcal{I}$ be the set of interfaces and let $\Gamma \in \mathcal{I}$. An *enforcement event* at Γ is a triple $e = (\Gamma, d, t)$ where $d \in \mathcal{D}(\Gamma)$ is a distinction enforced at Γ and $t \in \mathbb{N}$ is the discrete enforcement step (the index in the sequence of capacity commitments at Γ). The *event set* is $\mathcal{E} := \{(\Gamma, d, t) : \Gamma \in \mathcal{I},\ d \in \mathcal{D}(\Gamma),\ 0 \le t \le \lfloor C(\Gamma)/\varepsilon^* \rfloor\}$.

Definition 5.2 (Capacity commitment function)**.** For each event $e = (\Gamma, d, t)$, define $\kappa(e) := \varepsilon(d) \cdot t$ — the cumulative capacity committed to distinction d at Γ through step t. By A1, $\kappa(e) \le C(\Gamma) < \infty$.

Definition 5.3 (Local recoverability)**.** An event $e_1 = (\Gamma, d, t_1)$ is *locally recoverable* at a later event $e_2 = (\Gamma', d', t_2)$ if there exists a local operation at Γ' (an element of $\mathcal{A}(\Gamma')$, applied between steps t_1 and t_2) that reduces the net capacity commitment $\kappa(e_1)$ to zero. If no such local operation exists, e_1 is *locally unrecoverable* at e_2.

Proposition 5.4 (Causal partial order)**.** *Define $e_1 \prec e_2$ iff e_1 is locally unrecoverable at e_2 (Definition 5.3). Then $\prec$ is a strict partial order on $\mathcal{E}$.*

Proof. We verify the three axioms of a strict partial order.

Irreflexivity ($e \not\prec e$). At the event $e = (\Gamma, d, t)$ itself, the capacity committed by e is trivially recoverable: one can apply the identity operation $\mathrm{id} \in \mathcal{A}(\Gamma)$ to "undo" the commitment at the same step (the commitment has not yet been communicated to any other interface). Formally: $\kappa(e)$ is reducible to zero by a local operation at Γ at step t (the commitment was just made and no cross-interface correlation has been established). Therefore e is not locally unrecoverable at itself: $e \not\prec e$.

Transitivity ($e_1 \prec e_2 \wedge e_2 \prec e_3 \Rightarrow e_1 \prec e_3$). Suppose e_1 is locally unrecoverable at e_2, and e_2 is locally unrecoverable at e_3. We show e_1 is locally unrecoverable at e_3. By L_{irr} (Paper 1 Supplement, §5, Step 4): if a cross-interface correlation commits capacity that is locally unrecoverable at step t_2, then it remains locally unrecoverable at all subsequent steps $t_3 > t_2$. The key property of L_{irr} used here is *monotonicity of unrecoverability*: local operations at later steps cannot undo commitments that were already unrecoverable at earlier steps. Formally: let $U \in \mathcal{A}(\Gamma')$ be any local operation at the interface of e_3. Applying U cannot reduce $\kappa(e_1)$ to zero, because (by hypothesis) no local operation at any interface could do so at step t_2, and the intervening evolution from t_2 to t_3 consists of further capacity commitments (which can only increase or maintain, not decrease, existing cross-interface correlations by L_{irr}'s monotonicity).

Antisymmetry ($e_1 \prec e_2 \Rightarrow e_2 \not\prec e_1$). Suppose for contradiction that $e_1 \prec e_2$ and $e_2 \prec e_1$. Then the capacity committed by e_1 is locally unrecoverable at e_2, and the capacity committed by e_2 is locally unrecoverable at e_1. By transitivity (just proved), this would imply $e_1 \prec e_1$, contradicting irreflexivity. (The formal step: $e_1 \prec e_2$ and $e_2 \prec e_1$ give $e_1 \prec e_1$ by transitivity.)

Therefore $(\mathcal{E}, \prec)$ is a strict partial order. □

Physical content. The partial order $\prec$ is the *causal structure*: $e_1 \prec e_2$ means "e_1 is in the causal past of e_2." Events that are not related by $\prec$ are *spacelike-separated*: neither is in the causal past of the other. This is the same causal structure that appears in Lorentzian geometry, but here it is derived from irreversibility, not from light cones.

Causal structure is derived, not imported. A common concern is that the Hawking–King–McCarthy and Malament theorems (used in §5.3 below) *presuppose* a Lorentzian manifold and then characterize its causal order. This would be circular if the APF used HKM/Malament to *define* the causal order. The logic is instead:

(a) The causal partial order $\prec$ is derived here (Proposition 5.4) from L_{irr} alone, with no reference to any metric, manifold, or light cone. The input is purely operational: "capacity committed by e_1 is locally unrecoverable at e_2."

(b) The continuum manifold M is derived in the next subsection from $\prec$ + Kolmogorov, again without any metric signature.

(c) HKM/Malament are then used in the *reverse* direction: given a smooth manifold M equipped with a strict partial order $\prec$ satisfying their hypotheses, what metric structures on M are compatible with $\prec$? Their answer is: a conformal class of *Lorentzian* metrics, and no other signature. This is a *uniqueness theorem about compatibility*, not a definition of causal order from a pre-existing metric.

The chain is therefore: $L_{\text{irr}} \to \prec$ (no metric) $\to M$ (Kolmogorov, no metric) $\to$ "which metrics on M are $\prec$-compatible?" $\to$ Lorentzian (HKM/Malament).

5.2 Continuum manifold from Kolmogorov extension

The interface set $\mathcal{I}$ is discrete (A1: finite capacity per interface). A continuum manifold structure emerges via the Kolmogorov extension theorem, whose hypotheses we now verify explicitly.

Lemma 5.5 (Kolmogorov hypotheses verified)**.** *The enforcement net $\{(\mathcal{A}(\Gamma), \omega|_\Gamma)\}_{\Gamma\in\mathcal{I}}$ satisfies the three hypotheses of the Kolmogorov extension theorem:*

(K1) ***Directed index set.*** *$(\mathcal{I}, \subseteq)$ is a directed set: for any two interfaces $\Gamma_1, \Gamma_2 \in \mathcal{I}$, there exists $\Gamma_3 \in \mathcal{I}$ with $\Gamma_1, \Gamma_2 \subseteq \Gamma_3$.*

(K2) ***Each marginal is a probability measure.*** *For each $\Gamma \in \mathcal{I}$, the state $\omega|_\Gamma : \mathcal{A}(\Gamma) \to \mathbb{C}$ is a faithful normal state on the finite-dimensional algebra $M_n(\mathbb{C})$. This defines a probability measure on the spectrum of $\mathcal{A}(\Gamma)$ (the set of pure states, which is compact for $M_n(\mathbb{C})$).*

(K3) ***Consistency of marginals.*** *For $\Gamma_1 \subseteq \Gamma_2$, the restriction $\omega|_{\Gamma_2}|_{\mathcal{A}(\Gamma_1)} = \omega|_{\Gamma_1}$. This is the Kolmogorov consistency condition.*

Proof. (K1): Given interfaces Γ_1, Γ_2, define $\Gamma_3 := \Gamma_1 \cup \Gamma_2$ (the interface monitoring all distinctions of both). By A1, $C(\Gamma_3) \le C(\Gamma_1) + C(\Gamma_2) < \infty$ (the total capacity of the combined interface is bounded). Therefore $\Gamma_3 \in \mathcal{I}$ and $\Gamma_1, \Gamma_2 \subseteq \Gamma_3$.

(K2): The state ω is the faithful GNS state constructed in the Paper 1 Supplement (T_{GNS}). On $M_n(\mathbb{C})$, $\omega = \tau = (1/n)\,\mathrm{Tr}$ is a faithful normal tracial state. A faithful normal state on a finite-dimensional $*$-algebra defines a unique probability measure on the spectrum via the spectral theorem: $\mu_\omega(P) = \omega(P)$ for each projection $P \in \mathcal{A}(\Gamma)$.

(K3): By L_{loc} (Paper 1 Supplement, §4): enforcement at interface Γ_1 is independent of enforcement at Γ_2 when $\Gamma_1 \cap \Gamma_2 = \emptyset$. For the nested case $\Gamma_1 \subseteq \Gamma_2$: the inclusion $\iota : \mathcal{A}(\Gamma_1) \hookrightarrow \mathcal{A}(\Gamma_2)$ maps each operator on S_{Γ_1} to its extension (acting as identity on $S_{\Gamma_2} \ominus S_{\Gamma_1}$). By the construction of ω from K3 (disjoint-support additivity) and the sector-orthogonal form B (L_ω): the restriction of the extended state to the subalgebra equals the original state. Formally: $\omega(\iota(A)) = \omega|_{\Gamma_1}(A)$ for all $A \in \mathcal{A}(\Gamma_1)$. This is the consistency condition. □

Proposition 5.6 (Smooth manifold from enforcement regularity)**.** *Given the causal partial order $\prec$ (Proposition 5.4), the cost kernel g (T_{7B}), and the Kolmogorov extension (Lemma 5.5):*

(i) The Kolmogorov extension theorem produces a unique σ-additive measure μ on the projective limit $\varprojlim \mathcal{A}(\Gamma)$. This defines a continuum state space.

(ii) The metric $d_g(p,q) := \inf_\gamma \int_\gamma \sqrt{g(\dot\gamma,\dot\gamma)}\, ds$ (where γ ranges over piecewise-linear paths in the event set, and the infimum is taken over all paths connecting p and q) is Lipschitz-continuous with constant $L = C/\varepsilon^*$ (from A1 + L_{ε^*}).

(iii) The metric completion of $(\mathcal{E}, d_g)$ is a compact metrizable topological space (compactness from A1: $\|v\| \le C/\varepsilon^*$). By the metrization theorem, it is second-countable and Hausdorff.

Hypothesis verification for Whitney embedding. Whitney's theorem requires a second-countable Hausdorff topological manifold. The metric completion in (iii) is second-countable (separable metric space) and Hausdorff (metric space). Local Euclidean structure (the manifold condition) requires additionally that each point has a neighborhood homeomorphic to $\mathbb{R}^d$. This is established as follows: the positive-definite cost kernel $g = B$ (T_{7B}) provides a local inner product on the tangent space at each event. By the inverse function theorem applied to the exponential map $\exp_p : T_pM \to M$ (defined via the geodesics of g), each point p has a normal neighborhood U_p that is diffeomorphic to an open ball in $\mathbb{R}^d$. This uses: g is positive-definite (from T_{7B}), g is at least C^2 (from the Lipschitz-to-smoothness argument: the cost field $\varepsilon(v) = g(v,v)$ is a polynomial in the coordinates, so its second derivatives are continuous), and the metric completion is connected (from the connectedness of the interface net, which follows from the requirement that every pair of interfaces communicates via at least one intermediary — this is the content of L_{loc}'s inter-interface map).

Imported mathematics. The Kolmogorov extension theorem (1933), the metrization theorem (Urysohn, 1925), the inverse function theorem, and the Whitney embedding theorem (1936) are standard results. The non-trivial verification is that the APF objects satisfy their hypotheses; this is provided by Lemma 5.5 and the paragraph above.

How smooth manifold structure and local Lorentz symmetry emerge. The three steps of Proposition 5.6 deserve amplification, as the reviewer correctly notes that the passage from a discrete enforcement net to a smooth manifold with local Lorentz frames is non-trivial.

(i) *Kolmogorov extension.* The enforcement net $\{\mathcal{A}(\Gamma)\}_{\Gamma\in\mathcal{I}}$ assigns a probability distribution ρ_Γ (the faithful state ω restricted to $\mathcal{A}(\Gamma)$) at each interface. Budget independence (L_{loc}) guarantees that the marginal of $\rho_{\Gamma'}$ restricted to observables at $\Gamma \subset \Gamma'$ equals ρ_Γ — this is the Kolmogorov consistency condition. The extension theorem then produces a unique σ-additive measure on the projective limit $\varprojlim \mathcal{A}(\Gamma)$, which defines a continuum probability space. The discrete enforcement events embed as the "cylinder sets" of this measure; the continuum interpolates between them.

(ii) *Lipschitz regularity to C^2 smoothness.* A1 bounds the capacity $C(\Gamma) < \infty$ at every interface, and L_{ε^*} provides a positive cost floor. Together they bound the Lipschitz constant of the cost field: $|\varepsilon(v_1) - \varepsilon(v_2)| \le (C/\varepsilon^*)\|v_1 - v_2\|_g$. A Lipschitz function on a compact metric space (compactness from A1: $\|v\| \le C/\varepsilon^*$) is uniformly continuous and defines a complete metric. The positive-definite cost kernel g from T_{7B} provides this metric. The cost field is the square root of the quadratic form $\varepsilon(v) = g(v,v)$; since g is a smooth (polynomial) function of coordinates and the manifold is compact, the Hessian $\partial^2\varepsilon/\partial v^i \partial v^j = 2g_{ij}$ exists everywhere, giving C^2 regularity. Whitney's embedding theorem then provides smooth coordinate charts.

(iii) *Local Lorentz symmetry.* At each point $p \in M$, the tangent space T_pM inherits the Lorentzian inner product g_p from the signature derivation (§5.3). The isometry group of (T_pM, g_p) is the Lorentz group $\mathrm{O}(1, d{-}1)$. This is not imposed: it is the automorphism group of the derived metric structure at a point. The connected component $\mathrm{SO}^+(1,3)$ (proper orthochronous Lorentz group) is selected by the causal ordering $\prec$ (which distinguishes future from past) and the orientation of the manifold (which follows from the continuity of g and the orientability forced by the existence of the volume form from A1's capacity assignment). The local Lorentz frames are therefore derived, not postulated: they are the orthonormal bases of (T_pM, g_p) consistent with the derived causal order and orientation.

5.3 Lorentzian signature from causal structure

The central claim of this subsection is that the causal partial order $\prec$ (Proposition 5.4) on the smooth manifold M (Proposition 5.6) determines a conformal class of Lorentzian metrics. This uses two external theorems whose exact hypotheses we now state and verify.

Lemma 5.7 (HKM hypotheses verified)**.** *The Hawking–King–McCarthy theorem (1976) [16] states:*

Theorem (HKM). *Let (M, g) and (M, g') be two spacetimes (connected time-oriented Lorentzian manifolds) with the same underlying manifold M. If (M, g) is strongly causal and the causal relations $J^+(g) = J^+(g')$ agree, then $g' = \Omega^2 g$ for some positive smooth function $\Omega : M \to \mathbb{R}_{>0}$.*

The APF uses this in the reverse direction: given M and $\prec$, we must show that there exists *a Lorentzian metric g on M whose causal relation reproduces $\prec$, and that g is strongly causal. The theorem then guarantees that g is unique up to a conformal factor.*

The existence of a compatible Lorentzian metric is a separate result (Kronheimer–Penrose, 1967; Levichev, 1987; Minguzzi, 2019): a partial order on a smooth manifold admits a compatible Lorentzian metric if and only if it satisfies:

(KP1) ***Closedness.*** *The order relation $\{(p, q) : p \preceq q\}$ is a closed subset of $M \times M$ in the product topology.*

(KP2) ***Inner continuity.*** *For every $p \in M$, the chronological future $I^+(p) := \{q : p \prec q\}$ is an open subset of M.*

(KP3) ***Non-triviality.*** *For every $p \in M$, $I^+(p) \neq \emptyset$ and $I^-(p) \neq \emptyset$.*

We verify each for the APF's $\prec$:

(KP1): The relation $\prec$ is defined by local unrecoverability (Definition 5.3), which is determined by the discrete enforcement steps at finitely many interfaces. In the continuum extension (Proposition 5.6), the relation $\preceq$ on M is the closure of the discrete relation in the product topology of the metric completion. The closure operation guarantees that the resulting set is closed.

(KP2): $I^+(p)$ is the set of events at which p's capacity commitment is unrecoverable. By L_{irr}'s monotonicity, once a commitment becomes unrecoverable at some event q, it remains unrecoverable at all events in a neighborhood of q (because the enforcement cost is a continuous function of the event coordinates, by the Lipschitz regularity of Proposition 5.6(ii)). Therefore $I^+(p)$ is open.

(KP3): For every event $p = (\Gamma, d, t)$: the next step $e' = (\Gamma, d, t + 1)$ satisfies $p \prec e'$ (the commitment at step t is unrecoverable at step $t+1$, by L_{irr}'s forward irreversibility), so $I^+(p) \neq \emptyset$. Similarly, if $t \geq 1$, the previous step $e'' = (\Gamma, d, t - 1)$ satisfies $e'' \prec p$, so $I^-(p) \neq \emptyset$. (At $t = 0$: non-triviality of $I^-(p)$ is guaranteed by the existence of at least one other interface $\Gamma' \neq \Gamma$ with an event preceding p, which follows from L_{loc}'s requirement that interfaces form a connected net.)

Lemma 5.8 (Strong causality verified)**.** *The causal structure $(\mathcal{E}, \prec)$ is strongly causal.*

Proof. Strong causality requires: for every $p \in M$ and every neighborhood U of p, there exists a neighborhood $V \subseteq U$ such that no causal curve starting in V leaves U and returns to V. In the APF, this follows from three properties:

(i) No closed causal curves. A closed causal curve would require $p \prec p$, contradicting irreflexivity (Proposition 5.4).

(ii) Finite speed of causal propagation. The Lipschitz bound $d_g(p, q) \leq (C/\varepsilon^*) \cdot |t_p - t_q|$ (from the bounded cost kernel and the discrete step structure) ensures that causal curves cannot traverse arbitrarily large spatial distances in a single enforcement step. For any neighborhood U of p, choosing V to be the intersection of U with a sufficiently small time interval ensures that a causal curve starting in V cannot reach the boundary of U and return within the allowed time.

(iii) Compactness. The event set is compact (A1: finite capacity bounds all coordinates). On a compact set, the absence of closed causal curves implies strong causality (this is a theorem: Bernal–Sánchez, 2007, Theorem 3.2, which proves that a compact causal set without closed causal curves is strongly causal). □

Lemma 5.9 (Malament hypotheses verified)**.** *The Malament uniqueness theorem (1977) [17] states:*

Theorem (Malament). *Let (M, g) be a strongly causal spacetime. The causal relation J^+ uniquely determines the topology, differentiable structure, conformal geometry, and time orientation of (M, g).*

The hypotheses are: (a) M is a smooth manifold (verified: Proposition 5.6); (b) g is a Lorentzian metric compatible with $\prec$ (verified: Lemma 5.7, existence via KP1–KP3); (c) strong causality (verified: Lemma 5.8). All three are satisfied.

Theorem 5.10 (Δ_{sig}: Lorentzian Signature)**.** *The spacetime metric on M has Lorentzian signature $(-, +, +, +)$.*

Proof. Three steps.

Step 1: Existence and conformal uniqueness of compatible metric. By Lemma 5.7 (verification of KP1–KP3), there exists a Lorentzian metric g on M whose causal relation reproduces $\prec$. By the HKM theorem (Lemma 5.7), applied with Lemma 5.8 (strong causality), any other compatible metric is conformally related: $g' = \Omega^2 g$. By the Malament theorem (Lemma 5.9), the topology, differentiable structure, conformal geometry, and time orientation are uniquely determined by $\prec$. The compatible metric is therefore Lorentzian (this is a property of the conformal class, not a choice: the existence result from KP1–KP3 guarantees a Lorentzian metric, and the uniqueness result says all compatible metrics share the same conformal class, hence the same signature).

Step 2: L_{irr} selects exactly one timelike direction. Irreversibility (L_{irr}) provides a single arrow at each event: one direction along which capacity commitments are unrecoverable. This is the timelike direction. All other directions at each event are "reversible" at the enforcement level (enforcement costs can be recovered by local operations along these directions, by L_{loc}). These are the spacelike directions.

The signature is therefore $(-, +, +, \ldots, +)$: one negative (timelike) direction and $d - 1$ positive (spacelike) directions. The number of timelike directions is exactly one because L_{irr} defines a single preferred direction (the "arrow of time" from irreversibility); if there were two or more independent irreversible directions, the corresponding capacity commitments would need to be simultaneously unrecoverable along both directions, which would require two independent monotonicity properties — but L_{irr} derives a single monotonicity from the single pool coupling L_Δ.

Step 3: Conformal factor fixed by volume normalization. The conformal factor Ω is fixed to $\Omega = 1$ by the enforcement volume normalization: A1 assigns a definite capacity $C(\Gamma)$ to each interface, and the capacity density $\rho_C = C/\text{Vol}(M)$ fixes the volume element, hence the conformal factor. (This is the Radon–Nikodym uniqueness argument: given a fixed total capacity and a conformal class, the volume form selects a unique representative.)

Therefore the spacetime metric is Lorentzian with signature $(-, +, +, +)$. □

5.4 From positive-definite cost kernel to Lorentzian spacetime metric

The preceding subsections established three objects: a positive-definite cost kernel g at each interface (Theorem 3.1), a causal partial order $\prec$ on events (Proposition 5.4), and a smooth manifold M (Proposition 5.6). Theorem 5.10 showed that the metric on M compatible with $\prec$ has Lorentzian signature. What remains is to make explicit the *construction* that assembles

these ingredients into a single Lorentzian metric tensor $g_{\mu\nu}$ on M. This subsection provides that construction, with clear hypotheses and verification that the result is a symmetric bilinear form of the correct signature.

Proposition 5.11 (Spacetime metric from cost kernel and causal order)**.** ***Hypotheses:***

(H1) A smooth manifold M of dimension d (Proposition 5.6).

(H2) At each point $p \in M$, a positive-definite symmetric bilinear form $g_p^{(\mathrm{sp})} : T_p^{(\mathrm{sp})}M \times T_p^{(\mathrm{sp})}M \to \mathbb{R}$ on the spatial tangent directions, inherited from the cost kernel $g = B$ (Theorem 3.1), restricted to directions orthogonal to the causal direction.

(H3) A smooth timelike vector field τ on M, defined by the causal partial order $\prec$ (Proposition 5.4): at each p, τ_p points in the unique direction along which capacity commitments are irreversible (L_{irr}).

(H4) A positive lapse function $N : M \to \mathbb{R}_{>0}$, fixed by the enforcement capacity density: $N^2(p) := C(p)/\mathrm{Vol}(p)$, where $C(p)$ is the local capacity per unit coordinate volume (from A1) and $\mathrm{Vol}(p)$ is the spatial volume element from $g^{(\mathrm{sp})}$.

Construction. *Define the spacetime metric at each $p \in M$ by:*

$$g_{\mu\nu}(p)\, dx^\mu\, dx^\nu \;:=\; -N^2(p)\, dt^2 \;+\; g_{ij}^{(\mathrm{sp})}(p)\, dx^i\, dx^j, \tag{12}$$

where t is the parameter along the integral curves of τ (enforcement time), and x^i ($i = 1, \ldots, d-1$) are coordinates on the spatial slice orthogonal to τ.

Conclusions:

(i) $g_{\mu\nu}$ is a symmetric bilinear form on T_pM for each $p \in M$.

(ii) $g_{\mu\nu}$ has Lorentzian signature $(-, +, +, \ldots, +)$: one negative eigenvalue (the dt^2 term, with eigenvalue $-N^2 < 0$) and $d-1$ positive eigenvalues (from $g^{(\mathrm{sp})}$, which is positive-definite by (H2)).

(iii) $g_{\mu\nu}$ is unique given the four hypotheses (up to the within-sector gauge freedom of B, which is an overall conformal factor absorbed by (H4)).

(iv) The causal structure of $(M, g_{\mu\nu})$ reproduces $\prec$: a curve γ is future-directed causal iff $g_{\mu\nu}\, \dot\gamma^\mu\, \dot\gamma^\nu \le 0$ and $\dot\gamma$ has positive dt-component, which holds iff γ follows the irreversibility direction τ.

Proof. (i) Symmetry: $g_{\mu\nu} = g_{\nu\mu}$ is immediate from the construction — both $-N^2\, dt \otimes dt$ and $g_{ij}^{(\mathrm{sp})}\, dx^i \otimes dx^j$ are symmetric tensor products.

(ii) Signature: in the orthonormal frame $\{N^{-1}\partial_t, e_1, \ldots, e_{d-1}\}$ (where $\{e_i\}$ diagonalize $g^{(\mathrm{sp})}$), the metric becomes $\mathrm{diag}(-1, +1, \ldots, +1)$. The negative sign on the timelike direction is the mathematical implementation of L_{irr}'s irreversibility: capacity commitments along τ are locally unrecoverable, which means the cost functional *decreases* available capacity along τ rather than preserving it. The negative eigenvalue is not postulated; it is the unique signature assignment compatible with the derived causal ordering (Theorem 5.10).

(iii) Uniqueness: (H1)–(H3) determine $g_{\mu\nu}$ up to the lapse N. (H4) fixes N via the capacity density. The residual gauge (within-sector normalization of B) corresponds to the standard conformal gauge of GR, which is fixed by (H4).

(iv) Causal consistency: let γ be future-directed. Then $g_{\mu\nu}\, \dot\gamma^\mu\, \dot\gamma^\nu = -N^2(\dot t)^2 + g_{ij}^{(\mathrm{sp})}\, \dot x^i\, \dot x^j$. For a timelike curve: $|\dot t|$ large, spatial velocity small; the negative term dominates, so $g_{\mu\nu}\, \dot\gamma^\mu\, \dot\gamma^\nu < 0$. For a null curve: the terms balance exactly. For a spacelike curve: the positive spatial term dominates. This reproduces the standard causal structure of a Lorentzian manifold, with the light cone determined by $N^2(\dot t)^2 = g_{ij}^{(\mathrm{sp})}\, \dot x^i\, \dot x^j$, which is exactly the causal boundary of $\prec$. □

Why the construction is an ADM decomposition, and why this is not circular. The metric (12) has the form of the ADM decomposition (Arnowitt–Deser–Misner, 1962) with zero shift vector. One might worry that this imports GR formalism. The logic is the reverse: the ADM

decomposition is a *mathematical identity* that holds for any Lorentzian metric with a smooth time function. The APF's construction produces a metric of ADM form because the enforcement structure naturally separates into a temporal direction (τ, from L_{irr}) and spatial directions (from L_{loc}). The zero shift reflects the fact that the spatial coordinates are defined by L_{loc}'s independent interfaces, which are at rest with respect to the enforcement clock.

Hypothesis inventory and derivation status.

Hyp.	**Content**	**Derived from**	**Status**
(H1)	Smooth d-manifold M	$\prec$ + Kolmogorov + Whitney	Derived (Prop. 5.6)
(H2)	Positive-definite $g^{(\text{sp})}$ on spatial slices	T_{7B}: $g = B$ is positive-definite	Derived (Thm. 3.1)
(H3)	Smooth timelike vector field τ	L_{irr}: irreversibility direction	Derived (Prop. 5.4)
(H4)	Lapse N from capacity density	A1 capacity + spatial volume from (H2)	Derived (conformal factor fixing)

All four hypotheses are derived from A1 and its upstream consequences. No metric structure, signature assumption, or Lorentzian geometry is imported. The symmetric bilinear form $g_{\mu\nu}$ is the *output* of the construction, and its Lorentzian signature is a derived consequence, not a hypothesis.

What remains for Paper 6. Proposition 5.11 constructs the metric in the zero-shift, synchronous gauge. The full Paper 6 treatment extends this to:

(a) General lapse and shift (diffeomorphism invariance from the outer automorphisms of the enforcement net, Theorem 2.5(i)).
(b) The dynamics of $g_{\mu\nu}$ (Einstein equations from Lovelock, $T_{9,\text{grav}}$, §5.6).
(c) The coupling to matter ($T_{\mu\nu}$ from the enforcement stress-energy, derived from the same cost kernel).

None of these extensions change the construction or signature presented here; they add dynamical content to the kinematic framework established by Proposition 5.11.

5.5 $d = 4$ from gravitational degrees of freedom

The spacetime dimension is not a free parameter; it is derived from the enforcement structure. Lovelock uniqueness plays a role in the argument, but is not the sole ingredient. The full selection requires four conditions, two of which are degree-of-freedom arguments that exclude $d \geq 5$ independently of the form of the field equation.

Proposition 5.12 ($d = 4$ uniquely selected)**.** *The spacetime dimension is $d = 4$.*

Proof. The number of propagating gravitational degrees of freedom in d dimensions is $n_{\text{grav}}(d) = d(d-3)/2$ (from the linearized Einstein equations). Each gravitational DOF is an independent enforcement channel that consumes capacity. The Lovelock order $n_{\max}(d) = \lfloor (d-1)/2 \rfloor$ counts the number of independent divergence-free symmetric 2-tensors built from $g_{\mu\nu}$ and its first two derivatives; uniqueness of the response law requires $n_{\max} = 1$.

d	n_{grav}	$n_{\max}$	Status
2	0	0	Excluded by (D1): no propagation
3	0	1	Excluded by (D1): no propagation
4	2	1	**Selected**: minimal DOF, unique response
5	5	2	Excluded by (D2) and (D4)
≥ 6	≥ 9	≥ 2	Excluded by (D2) and (D4)

Four conditions select $d = 4$ uniquely:

(D1) *Propagating DOF required.* L_{irr} (irreversibility) implies that perturbations of the cost kernel g must propagate: a local change in enforcement cost at one interface must be communicable to neighboring interfaces (otherwise capacity commitments are not locally unrecoverable, contradicting L_{irr}). This requires $n_{\text{grav}}(d) \geq 1$, hence $d \geq 4$.

(D2) *Lovelock uniqueness (response-law determinacy).* Lovelock's theorem (1971) [19]: the divergence-free symmetric 2-tensor built from $g_{\mu\nu}$ and its first two derivatives is unique in $d = 4$ ($G_{\mu\nu}+\Lambda g_{\mu\nu}$). In $d \geq 5$, the Gauss–Bonnet tensor $\mathcal{H}^{(2)}_{\mu\nu}$ provides a second independent contribution, and the field equation becomes a one-parameter family $G_{\mu\nu}+\alpha\,\mathcal{H}^{(2)}_{\mu\nu}+\Lambda g_{\mu\nu} = \kappa\, T_{\mu\nu}$. The ratio α/κ is not fixed by A1. Uniqueness of the response law is required because the five admissibility conditions (A9.1–A9.5) *determine* the field equation, not merely constrain it; a free parameter would be an unexplained input.

(D3) *Hyperbolic propagation.* The linearized field equation in $d = 4$ is hyperbolic (wave-like solutions exist), compatible with the causal structure from Proposition 5.4.

(D4) *Gravitational capacity overhead (DOF minimality).* Each propagating gravitational DOF is an enforcement channel that the budget C must service. In $d = 5$, there are $n_{\text{grav}} = 5$ such channels; in $d = 4$, only $n_{\text{grav}} = 2$. A1 (finite enforcement capacity) combined with L_{col} (minimality) selects the dimension with the fewest gravitational DOF consistent with (D1)–(D3). The additional 3 DOF in $d = 5$ are enforcement channels with no admissibility-derived role: they carry capacity cost but serve no function required by the derivation chain. More concretely, the capacity accounting that yields $C_{\text{total}} = 61$ (gauge generators + fermion DOF + Higgs components) assumes the gravitational sector is serviced by exactly 2 propagating modes; $d = 5$ would require reallocating capacity to 5 gravitational modes, disrupting the matter-content budget and breaking the fermion-scan closure that selects the Standard Model field content.

$d = 4$ is the unique dimension satisfying all four. □

Why Lovelock alone does not suffice, and why "setting $\alpha = 0$ in $d \geq 5$" does not rescue higher dimensions. Two objections to the $d = 4$ selection merit explicit treatment.

The first objection is that Lovelock's theorem does not make gravity *impossible* in $d \geq 5$; it merely makes the field equation *non-unique*. One could accept the most general Lovelock Lagrangian in $d \geq 5$ and treat α/κ as a measurable coupling. The APF's answer is that this would constitute a *derivation failure*, not a physical inconsistency: the framework claims to derive all couplings from A1, and an undetermined ratio is a gap in that derivation. The argument is not that $d \geq 5$ gravity is inconsistent, but that the derivation chain terminates at a parametric family rather than a unique answer, breaking the zero-free-parameter architecture.

The second objection is subtler: one might propose that in $d \geq 5$, the Gauss–Bonnet coupling happens to vanish ($\alpha = 0$), recovering a unique Einstein equation even though Lovelock permits more. This fails for two reasons:

(a) Setting $\alpha = 0$ is an *additional constraint* not derived from A1 or any admissibility condition. It must be postulated, which violates the framework's derivation discipline: if $\alpha = 0$ cannot be derived from the enforcement structure, then it is a free choice, and the derivation is incomplete.

(b) Even with $\alpha = 0$, the $d = 5$ gravitational sector still carries 5 propagating DOF, not 2. These 5 channels consume enforcement capacity regardless of the field-equation form. The DOF count is a property of the *kinematics* (the linearized mode structure of a metric theory in d dimensions), not of the specific *dynamics* (the choice of Lovelock couplings). Setting the Gauss–Bonnet coupling to zero changes the dynamics but does not remove the kinematic overhead: there are still 5 independent polarizations of the graviton in $d = 5$, each requiring capacity allocation. Condition (D4) excludes $d = 5$ on this kinematic ground alone, independently of the Lovelock-uniqueness argument.

In summary, (D2) and (D4) provide *independent* exclusions of $d \geq 5$: (D2) via response-law non-uniqueness (a derivation-chain failure), and (D4) via excess gravitational capacity overhead (a capacity-budget failure). Either suffices; both apply.

5.6 Einstein equations from Lovelock

Theorem 5.13 ($T_{9,\mathrm{grav}}$: Einstein Field Equations)**.** *The dynamics of the cost kernel $g_{\mu\nu}$ are governed by the Einstein equations:*

$$G_{\mu\nu} + \Lambda\, g_{\mu\nu} = \kappa\, T_{\mu\nu},$$

where $G_{\mu\nu}$ is the Einstein tensor, Λ is the cosmological constant, and $T_{\mu\nu}$ is the stress-energy tensor of the matter content.

The proof uses Lovelock's theorem, whose exact hypotheses we verify.

Lemma 5.14 (Lovelock hypotheses verified)**.** *Lovelock's theorem (1971) [19] states:*

Theorem (Lovelock). *Let (M, g) be a d-dimensional pseudo-Riemannian manifold. The most general symmetric, divergence-free $(0, 2)$-tensor field $\mathcal{G}_{\mu\nu}$ constructed from $g_{\mu\nu}$ and its derivatives up to second order is:*

$$\mathcal{G}_{\mu\nu} = \sum_{p=0}^{\lfloor (d-1)/2 \rfloor} c_p\, \mathcal{H}^{(p)}_{\mu\nu},$$

where $\mathcal{H}^{(0)}_{\mu\nu} = g_{\mu\nu}$ (cosmological term), $\mathcal{H}^{(1)}_{\mu\nu} = G_{\mu\nu}$ (Einstein tensor), and $\mathcal{H}^{(p)}_{\mu\nu}$ for $p \geq 2$ are the higher Lovelock tensors (with $\mathcal{H}^{(2)}$ being the Gauss–Bonnet tensor). In $d = 4$: $\mathcal{H}^{(2)}_{\mu\nu}$ is topological (vanishes identically from the equations of motion), so the unique solution is $\mathcal{G}_{\mu\nu} = c_0\, g_{\mu\nu} + c_1\, G_{\mu\nu}$.

The hypotheses are:

(LV1) ***Pseudo-Riemannian manifold.*** *(M, g) is a smooth manifold with a non-degenerate symmetric bilinear form of constant signature.*

(LV2) ***Symmetry.*** *$\mathcal{G}_{\mu\nu} = \mathcal{G}_{\nu\mu}$.*

(LV3) ***Divergence-free.*** *$\nabla^\mu \mathcal{G}_{\mu\nu} = 0$.*

(LV4) ***Second-order.*** *$\mathcal{G}_{\mu\nu}$ depends on $g_{\alpha\beta}$ and its first and second derivatives $\partial_\gamma g_{\alpha\beta}$, $\partial_\gamma \partial_\delta g_{\alpha\beta}$, but not on higher derivatives.*

(LV5) ***Dimension.*** *$d = 4$.*

We verify each for the APF's cost-kernel metric:

(LV1): M is a smooth manifold (Proposition 5.6); $g_{\mu\nu}$ is Lorentzian with signature $(-, +, +, +)$ (Theorem 5.10, Proposition 5.11). Non-degeneracy: $g_{\mu\nu}$ has no zero eigenvalues (the timelike eigenvalue is $-N^2 < 0$ and the spatial eigenvalues are positive by the positive-definiteness of $g^{(\mathrm{sp})}$ from T_{7B}). Constant signature: the signature is $(-, +, +, +)$ everywhere, because $N > 0$ everywhere (from H4: $N^2 = C/\mathrm{Vol} > 0$) and $g^{(\mathrm{sp})}$ is positive-definite everywhere (from T_{7B}: $B(v, v) > 0$ for all $v \neq 0$ at every interface).

(LV2): Symmetry of the response tensor $\mathcal{G}_{\mu\nu}$ is required by the physical interpretation: $\mathcal{G}_{\mu\nu}$ is equated to $\kappa\, T_{\mu\nu}$, and the stress-energy tensor is symmetric (from the conservation of angular momentum, which follows from the enforcement algebra's isotropy: MD applied to the spacetime sector).

(LV3): Divergence-freeness $\nabla^\mu \mathcal{G}_{\mu\nu} = 0$ is required by the conservation of the source: $\nabla^\mu T_{\mu\nu} = 0$ (energy-momentum conservation, from A1: total capacity is conserved). Since $\mathcal{G}_{\mu\nu} = \kappa\, T_{\mu\nu}$, the left-hand side must also be divergence-free. This is the contracted Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$ (a mathematical identity of the Riemann tensor, imported), combined with $\nabla^\mu(\Lambda\, g_{\mu\nu}) = 0$ (metric compatibility).

(LV4): Second-order dependence on the metric follows from the C^2 regularity of $g_{\mu\nu}$: the cost kernel is a polynomial in the sector coordinates (from T_{7B}: $\varepsilon(v) = B(v, v)$), so its second derivatives are continuous. Higher derivatives of g are not available from the enforcement structure at this order of regularity. This is not a choice but a consequence: the enforcement cost is a quadratic form (T_{7B}), so the response law can depend on at most two derivatives (the Hessian of the cost).

(LV5): $d = 4$ is derived in Proposition 5.12.

Proof of Theorem 5.13. The five admissibility conditions (A9.1–A9.5) are:

(A9.1) *Locality*: the response depends on g and finitely many derivatives (from L_{loc}: enforcement is local to each interface). This establishes (LV4).

(A9.2) *General covariance*: the response is tensorial (from the interface-independence of the enforcement algebra: a coordinate relabeling of $\mathcal{I}$ is a kinematic automorphism, Lemma 2.4). Combined with the symmetry of $T_{\mu\nu}$, this establishes (LV2).

(A9.3) *Conservation*: $\nabla_\mu T^{\mu\nu} = 0$ identically (from A1: total capacity is conserved; the stress-energy tensor is the capacity current). This establishes (LV3).

(A9.4) *Second-order stability*: the response involves at most second derivatives of g (from T_{7B}: the cost kernel is C^2; higher derivatives would require differentiability not established). This reinforces (LV4).

(A9.5) *Hyperbolic propagation*: the linearized response admits wave-like solutions (from D3 of Proposition 5.12).

By Lemma 5.14, all five Lovelock hypotheses are satisfied. In $d = 4$ (Proposition 5.12), Lovelock's theorem gives $\mathcal{G}_{\mu\nu} = c_0\, g_{\mu\nu} + c_1\, G_{\mu\nu}$ as the unique solution. Identifying $c_0 = \Lambda$ and $c_1 = 1$ (with appropriate normalization via Newton's constant $\kappa = 8\pi G$, fixed by the weak-field limit matching condition A9.5) yields $G_{\mu\nu} + \Lambda\, g_{\mu\nu} = \kappa\, T_{\mu\nu}$. □

Imported mathematics. Lovelock's theorem (1971) is the sole external input; its hypotheses are verified in Lemma 5.14. The Hawking–King–McCarthy (1976) and Malament (1977) theorems, used in the signature derivation, have their hypotheses verified in Lemma 5.7, Lemma 5.8, and Lemma 5.9. The Kolmogorov extension theorem (1933) has its hypotheses verified in Lemma 5.5. No external theorem is invoked without explicit hypothesis verification.

Code reference: `check_Delta_signature, check_T8, check_T9_grav` in `spacetime.py, gravity.py`.

Part II
Gauge Template and Matter Content

6 Gauge template uniqueness: exhaustive Lie algebra classification

Paper 2 §5.1 states $L_{\text{gauge,unique}}$ in compact form. This section provides the full classification with explicit exclusion reasons for all 17 compact simple Lie algebras, quantified capacity costs, and closed-form proofs for every elimination.

The capacity cost of a gauge group G is $C(G) = \dim(G) \cdot F(1)$ where F is the unique cost function from L_{Cauchy} (Theorem 1.3). Since $F(1) = 1$ (unit normalization), $C(G) = \dim(G)$. This is the quantity minimized in the classification below.

Proposition 6.1 (Gauge generators are separated enforcement interfaces)**.** *For any compact Lie group G with Lie algebra $\mathfrak{g}$ of dimension n, the n independent generators $\{T^a\}_{a=1}^n$ constitute n* separated *enforcement interfaces in the sense of L_{Cauchy}'s additivity premise (C1): for any pair $a \neq b$, committing enforcement to T^a does not reshape the channel landscape seen by T^b, so the superadditive surplus $\Delta(T^a, T^b) = 0$. Therefore the total cost is additive: $C(G) = \sum_{a=1}^n F(1) = \dim(G)$.*

Proof. The generators of a compact Lie algebra are Killing-orthogonal with respect to the negative-definite Killing form: $K(T^a, T^b) \propto \delta^{ab}$. By T_M (interface monogamy, Paper 1 Supplement) and L_{loc} (locality), two distinct enforcement channels whose anchors are mutually orthogonal under a sector-orthogonal bilinear form are *separated interfaces*: commitment to one does not rearrange the other. This is the condition under which L_{Cauchy}'s C1 (additivity of cost at separated interfaces) applies directly. Hence the cost of enforcing all n generators is $\sum_a F(1) = n$ by C1 applied $(n-1)$ times. Non-commuting generators within a simple factor (e.g., $[T^1, T^2] = if^{12c}T^c$ in SU(3)) are Killing-orthogonal despite their non-commutativity: the separated-interface condition is stated in terms of the B-orthogonality of the enforcement projections, not in terms of Lie-algebra commutativity. Therefore the cost is linear in $\dim(G)$ for every compact simple Lie group, including non-abelian simple groups such as E_6 whose 78 generators form an irreducible Killing-orthogonal basis. □

Remark. This proposition closes an otherwise tacit step: without it, the cost formula $C(G) = \dim(G)$ would implicitly assume that gauge generators are C1-admissible without derivation. The capacity-minimality comparisons throughout this section (E_6: 78 vs. SU(3): 8, etc.) all rely on Proposition 6.1; it is the bridge from L_{Cauchy}'s $\mathbb{N}$-domain additivity to the linearity of cost in $\dim(G)$.

6.1 Setup: the classification problem

Theorem R (§3.5) provides three abstract carrier requirements:

R1 A faithful complex carrier of dimension ≥ 3 with irreducible totally antisymmetric invariant of rank $k \geq 3$.

R2 A faithful pseudoreal carrier of dimension exactly 2.

R3 A single abelian factor (U(1)) with distinct charge assignments for all physical multiplets.

R1–R3 are derived from A1 without reference to any specific Lie group. The problem is now purely mathematical: *which compact Lie group(s) can host all three carriers simultaneously, at minimum total capacity cost?*

The capacity cost of a gauge group G is $\dim(G) \cdot \varepsilon^*$ (L_{Cauchy}, Theorem 1.3): the number of independent gauge generators, each costing one enforcement channel. This cost functional is unique (no free parameter) and is the quantity to be minimized.

6.2 Exhaustive classification of the confining factor (R1)

Input. The classification of compact simple Lie algebras is a completed theorem of pure mathematics. There are exactly four infinite families (A_n, B_n, C_n, D_n for $n \geq 1, 2, 3, 4$ respectively) and five exceptional algebras (G_2, F_4, E_6, E_7, E_8). For each, the fundamental (lowest-dimensional faithful) representation has a known dimension and reality type (real, pseudoreal, or complex). This data is imported from standard references [3, 4] and is not framework-dependent.

Filters. R1 imposes three necessary conditions on the confining factor G_{conf}:

(F1) The fundamental representation must be *complex* ($V \not\cong V^*$ as G-modules). This is forced by $B1'$ (Theorem 3.6): real and pseudoreal carriers cannot sustain oriented composites.

(F2) The representation must admit a totally antisymmetric invariant of rank $k \geq 3$. This is forced by the oriented-composite requirement: bilinear ($k = 2$) invariants produce self-conjugate composites ($B = \bar{B}$).

(F3) The representation must be *faithful*: every non-identity group element acts non-trivially. This is forced by enforcement completeness (A1): if a gauge transformation acts trivially on all matter, it is physically undetectable and should not be counted.

Theorem 6.2 ($L_{\text{gauge,unique}}$ — Step 2: Confining Factor Classification)**.** *The only compact simple Lie algebra whose fundamental representation satisfies (F1)–(F3) is $A_n = \mathfrak{su}(n+1)$ with $n \geq 2$ (equivalently, $\mathrm{SU}(N_c)$ with $N_c \geq 3$).*

Proof. We test each family and each exceptional algebra against (F1)–(F3). The full data is:

Algebra	**Group**	$\dim_{\text{fund}}$	**Type**	**(F1)**	**(F2)**	**Verdict**
Family $A_n = \mathfrak{su}(n+1)$						
A_1	SU(2)	2	Pseudoreal	Fail	—	Excluded (F1)
A_2	SU(3)	3	Complex	Pass	k=3: Pass	**Pass**
A_3	SU(4)	4	Complex	Pass	k=4: Pass	Pass*
A_4	SU(5)	5	Complex	Pass	k=5: Pass	Pass*
$A_{n\geq 5}$	SU(n+1)	n+1	Complex	Pass	k=n+1: Pass	Pass*
Family $B_n = \mathfrak{so}(2n+1)$, $n \geq 2$						
B_2	SO(5)	5	Real	Fail	—	Excluded (F1)
B_3	SO(7)	7	Real	Fail	—	Excluded (F1)
$B_{n\geq 2}$	SO(2n+1)	2n+1	Real	Fail	—	Excluded (F1)
Family $C_n = \mathfrak{sp}(2n)$, $n \geq 3$						
C_3	Sp(6)	6	Pseudoreal	Fail	—	Excluded (F1)
$C_{n\geq 3}$	Sp(2n)	2n	Pseudoreal	Fail	—	Excluded (F1)
Family $D_n = \mathfrak{so}(2n)$, $n \geq 4$						
D_4	SO(8)	8	Real	Fail	—	Excluded (F1)
$D_{n\geq 4}$	SO(2n)	2n	Real	Fail	—	Excluded (F1)
Exceptional algebras						
G_2		7	Real	Fail	—	Excluded (F1)
F_4		26	Real	Fail	—	Excluded (F1)
E_6		27	Complex	Pass	k=3: Pass	Excluded†
E_7		56	Pseudoreal	Fail	—	Excluded (F1)
E_8		248	Real	Fail	—	Excluded (F1)

Pass: satisfies (F1)–(F3) but is non-minimal (higher $\dim(G)$ than SU(3); eliminated in Step 3 below).

†E_6 satisfies (F1) and (F2) but is excluded by capacity minimality: $\dim(E_6) = 78$ vs. $\dim(\mathrm{SU}(3)) = 8$. The fundamental representation of E_6 has dimension 27, so even if one ignored cost, the matter field degrees of freedom per generation would be vastly larger than the SM.

Mechanism of exclusion. The B_n, C_n, D_n families and the exceptionals G_2, F_4, E_7, E_8 are all excluded by (F1) alone: their fundamental representations are real or pseudoreal, and $B1'$ (Theorem 3.6) proves that real/pseudoreal carriers cannot sustain oriented composites. This is not a numerical coincidence — it reflects the deep mathematical fact that only A_n ($n \geq 2$) and E_6 have complex fundamental representations among compact simple Lie algebras.

E_6 passes (F1) and (F2) but is excluded on capacity grounds: $\dim(E_6) = 78$ requires 78 gauge generators, each costing one enforcement channel. The SU(3) alternative requires 8 generators. By L_{Cauchy} (Theorem 1.3), the cost functional is linear in generator count, so E_6 costs $78/8 = 9.75$ times as much as SU(3). A1 (capacity minimality) selects the cheapest adequate structure.

The only surviving family is $A_n = \mathrm{SU}(N_c)$ with $N_c \geq 3$. □

Where the E_6 cut actually happens. E_6 satisfies every representation-theoretic filter: it has complex fundamental representations (F1) and antisymmetric invariants of rank ≥ 3 (F2). The difference between "$\mathrm{SU}(N_c)$ survives" and "E_6 is rejected" is not a representation-theory result; it is a capacity-minimality result under A1. A reader who accepts the carrier requirements R1–R3 but doubts the capacity-minimality clause of A1 has not yet been given a reason to prefer SU(3) over E_6. The minimality clause is what closes the gap, and it is derived from A1 in L_{col} and the enforcement-cost ledger. The E_6 exclusion is therefore a direct consequence of the minimum-cost ontology: E_6 is admissible (its cost fits below the capacity bound) but is not minimum-cost; SU(3) is. "E_6 is rejected" is shorthand for "E_6 has higher cost than SU(3) and both are admissible," not for an act of rejection.

6.3 Capacity optimization: $N_c = 3$

Within the surviving $\mathrm{SU}(N_c)$ family, all $N_c \geq 3$ satisfy (F1)–(F3). The antisymmetric invariant has rank $k = N_c$ (the ε-tensor $\varepsilon_{i_1 \cdots i_{N_c}}$). The oriented-composite requirement forces $k \geq 3$ ($B1'$ excludes $k = 2$), so $N_c \geq 3$ is already established.

Two further filters apply:

Proposition 6.3 (Witten anomaly excludes even N_c)**.** *For gauge group* $\mathrm{SU}(N_c) \times \mathrm{SU}(2) \times \mathrm{U}(1)$*, the number of* SU(2) *doublets per generation is* $N_c + 1$ *(one quark doublet contributing* N_c *colors, plus one lepton doublet). The Witten anomaly [5] requires this number to be even. Therefore* N_c *must be odd.*

Joint consistency with T_{field}. The statement "one quark doublet plus one lepton doublet per generation" is the SM-like template structure that emerges from T_{field} (§7) rather than an input independent of it. The Witten-anomaly cut applied in the current section therefore functions as a *joint consistency* check with the fermion scan: for any odd $N_c \geq 3$, the minimum-DOF anomaly-free template has the (quark-doublet, lepton-doublet) structure, and the Witten filter is satisfied; for any even N_c, no admissible template of this structure survives. The logical ordering is not "N_c is fixed in §5 then templates are scanned in §6." Rather, the admissible N_c's and admissible fermion templates are a joint optimum, and the supplement exhibits that joint optimum by first narrowing N_c to odd values under the generic template ansatz (this section) and then closing $\mathrm{N}_c = 3$ by the full scan (§7). Readers who prefer a single-step derivation may take Propositions 6.3 and 6.4 together with Theorem 7.7 as a simultaneous result: ($N_c = 3$, SM template) is the unique joint minimum-cost admissible pair.

Proposition 6.4 (Capacity minimality selects $N_c = 3$)**.** *The total gauge group dimension is* $\dim(G) = (N_c^2 - 1) + 3 + 1 = N_c^2 + 3$*. For odd* $N_c \geq 3$*:*

N_c	*Witten*	$\dim(G)$	C_{total}	*Verdict*
3	*Pass (4 doublets)*	*12*	*61*	***Minimum cost***
5	*Pass (6 doublets)*	*28*	*97*	*Non-minimal (2.3×)*
7	*Pass (8 doublets)*	*52*	*141*	*Non-minimal (4.3×)*
9	*Pass (10 doublets)*	*84*	*193*	*Non-minimal (7×)*

$C_{\text{total}} = (4N_c + 3) \cdot 3 + 4 + (N_c^2 + 3)$*: fermions + Higgs + gauge bosons. The cost is strictly increasing and unbounded.* $N_c = 3$ *is the unique minimum.*

6.4 Chiral factor classification (R2)

Proposition 6.5 (SU(2) is the unique chiral factor)**.** *R2 requires a compact simple Lie group with a faithful pseudoreal representation of dimension exactly* 2*. Among all compact simple Lie algebras:*

(i) $\mathfrak{su}(2)$ *has a 2-dimensional irreducible representation (the fundamental doublet) which is pseudoreal (*$J = i\sigma_2 K$*,* $J^2 = -1$*).*

(ii) All other compact simple Lie algebras have minimum faithful dimension ≥ 3*:* $\mathfrak{su}(n+1)$ *for* $n \geq 2$ *has* $\dim_{\text{fund}} = n + 1 \geq 3$*;* $\mathfrak{so}(2n+1)$ *has* $\dim_{\text{fund}} \geq 5$*;* $\mathfrak{sp}(2n)$ *for* $n \geq 2$ *has* $\dim_{\text{fund}} \geq 4$*; all exceptionals have* $\dim_{\text{fund}} \geq 7$*.*

Therefore SU(2) *is the unique compact simple Lie group satisfying R2.*

At the group level, the gauge factor must be SU(2) *rather than its quotient* $\mathrm{SO}(3) = \mathrm{SU}(2)/\mathbb{Z}_2$*: the doublet representation does not descend to* SO(3) *(it is not single-valued under* 2π *rotations), so faithfulness requires the simply connected cover* SU(2)*.*

6.5 Abelian factor and simple-group alternatives (R3)

Proposition 6.6 (U(1) is the unique abelian factor)**.** *R3 requires exactly one abelian factor.* $\mathrm{U}(1) \cong \mathbb{R}/\mathbb{Z}$ *is the unique connected compact 1-dimensional abelian Lie group. A second* U(1) *would be non-minimal under A1: one* U(1) *with distinct hypercharge assignments already resolves all multiplet degeneracies (Theorem R, R3 derivation in §3). Additional* U(1) *factors add* ≥ 1 *gauge generator each with no enforcement gain.*

Proposition 6.7 (Simple-group alternatives are excluded)**.** *Any simple compact Lie group* G_{simple} *containing* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ *as a subgroup has* $\dim(G_{\text{simple}}) \geq 24$*:*

Group	dim	***Cost ratio***	***Contains SM?***
$\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$	*12*	1.0×	*(Product)*
SU(5)	*24*	2.0×	*Yes (Georgi–Glashow)*
SO(10)	*45*	3.75×	*Yes*
E_6	*78*	6.5×	*Yes*

Every simple envelope costs at least 2× *the product structure. A1 (capacity minimality) selects the product. Additionally, enforcement independence (Step 1 of* $L_{\text{gauge,unique}}$*) requires factorization:* T_M *(monogamy) and* L_{loc} *(locality) force independent enforcement channels to live in commuting subgroups with independent couplings. A simple group has a single coupling constant, contradicting the requirement that the confining factor (which confines at low energy) and the chiral factor (which must* not *confine, to preserve chirality) run independently.*

6.6 Consolidation: unique gauge template

Combining Steps 1–6:

Theorem 6.8 ($L_{\text{gauge,unique}}$: Gauge Template Uniqueness)**.** *The unique gauge group satisfying R1–R3 at minimum capacity cost is*

$$G \;=\; \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1), \qquad \dim(G) = 12.$$

No alternative compact Lie group structure satisfies all three carrier requirements. No simple-group envelope is capacity-competitive. No additional abelian factor is needed.

Derivation chain: A1 → Theorem R (R1+R2+R3) → Lie classification $\to \mathrm{SU}(N_c) \times \mathrm{SU}(2) \times \mathrm{U}(1) \to$ *Witten + minimality* $\to N_c = 3$.

Imported mathematics: The classification of compact simple Lie algebras (Killing 1888, Cartan 1894) and the Witten anomaly theorem (Witten 1982) are used as external mathematical inputs. Both are established results with independent proofs; neither is framework-specific. All physical inputs (R1–R3) are derived from A1 in §3.

Code reference: `check_L_gauge_template_uniqueness` in `gauge.py`.

7 Fermion content: exhaustive scan

Descriptive reading of the scan. The enumeration in this section is *not* a physical process by which nature "chooses" the Standard Model fermion content from a space of candidates. The admissibility conditions (five exclusion proofs bounding the search space, plus seven sequential filters encoding anomaly cancellation, asymptotic freedom, and chirality) define a predicate $\mathrm{Adm}(\mathcal{T})$ on chiral fermion templates. The scan is an *algorithmic verification* that the set $\{\mathcal{T} : \mathrm{Adm}(\mathcal{T})\}$ has a unique member up to the stated equivalences — namely the 45-Weyl-fermion SM template. The 1,680 distinct templates (4,680 ordered iterations) are artefacts of the verification procedure, not candidates in a dynamical competition. The claim of this section is therefore descriptive: "the admissibility predicate is satisfied by exactly one template," not "the framework selects one template from among 1,680."

This section specifies the 1,680-template search space, defines the seven sequential filters, provides the survivor count after each filter, and proves that no candidates exist outside the explicitly bounded set.

7.1 Template definition

A *chiral fermion template* is a finite multiset $\mathcal{T} = \{(r_i^{(3)}, r_i^{(2)})\}_{i=1}^{|\mathcal{T}|}$ where $r_i^{(3)}$ is an irreducible representation of SU(3) and $r_i^{(2)}$ is an irreducible representation of SU(2). Each entry represents a left-handed Weyl fermion field. The total number of Weyl degrees of freedom per generation is $\mathrm{DOF} = \sum_i \dim(r_i^{(3)}) \cdot \dim(r_i^{(2)})$. The template describes one generation; the full content is $N_{\mathrm{gen}} \times \mathrm{DOF}$ with $N_{\mathrm{gen}} = 3$ (derived in Paper 2, §9).

The search space is bounded by five closed-form exclusion proofs that restrict both the allowed representations and the number of field types.

7.2 Search space bounds: five exclusion proofs

Proposition 7.1 (P1: Large SU(3) representations are AF-excluded)**.** *Any* SU(3) *representation with dimension* ≥ 10 *(i.e.,* $\mathbf{10}$*,* $\mathbf{15}$*, ...) exceeds the* SU(3) *asymptotic freedom budget as a single field. For the* $\mathbf{10}$*: the Dynkin index is* $T(\mathbf{10}) = 15/2$*; the AF contribution of a single field in the* $\mathbf{10}$ *with* SU(2) *singlet and 3 generations is* $(4/3)(15/2)(1)(3) = 30 > 11 = (11/3) \cdot 3$*. Therefore no template containing any* SU(3) *rep of dimension* ≥ 10 *can be asymptotically free. The allowed* SU(3) *representations are:* $\mathbf{1}, \mathbf{3}, \bar{\mathbf{3}}, \mathbf{6}, \bar{\mathbf{6}}, \mathbf{8}$*.*

Proposition 7.2 (P2: Colored SU(2) triplets and higher are AF-excluded)**.** *Any colored field (*$\dim(r^{(3)}) \geq 3$*) in an* SU(2) *representation of dimension* ≥ 3 *exceeds the* SU(2) *AF budget. For a* $(\mathbf{3}, \mathbf{3})$ *field:* $T(\mathbf{3}_{\mathrm{SU}(2)}) = 2$*,* $\dim(\mathbf{3}_{\mathrm{SU}(3)}) = 3$*, contribution* $= (4/3)(2)(3)(3) = 24 > 22/3 \cdot 2 = 44/3 \approx 14.7$*. Therefore colored fields are restricted to* SU(2) *representations* $\{\mathbf{1}, \mathbf{2}\}$*.*

Proposition 7.3 (P3: Colorless SU(2) triplets are DOF-excluded)**.** *Replacing a colorless* SU(2) *doublet* $(\mathbf{1}, \mathbf{2})$ *with a triplet* $(\mathbf{1}, \mathbf{3})$ *adds* $(3 - 2) \times 3 = 3$ *DOF per 3 generations. Since the SM template already has 45 DOF and is the minimum survivor (as verified by the scan), any template with colorless* SU(2) *triplets has DOF* $\geq 48 > 45$ *and is non-minimal.*

Proposition 7.4 (P4: Multiple colored doublets are DOF-excluded)**.** *A template with two colored* SU(2) *doublets requires at least two colored singlets for* $[\mathrm{SU}(3)]^3$ *anomaly cancellation plus lepton content. The tightest such template is* $(\mathbf{3}, \mathbf{2}) + (\bar{\mathbf{3}}, \mathbf{2}) + (\mathbf{3}, \mathbf{1}) + (\bar{\mathbf{3}}, \mathbf{1}) + (\mathbf{1}, \mathbf{2}) + (\mathbf{1}, \mathbf{1})$*, with DOF* $= (6 + 6 + 3 + 3 + 2 + 1) \times 3 = 63 > 45$*.*

Proposition 7.5 (P5: More than 5 field types are DOF-excluded)**.** *Each field type contributes at least* $1 \times 1 \times 3 = 3$ *DOF (the smallest possible: a colorless* SU(2) *singlet with 3 generations). A template with* > 5 *field types has DOF* $\geq 45 + 3 > 45$ *and is non-minimal.*

These five proofs establish that the search space is *finite and bounded*: SU(3) reps $\in \{3, \bar{3}, 6, \bar{6}, 8\}$ (excluding singlets, which are colorless), SU(2) reps $\in \{1, 2\}$, and $|\mathcal{T}| \leq 5$. Combined with multiplicity choices (0–3 colored singlets drawn with replacement, 0–2 lepton singlets, doublet present or absent), this yields 1,680 distinct candidate templates.[1]

7.3 Seven filters

Each filter is derived from the framework. They are applied sequentially; each eliminates candidates that passed all previous filters.

F1 SU(3) **asymptotic freedom.** $b_0^{(3)} = 11 - (4/3)\sum_i T(r_i^{(3)})\dim(r_i^{(2)}) \cdot N_{\text{gen}} > 0$. Derived from non-abelian self-interaction dominance (§4).

F2 SU(2) **asymptotic freedom.** $b_0^{(2)} = 22/3 - (4/3)\sum_i T(r_i^{(2)})\dim(r_i^{(3)}) \cdot N_{\text{gen}} > 0$.

F3 Chirality. The template must contain both colored doublets and colored singlets. A template with only colored doublets or only colored singlets is vector-like and has zero intrinsic gauge irreversibility (R2).

F4 $[\text{SU}(3)]^3$ **anomaly cancellation.** $\sum_i A(r_i^{(3)}) \cdot \dim(r_i^{(2)}) = 0$, where A is the cubic anomaly coefficient ($A(\mathbf{3}) = 1/2$, $A(\bar{\mathbf{3}}) = -1/2$, $A(\mathbf{6}) = 5/2$, $A(\bar{\mathbf{6}}) = -5/2$, $A(\mathbf{8}) = 0$).

F5 Witten anomaly freedom. The total number of SU(2) doublets (counted with SU(3) multiplicity) must be even: $\sum_i \dim(r_i^{(3)}) \cdot [r_i^{(2)} = \mathbf{2}] \equiv 0 \pmod 2$.

F6 Full anomaly system solvable. The combined system of $[\text{SU}(2)]^2[\text{U}(1)]$, $[\text{SU}(3)]^2[\text{U}(1)]$, $[\text{grav}]^2[\text{U}(1)]$, and $[\text{U}(1)]^3$ anomaly cancellation conditions must admit rational hypercharge solutions. Templates containing colored singlets with $A(R) = 0$ (e.g. the $\mathbf{8}$) are handled by spectator reduction: such fields carry $Y = 0$ and decouple from all anomaly conditions; the reduced template is then tested analytically. For the spectator-free core, the system reduces to a quadratic in $z = Y_u/Y_Q$ having rational roots (§8).

F7 CPT quotient. Templates related by charge conjugation (replacing every representation with its conjugate) describe the same physics. The CPT equivalence class is identified and only the canonical representative is counted.

7.4 Waterfall table

Filter	Survivors	Eliminated	Mechanism
Search space (P1–P5)	1,680	—	Bounded by exclusion proofs
F1+F2: Asymptotic freedom	348	1,332	$b_0^{(3)} > 0$ and $b_0^{(2)} > 0$
F3: Chirality	324	24	Colored doublets + singlets required
F4: $[\text{SU}(3)]^3$ anomaly	24	300	$\sum A_3 d_2 = 0$
F5: Witten anomaly	12	12	#doublets even
F6: Full anomaly system	8	4	Rational hypercharges exist
F7: CPT quotient	4	4	Conjugate pairs identified
Minimality	**1**	3	Min DOF = 45 selected

Counting convention (narrative vs. enumeration). The counts in the table above are the number of *distinct* templates surviving each filter, with unordered colored-singlet positions. The reference implementation (`check_T_field` in `gauge.py` and `fermion_scan_standalone.py`) uses an ordered-tuple enumeration (`itertools.product`) and therefore tests each distinct template multiple times, producing a raw iteration count of 4,680 rather than 1,680, and downstream stage counts $48 \to 24 \to 4 \to 2 \to 1$ after quotienting by the ordering redundancy $48/2 = 24$,

[1] Colored singlets are unordered within a template, so the correct enumeration uses combinations with replacement: $5 \times (1 + 5 + 15 + 35) \times 2 \times 3 = 1{,}680$. A standalone verification script with exact rational arithmetic and four built-in red-team challenges is archived at `https://doi.org/10.5281/zenodo.19154197`; an interactive Colab walkthrough is at `https://colab.research.google.com/drive/1otO1Qwp_Lr3OsK6ykkeMRATanecPkKg6`; source code is at `https://github.com/Ethan-Brooke/APF-Paper-2-The-Structure-of-Admissible-Physics`.

etc. The survivor *set* (as an unordered collection of templates) is identical under either convention, and the unique minimum-DOF winner is the Standard Model in both cases. A reviewer running the code will see the enumeration multiplicities; the table reports the unordered count because that is what the five exclusion proofs (P1–P5) actually bound.

Spectator octet handling. Templates #5–#8 in the post-F6 listing below include the adjoint $\mathbf{8}$ as a colored SU(2) singlet. Because $A(\mathbf{8}) = 0$, spectator-reduction analysis (removing the anomaly-free spectator field and testing the reduced template) keeps these templates through F6 in the prose derivation, where they fail only at the minimality stage (DOF $\geq 69 > 45$). The reference implementation takes a shorter path: its F6 predicate (`passes_full_anomaly`) requires every colored singlet to have SU(3) dimension equal to $N_c = 3$, which eliminates the $(\mathbf{8}, \mathbf{1})$ at F6 rather than at minimality. Both treatments converge on the same survivor set (the SM at DOF = 45), and the two are equivalent for admissibility bookkeeping: spectator octets are eliminated either by analytical reduction plus minimality, or by the uniform-dimension test, at no change to the final result. The table's F6 count of 8 is the analytical count before spectator reduction; the code's F6 count of 4 is the count after uniform-dimension filtering. Both routes agree after CPT quotient and minimality.

The final winner is exact and reproduced by `check_T_field` in `gauge.py` and independently by `fermion_scan_standalone.py` [20].

Proposition 7.6 (Filter-order independence)**.** *The survivor set $\mathcal{S} = \{\mathcal{T} : \mathcal{T}$ passes F1–F7$\}$ is independent of the order in which the seven filters are applied.*

Proof. Each filter F_k defines a predicate $P_k : \mathcal{T} \to \{0, 1\}$ on templates. A template survives all filters iff it satisfies the conjunction $P_1 \wedge P_2 \wedge \cdots \wedge P_7$. Logical conjunction is commutative and associative: $P_{\sigma(1)} \wedge \cdots \wedge P_{\sigma(7)} = P_1 \wedge \cdots \wedge P_7$ for any permutation $\sigma \in S_7$. Therefore the survivor set $\mathcal{S} = \{\mathcal{T} : \bigwedge_k P_k(\mathcal{T}) = 1\}$ is identical regardless of evaluation order.

What changes with filter order is the *waterfall count* (number of templates eliminated at each stage), because a template eliminated by F_k in the standard order might instead be eliminated by F_j with $j > k$ if the order is reversed. The final survivor set, however, is order-invariant. The standalone verification script confirms this: running all $7! = 5{,}040$ permutations on the 1,680-template set yields the same 4 post-CPT survivors (and the same unique DOF-45 winner) in every case. □

Search-space bias and completeness. A distinct concern is whether the search space itself (the 1,680 templates bounded by P1–P5) introduces bias by excluding candidates that might survive the filters. This is addressed by the five exclusion proofs, each of which proves an *unconditional* bound: P1 excludes SU(3) reps of dimension ≥ 10 because a single such field exceeds the AF budget regardless of what other fields are present; P2–P5 similarly establish bounds that depend only on the AF constraints and the DOF lower bound, not on the other filters. Crucially, the exclusion proofs use only AF and DOF bounds, which are *monotone*: adding fields to a template can only increase the DOF and decrease the AF margin. Therefore any template outside the search space fails either AF or minimality, and would be eliminated by F1/F2 or the DOF bound regardless of the remaining filters. The search space is the exact set of templates that *could* survive; no candidates are hidden outside it.

Explicit listing of all post-F6 survivors. The 8 templates surviving F1–F6 (before CPT quotient) are:

#	Template	DOF	Status
1	$(3,2)+(\bar{3},1)+(\bar{3},1)+(1,2)+(1,1)$	45	SM
2	$(\bar{3},2)+(3,1)+(3,1)+(1,2)+(1,1)$	45	CPT of #1
3	$(3,2)+(\bar{3},1)+(\bar{3},1)+(1,2)+(1,1)+(1,1)$	48	SM + extra singlet
4	$(\bar{3},2)+(3,1)+(3,1)+(1,2)+(1,1)+(1,1)$	48	CPT of #3
5	$(3,2)+(\bar{3},1)+(\bar{3},1)+(8,1)+(1,2)+(1,1)$	69	SM + spectator octet
6	$(\bar{3},2)+(3,1)+(3,1)+(8,1)+(1,2)+(1,1)$	69	CPT of #5
7	$(3,2)+(\bar{3},1)+(\bar{3},1)+(8,1)+(1,2)+(1,1)+(1,1)$	72	SM + octet + singlet
8	$(\bar{3},2)+(3,1)+(3,1)+(8,1)+(1,2)+(1,1)+(1,1)$	72	CPT of #7

Templates #5–#8 contain the adjoint $\mathbf{8}$ as a colored SU(2) singlet. Since $A(\mathbf{8}) = 0$, this field carries $Y = 0$ and decouples from all anomaly conditions—it is an anomaly-free spectator. Its DOF cost ($8 \times 3 = 24$ per generation) is far above the minimum.

After the CPT quotient (F7), templates #2, #4, #6, and #8 are identified with #1, #3, #5, and #7 respectively, leaving 4 CPT-inequivalent survivors. Minimality selects DOF = 45, giving exactly one winner: the Standard Model. The second-lightest survivor (#3, DOF = 48) differs from the SM by one additional right-handed neutrino-like singlet per generation; its exclusion by minimality is the $C = 64 \to 5\sigma$ result of §9.

7.5 The unique survivor

Theorem 7.7 (T_{field}: SM Fermion Content)**.** *The unique minimal chiral fermion template consistent with* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$*, three generations, and all admissibility constraints is:*

$$\{Q(\mathbf{3},\mathbf{2}),\ u^c(\bar{\mathbf{3}},\mathbf{1}),\ d^c(\bar{\mathbf{3}},\mathbf{1}),\ L(\mathbf{1},\mathbf{2}),\ e^c(\mathbf{1},\mathbf{1})\} \times 3 \textit{ generations},$$

with $\mathrm{DOF} = (6+3+3+2+1) \times 3 = 45$ *Weyl fermions.*

No other template among the 1,680 candidates has DOF ≤ 45 and passes all seven filters. The five exclusion proofs (P1–P5) guarantee that no candidate outside the search space can have DOF < 45. The derivation is therefore exhaustive: the SM fermion content is the unique solution.

Code reference: `check_T_field` in `gauge.py`.; standalone verification: `fermion_scan_standalone.py` [20]

8 Anomaly cancellation and hypercharge uniqueness

8.1 The four anomaly conditions

The quantum consistency of a chiral gauge theory requires the cancellation of all gauge anomalies. For SU(3)×SU(2)×U(1) with the surviving template, there are four independent conditions on the five hypercharges $\{Y_Q, Y_u, Y_d, Y_L, Y_e\}$. Three are linear:

$$[\mathrm{SU}(2)]^2[\mathrm{U}(1)] = 0: \qquad N_c Y_Q + Y_L = 0, \tag{13}$$

$$[\mathrm{SU}(3)]^2[\mathrm{U}(1)] = 0: \qquad 2Y_Q - Y_u - Y_d = 0, \tag{14}$$

$$[\mathrm{grav}]^2[\mathrm{U}(1)] = 0: \qquad 2N_c Y_Q + 2Y_L - N_c Y_u - N_c Y_d - Y_e = 0. \tag{15}$$

The fourth is the cubic anomaly:

$$[\mathrm{U}(1)]^3 = 0: \quad 2N_c Y_Q^3 + 2Y_L^3 - N_c Y_u^3 - N_c Y_d^3 - Y_e^3 = 0. \tag{16}$$

8.2 Reduction to a quadratic

Solving (13)–(15) gives $Y_L = -N_c Y_Q$, $Y_d = 2Y_Q - Y_u$, $Y_e = -2N_c Y_Q$. Substituting into (16) and dividing by $Y_Q^3 \neq 0$ (non-trivial grading), define $z := Y_u/Y_Q$:

$$z^2 - 2z - (N_c^2 - 1) = 0. \tag{17}$$

Why the roots are rational. Equation (17) is a *monic* polynomial (leading coefficient = 1) with integer coefficients (-2 and $-(N_c^2 - 1)$, where $N_c \in \mathbb{Z}$). By the rational root theorem for monic integer polynomials: if $z = p/q$ in lowest terms is a root, then $q \mid 1$, so $z \in \mathbb{Z}$. The discriminant $\Delta = 4 + 4(N_c^2 - 1) = 4N_c^2$ is a perfect square, confirming integer roots:

$$z = 1 \pm N_c. \tag{18}$$

Since $Y_L = -N_c Y_Q$, $Y_d = 2Y_Q - Y_u = (2 - z)Y_Q$, and $Y_e = -2N_c Y_Q$ are all integer multiples of Y_Q, and Y_Q itself is a free normalization (any nonzero rational value is admissible, since it sets the overall scale of the U(1) generator), choosing $Y_Q = 1/6$ gives $Y_i \in \mathbb{Q}$ for all five multiplets. Charge quantization — the fact that all hypercharges are rational multiples of a single unit — is thus a *consequence* of anomaly cancellation, not an input. For $N_c = 3$: $z \in \{4, -2\}$. These are related by $z \to 2 - z$ (the $u \leftrightarrow d$ relabeling), so they describe the same physics. Choosing $z = 4$ and normalizing $Y_Q = 1/6$:

Theorem 8.1 ($L_{\text{hypercharge}}$: Unique Hypercharge Assignment)**.**

$$Y_Q = \tfrac{1}{6}, \quad Y_u = \tfrac{2}{3}, \quad Y_d = -\tfrac{1}{3}, \quad Y_L = -\tfrac{1}{2}, \quad Y_e = -1.$$

These are the unique hypercharges (up to normalization and $u \leftrightarrow d$) consistent with anomaly cancellation for the derived template with $N_c = 3$.

Consequences. Electric charges $Q = T_3 + Y$ give $Q(u) = 2/3$, $Q(d) = -1/3$, $Q(e) = -1$, $Q(\nu) = 0$. Charge quantization (all charges are multiples of 1/3) is derived from the anomaly structure, not assumed. The quark–lepton relation $Y_L = -N_c Y_Q$ ties lepton hypercharge to the number of colors.

Code reference: `check_T_field (hypercharge section)` in `gauge.py`.

9 Neutrino masses and the 45-fermion prediction

The derived fermion content (45 Weyl fermions per 3 generations) contains no right-handed neutrino. This is a prediction, not a gap.

The 45-fermion count is the unique minimum-DOF survivor of the 7-filter scan. It contains no right-handed neutrino ν_R. This is a *prediction*: the framework derives the minimal fermion content and finds that ν_R is not among it.

Proposition 9.1 (Neutrino mass mechanism)**.** *Neutrino masses are accommodated within the 45-fermion framework via the dimension-5 Weinberg operator:*

$$\mathcal{O}_5 \;=\; \frac{c_{ij}}{\Lambda}\,(L_i H)(L_j H),$$

where L_i is the lepton doublet of generation i, H is the Higgs doublet, and Λ is a high scale. After electroweak symmetry breaking ($\langle H\rangle = v$), this generates Majorana masses $m_\nu \sim c\,v^2/\Lambda$ for left-handed neutrinos.

Why ν_R is not counted. The Weinberg operator can be UV-completed by a heavy Majorana fermion at scale Λ (the type-I seesaw). This heavy state is above the IR saturation scale ($T_{\text{confinement}}$) and is not a distinct enforcement channel at low energies. The capacity count ($C_{\text{total}} = 61$) reflects the *low-energy* enforcement ledger; states above the saturation scale contribute to the UV completion but not to the structural partition.

Prediction. Neutrino masses are Majorana (lepton number violated by 2 units). Neutrinoless double-beta decay ($0\nu\beta\beta$) is predicted to occur at a rate determined by the Majorana mass matrix.

Observational constraint. If ν_R were added as a low-energy enforcement channel (3 species for 3 generations), $C_{\text{total}} = 64$ and $\Omega_\Lambda = 42/64 = 0.656$. Planck measures $\Omega_\Lambda = 0.6889 \pm 0.0056$ [9]: the 64-type prediction is $> 5\sigma$ from observation. The 45-fermion count is therefore observationally constrained, not merely a theoretical preference for minimality.

Code reference: `check_T_field` in `gauge.py`.

Part III
Capacity Counting and Cosmology

10 Capacity counting and the BRST question

10.1 Operational definition of capacity types

A *capacity type* is a structurally distinct enforcement role in the low-energy ledger. The count is:

$$C_{\text{total}} = \underbrace{45}_{\text{fermion types}} + \underbrace{4}_{\text{Higgs real components}} + \underbrace{12}_{\text{gauge generators}} = 61. \tag{19}$$

Fermion types (45): the 15 Weyl fermions per generation ($Q = 6$ colored-doublet components + $u^c = 3$ colored-singlet + $d^c = 3$ + $L = 2$ weak-doublet components + $e^c = 1$) times 3 generations. Color and weak-isospin multiplicity *are* counted here as independent matter-sector enforcement roles, because each distinct quantum-number configuration is a separately enforceable distinction in the capacity ledger. The arithmetic is $(6+3+3+2+1)\times 3 = 15\times 3 = 45$, consistent with `check_L_count` in `gauge.py` (`per_gen = 6+3+3+2+1 = 15`). Internal-structure gauge channels (the 27 fermion-internal channels counted in L_{vac}, Paper 2 §7.1) are a distinct bookkeeping allocated to the vacuum sector; they are not the matter-sector fermion types counted here. This resolves an earlier prose formulation that read "the 5 multiplet types" and counted $5 \times 3 = 15$ rather than 45; the correct count is $(6 + 3 + 3 + 2 + 1) \times 3 = 45$.
Higgs components (4): the 4 real degrees of freedom of the Higgs doublet. After SSB, 3 become Goldstone bosons (absorbed into $W^{\pm}, Z$) and 1 is the physical Higgs. All 4 are counted because the capacity ledger operates at the structural level, before SSB selects a vacuum.
Gauge generators (12): $\dim(\mathrm{SU}(3)) + \dim(\mathrm{SU}(2)) + \dim(\mathrm{U}(1)) = 8 + 3 + 1$. Each generator is a structurally distinct enforcement channel.

10.2 Why gauge constraints are not subtracted (the BRST question)

In perturbative quantum field theory, the physical Hilbert space is the BRST cohomology: gauge constraints subtract unphysical polarizations, reducing the propagating degrees of freedom. A natural question is: why does the capacity count not perform a similar subtraction?

The answer is that capacity types count *enforcement roles*, not *propagating degrees of freedom*. The distinction is:

Propagating DOF (QFT accounting): count on-shell polarization states. A massless gauge boson has 2 polarizations; BRST subtracts the longitudinal and timelike modes. This gives $2 \times 12 = 24$ propagating gauge DOF, not 12.

Enforcement roles (capacity accounting): count structurally distinct channels in the enforcement ledger. Each gauge generator $T^a \in \mathfrak{g}$ is one independent enforcement channel: it acts as a derivation on the algebra of observables, routing enforcement between matter fields. The number of generators is $\dim(\mathfrak{g}) = 12$, regardless of how many polarization states each generator supports.

The 12 gauge generators are assigned to the *vacuum* sector (Q1 = 0 in P_{exhaust}) precisely because they are enforcement *infrastructure*, not enforcement *consumers*. A gauge generator cannot route through itself; it has no upstream pathway to enter as a matter excitation. BRST is irrelevant to this accounting: it subtracts polarizations from the propagator, which is a kinematic object; the capacity count is a structural object.

Code reference: `check_L_count` in `gauge.py`.

11 The MECE partition

The 61 capacity types are partitioned by two binary predicates into three mutually exclusive, collectively exhaustive sectors.

Definition 11.1 (Predicates Q1 and Q2)**.**

- **Q1 Gauge addressability** (from T3): does the capacity type route through the gauge channels SU(3) × SU(2) × U(1)?
- **Q2 Confinement** (within Q1=1): does the gauge-addressable type carry conserved labels protected by SU(3) confinement?

Theorem 11.2 (P_{exhaust}: Predicate Exhaustion)**.**

$$\underbrace{61}_{C_{\text{total}}} = \underbrace{42}_{\textit{vacuum (Q1=0)}} + \underbrace{3}_{\textit{baryonic (Q1=1, Q2=1)}} + \underbrace{16}_{\textit{dark (Q1=1, Q2=0)}}.$$

No third predicate can refine this partition.

Proof. The vacuum sector (42 types) is defined by the absence of gauge-addressable labels. Any splitting predicate would introduce an addressable distinction, moving types to Q1=1 — a contradiction. The baryonic sector (3 types) is indexed by $N_c = 3$ color charges, the minimal confining carrier. Sub-ternary splitting would require SU($n < 3$) confinement, which does not exist. The dark sector (16 types) consists of gauge-singlet enforcement channels; further splitting requires a gauge pathway outside the derived group, which $L_{\text{gauge,unique}}$ excludes.

The vacuum count 42 is proved via the cross-match L_{vac} (Paper 2, §7.1): 12 gauge generators + 27 fermion internal-structure channels + 3 absorbed Higgs Goldstones = 42. The matter count is $61 - 42 = 19 = 3 + 16$. □

Code reference: `check_P_exhaust` in `core.py`.

12 Horizon equipartition (L_{equip})

12.1 Strengthened equal-priors argument

The key step in L_{equip} is the claim that the 61 capacity types receive equal capacity at the horizon. Paper 2 states this via "the principle of indifference over structurally equivalent microstates." We sharpen this to a mathematical theorem.

Theorem 12.1 (L_{equip}: Horizon Equipartition)**.** *At the causal horizon, the maximum-entropy capacity distribution over $C_{\text{total}} = 61$ types, subject to the constraint $\sum_i \varepsilon_i = C$ and the floor $\varepsilon_i \geq \varepsilon^* > 0$, is the uniform distribution $\varepsilon_i = C/C_{\text{total}}$ for all i. The density fraction of sector X is $\Omega_X = |X|/C_{\text{total}}$.*

Proof. The entropy functional $S(\{\varepsilon_i\}) = -\sum_i \varepsilon_i \ln \varepsilon_i$ is strictly concave on the simplex $\{\varepsilon_i \geq \varepsilon^* : \sum \varepsilon_i = C\}$. A strictly concave function on a compact convex set has a unique maximum. By Lagrange multipliers: $\partial S/\partial \varepsilon_i = -(1 + \ln \varepsilon_i) = \lambda$ for all i, giving $\varepsilon_i = e^{-(1+\lambda)}$ for all i. The constraint $\sum \varepsilon_i = C$ then gives $\varepsilon_i = C/61$. The floor $\varepsilon_i \geq \varepsilon^*$ is satisfied when $C \geq 61\varepsilon^*$, which holds at any interface above the critical saturation s_{crit}.

The density fraction of sector X is: $\Omega_X = \sum_{i \in X} \varepsilon_i / \sum_{\text{all}} \varepsilon_i = |X| \cdot (C/61)/C = |X|/61$. The total capacity C cancels. The fractions depend only on the integer type counts:

$$\Omega_\Lambda = 42/61 = 0.6885, \quad \Omega_m = 19/61 = 0.3115.$$

□

This is not an appeal to the "principle of indifference" — it is a constrained optimization on a finite discrete set. The strict concavity of S is a standard result; the uniqueness of the maximum is guaranteed by the compactness of the feasible set and the strict concavity of the objective.

12.2 Bridge to the Friedmann equations

The structural result above establishes the partition fractions. The cosmological interpretation requires identifying the enforcement ledger's capacity distribution with the energy budget in the Friedmann equation:

$$H^2 = \frac{8\pi G}{3}\rho_{\text{total}}, \qquad \rho_{\text{total}} = \rho_\Lambda + \rho_m. \tag{20}$$

The bridge is: at the causal horizon (Bekenstein saturation), the enforcement ledger's total committed capacity *is* the stress-energy budget. Each capacity type contributes equally to the total energy density (by L_{equip}), so $\rho_\Lambda/\rho_{\text{total}} = 42/61$ and $\rho_m/\rho_{\text{total}} = 19/61$.

Proof boundary. The identification of the enforcement horizon with the cosmological de Sitter horizon is established in Paper 6 (*Cosmological Evolution*), which derives the full de Sitter trajectory and confirms that the relevant horizon is the one at which the enforcement ledger reaches Bekenstein saturation. Without Paper 6, the cosmological claim is conditional: *if the enforcement horizon is the de Sitter horizon, then* $\Omega_\Lambda = 42/61$.

Code reference: `check_L_equip` in `cosmology.py`.

12.3 Horizon choice and departures from equilibrium

L_{equip} holds at a specific horizon: the one at which the enforcement ledger reaches Bekenstein saturation. This subsection examines (a) which horizon this is, (b) whether the prediction is sensitive to the choice, and (c) how departures from equilibrium affect the density fractions.

(a) Which horizon. Three horizons appear in cosmological contexts:

(i) The *event horizon*: the boundary of the causal future of an observer. In de Sitter space, this is the cosmological event horizon at proper distance $r_{\text{EH}} = c/H$.

(ii) The *apparent horizon*: the outermost marginally trapped surface. In a flat FRW universe, this is the Hubble radius $r_{\text{AH}} = c/H(t)$.

(iii) The *particle horizon*: the boundary of the causal past. Not relevant for the saturation argument (it concerns the past, not the enforcement budget for future commitments).

The APF's enforcement horizon is the surface at which the Bekenstein bound $S \leq A/4G\hbar$ is saturated: the entropy of the enclosed region equals the holographic capacity of the boundary. In de Sitter space, the event horizon and the apparent horizon coincide (both equal c/H_Λ in the late-time limit). At earlier epochs, the apparent horizon is time-dependent and the event horizon may not yet be well-defined. The prediction $\Omega_\Lambda = 42/61$ applies at *saturation*, which the APF identifies with the late-time de Sitter attractor. At this fixed point, the distinction between event and apparent horizons vanishes.

(b) Sensitivity to horizon choice. Suppose one evaluates the capacity partition at a different surface. The fractions $\Omega_X = |X|/C_{\text{total}}$ depend on the integer type counts, not on the area of the horizon. The total capacity C cancels in the ratio (as shown in the proof of Theorem 12.1). Therefore:

- If the type count $C_{\text{total}} = 61$ is the same at the new surface, the fractions are unchanged.
- If the type count changes (e.g., because some channels are frozen out below a mass threshold), the fractions shift. This is the sensitivity captured by the table in §15.1.

The prediction is therefore *robust to horizon choice* provided the type count is evaluated at the scale where all 61 channels are active. The Bekenstein saturation scale is precisely this scale: it is the coarsest (lowest-energy) scale at which the full enforcement ledger is committed.

(c) Departures from equilibrium. L_{equip} is a maximum-entropy result. If the system is out of equilibrium, the capacity distribution $\{\varepsilon_i\}$ need not be uniform. The density fractions would then depend on the specific non-equilibrium distribution, and the prediction 42/61 would not hold exactly. Three considerations bound the deviation:

(a) *Approach to equilibrium.* The second law of thermodynamics (here: monotonic increase of $S(\{\varepsilon_i\})$ under capacity redistribution, which follows from L_{irr}) guarantees that the distribution approaches uniformity over time. At the de Sitter fixed point, the cosmological evolution has had infinite proper time to equilibrate; the system is at or arbitrarily close to maximum entropy.

(b) *Perturbative corrections.* Near the uniform distribution, write $\varepsilon_i = (C/61)(1+\delta_i)$ with $\sum \delta_i = 0$. The density fraction of sector X becomes $\Omega_X = (|X|/61)(1+\langle\delta\rangle_X)$, where $\langle\delta\rangle_X = (1/|X|)\sum_{i\in X}\delta_i$. The correction $\langle\delta\rangle_X$ vanishes when δ_i is uncorrelated with sector membership. For the correction to shift Ω_Λ outside the Planck 1σ band (± 0.0056), one would need $|\langle\delta\rangle_\Lambda| \gtrsim 0.008$, i.e., a systematic 0.8% bias toward or against vacuum-sector channels — a large departure from equilibrium at the de Sitter fixed point.

(c) *Saturation independence.* The integer type counts (42 vacuum, 3 baryonic, 16 dark) are topological invariants of the MECE partition (§11); they do not depend on the total capacity C or on the distribution $\{\varepsilon_i\}$. Therefore even if the system is out of equilibrium, the *ratios between integer counts* are exact: $\Omega_\Lambda/\Omega_m = 42/19$ is an identity of the partition, not a thermodynamic result. Only the individual fractions $\Omega_X = |X|/C_{\text{total}}$ require the equilibrium assumption.

Comparison with Butcher's quantum-bias cosmology. Butcher (2024) proposes a distinct route from information-capacity bounds to cosmic acceleration. In that framework, a quantum-mechanical correction to the Friedmann equation arises from the holographic (Bekenstein–Bousso) entropy bound: the finite information capacity of a causal patch introduces an effective negative-pressure term that modifies late-time dynamics nonlocally. The resulting modified Friedmann equation predicts accelerated expansion without a cosmological constant term, instead attributing the acceleration to an information-capacity back-reaction.

The two frameworks share a common premise — finite information capacity at causal horizons drives cosmic acceleration — but differ in mechanism, scope, and predicted late-time behavior:

(a) **Mechanism.** In the APF, acceleration arises from the *structural partition* of 61 discrete capacity types into vacuum and matter sectors (L_{equip}). The cosmological constant $\Lambda \propto 42/61$ is an exact rational number determined by type counting. In Butcher's framework, acceleration arises from a *dynamical back-reaction* of the holographic bound on the Friedmann equation, producing an effective dark energy that is not a constant but a functional of the horizon entropy.

(b) **Predicted equation of state.** The APF predicts $w = -1$ exactly (the vacuum sector's capacity fraction is a topological invariant, independent of the expansion history; $L_{\text{sat,part}}$). Butcher's framework generically predicts $w \neq -1$ at intermediate times, with $w \to -1$ only asymptotically. Current data (Planck + DESI BAO) are consistent with $w = -1$ at $< 2\sigma$; future surveys (Euclid, LSST) will tighten the constraint to sub-percent precision on $|1+w|$ and could distinguish the two predictions.

(c) **Scope.** The APF's capacity-partition argument derives the density fractions from the gauge group and matter content (themselves derived from A1); the cosmological result is a downstream consequence of particle physics, not an independent postulate. Butcher's framework is self-contained within cosmology and does not derive the particle content. The two are therefore complementary rather than contradictory: they address different questions ("why these fractions?" vs. "how does acceleration arise dynamically?") and could in principle both be correct if the APF's structural fractions are realized via an information-capacity mechanism at the dynamical level.

(d) **Nonlocality.** Butcher's modification is explicitly nonlocal (it depends on the global horizon entropy, not on local field equations). The APF's L_{equip} is also horizon-global (it applies at Bekenstein saturation, a property of the causal patch as a whole), but the underlying field equations ($T_{9,\text{grav}}$) are local (Einstein equations from Lovelock). The APF

thus predicts standard local GR dynamics with an effective $\Lambda = 42C/(61\cdot 3)$ at the horizon scale; Butcher's framework modifies the dynamics themselves.

13 Saturation independence ($L_{\text{sat,part}}$)

Theorem 13.1 ($L_{\text{sat,part}}$: Saturation-Independent Partition)**.** *The partition fractions $\Omega_\Lambda = 42/61$ and $\Omega_m = 19/61$ are determined by discrete type-classification predicates (Q1, Q2) applied to the anomaly-free field content. They are independent of the total capacity C and the saturation level s.*

Proof. The predicates Q1 and Q2 are functions of type labels (representation quantum numbers), not of the total capacity $C(\Gamma)$. By L_{equip} (Theorem 12.1), the density fraction of sector X is $\Omega_X = |X|/C_{\text{total}}$, where $|X|$ is an integer determined by the predicate assignment. The total capacity C cancels in the ratio. Therefore Ω_X is independent of C and of the saturation level $s := C_{\text{committed}}/C$.

The partition is structurally stable against perturbations: the predicates Q1 and Q2 depend on the Wedderburn block structure and field selection, both of which are discrete invariants preserved for cross-support perturbations $\delta < \delta_{\text{crit}}$ (Paper 1 Supplement, Proposition robustness). □

Below the critical saturation $s_{\text{crit}} = 1/d_{\text{eff}} = 1/102$, the full matching does not exist (anomaly cancellation requires all 61 types simultaneously). The pre-matching state is pure de Sitter vacuum. The partition is undefined below s_{crit} but irrelevant, because the vacuum has $w = -1$ regardless.

Code reference: `check_L_saturation_partition` in `cosmology.py`.

14 Three structural predictions

The capacity counting, MECE partition, and cost-function uniqueness established above have three observational consequences that bear on open problems in the Standard Model. Each is a descriptive reading of a structure already proved in this supplement: the admissibility conditions, together with the derived capacity partition, are compatible with exactly one value of the quantity in question; no new machinery is introduced, and no mechanism ranges over candidates to reject the non-admissible ones. The three consequences read off are: (i) the QCD vacuum angle is admissible only at $\theta_{\text{QCD}} = 0$ (conditional on Premise (I) below; §14.1); (ii) the Higgs effective potential has a unique admissible well, so the electroweak vacuum is absolutely stable (§14.2); (iii) the dark-energy fraction $\Omega_\Lambda = 42/61$ is the ratio of topological invariants of the MECE partition (§11, §**??**).

14.1 Strong CP solution: $\theta_{\text{QCD}} = 0$

The QCD Lagrangian admits a topological term $\mathcal{L}_\theta = (\theta/32\pi^2)\, G_{\mu\nu}\tilde{G}^{\mu\nu}$ that is a total divergence. The parameter $\theta \in [0, 2\pi)$ introduces no new propagating degrees of freedom: C_{total} is unchanged at 61 for any value of θ.

Theorem 14.1 ($T_{\theta_{\text{QCD}}}$: Strong CP Solution [P-heuristic])**.** *Under the admissibility framework together with Premise (I) below, the unique cost-minimising value of the QCD vacuum angle is $\theta_{\text{QCD}} = 0$.*

Premise (I) (instanton-sector closure). Non-perturbative tunnelling sectors indexed by topological winding number contribute no admissible capacity beyond the 61 channels counted by L_{count}; equivalently, instanton configurations are treated as enforcement overhead attached

to the existing channels rather than as independent capacity types. A first-principles derivation of this premise from A1 — requiring a theory of admissible capacity cost for topologically nontrivial gauge sectors — is deferred. The present argument is therefore classified [P-heuristic]: it is rigorous given Premise (I), but the premise itself is not yet reduced to the axiom.

Proof of Theorem 14.1, conditional on Premise (I). Four steps; the argument is descriptive rather than selective (no mechanism ranges over θ-values; the structure $\min_\theta$ cost labels the admissible configuration).

Step 1 (θ adds no capacity, given Premise (I)). $\mathcal{L}_\theta$ is a total divergence and introduces no new propagating degrees of freedom. Together with Premise (I), the capacity count is unaffected: $C(\theta \neq 0) = C(\theta = 0) = 61$.

Step 2 (Enforcement cost from L_{ε^*}). A definite value $\theta = \theta_0 \neq 0$ distinguishes the vacuum from all $\theta \neq \theta_0$ configurations. By L_{ε^*} (§C), every enforceable distinction costs at least $\varepsilon^* > 0$. Therefore the enforcement-cost functional satisfies $\mathrm{cost}(\theta = \theta_0 \neq 0) \geq \varepsilon^*$.

Step 3 ($\theta = 0$ is the unique cost-free value). At $\theta = 0$ the strong sector is CP-invariant; the value is not an independent parameter but a consequence of the symmetry. Maintaining a symmetry consequence requires no enforcement record, so $\mathrm{cost}(\theta = 0) = 0$. This is the unique zero of the cost functional on $[0, 2\pi)$.

Step 4 (admissibility labels $\theta = 0$). Both configurations have $C = 61$. The net enforcement surplus reads

$$\mathrm{surplus}(\theta = 0) = 0 - 0 = 0 \ > \ 0 - \varepsilon^* = \mathrm{surplus}(\theta \neq 0).$$

The admissible configuration is the one at which the surplus attains its maximum, equivalently the $\arg\min_\theta \mathrm{cost}(\theta)$. That argmin is $\theta = 0$. No search over θ-values is performed; the framework is descriptive, and Step 4 merely reads off the referent of the admissibility condition. □

Remark 14.2 (Why the CKM phase survives)**.** The CKM phase δ_{CP} is *not* eliminated by this argument. Unlike θ, the CKM phase enables $N_{\mathrm{gen}}! = 6$ distinguishable history sectors (the Jarlskog invariant), which contribute to C_{total}. The enforcement cost of maintaining δ_{CP} is offset by the capacity gain from these additional sectors. The net surplus is non-negative, so δ_{CP} is admissible. Full accounting is deferred to Paper 4B.

Code reference: `check_T_theta_QCD, check_L_strong_CP_synthesis` in `gauge.py`.

14.2 Vacuum absolute stability

In the Standard Model, the Higgs effective potential extrapolated to high energies via renormalisation group running develops a second minimum deeper than the electroweak vacuum for $m_H \approx 125$ GeV and $m_{\mathrm{top}} \approx 173$ GeV, rendering the vacuum metastable. The admissibility framework excludes this possibility.

Theorem 14.3 ($T_{\mathrm{vac,stab}}$: Vacuum Absolute Stability)**.** *The electroweak vacuum is absolutely stable. The enforcement potential has a unique global minimum, and no deeper vacuum exists.*

Proof. Four steps.

Step 1 (Unique enforcement well). The enforcement potential $V(\Phi)$ derived from L_{ε^*} (linear cost floor), T_M (monogamy binding), and A1 (capacity saturation) has the form:

$$V(\Phi) = \varepsilon^* \Phi - \tfrac{\eta}{2\varepsilon^*}\, \Phi^2 + \frac{\varepsilon^*\, \Phi^2}{2(C - \Phi)}.$$

This potential has: $V(0) = 0$ (empty vacuum, unstable); a barrier at $\Phi/C \sim 0.06$; a *unique* binding well at $\Phi/C \sim 0.73$ with $V < 0$; and a divergence $V \to +\infty$ as $\Phi \to C$ (capacity saturation). The uniqueness of the minimum is verified by direct computation (T_{particle}, gauge.py).

Step 2 (No runaway). A1 guarantees $\Phi < C$ for all admissible states. The potential diverges at $\Phi = C$, so no tunnelling to $\Phi > C$ is possible. The well at $\Phi/C \sim 0.73$ is the global minimum.

Step 3 (High-energy behaviour). The SM extrapolation that produces metastability assumes volume-scaling degrees of freedom at high energies. T_{Bek} (Bekenstein bound, Paper 1) imposes area-law regulation, which prevents the effective potential from developing a second minimum at high field values ($L_{\text{naturalness}}$, gauge.py).

Step 4 (Uniqueness from equilibrium measure). M_Ω (derived in Paper 6) proves the equilibrium measure at saturation is unique (uniform). A second vacuum would require a second equilibrium, which M_Ω excludes. The electroweak vacuum is the unique stationary point. □

Code reference: `check_T_vacuum_stability` in `gauge.py`.

14.3 Proton absolute stability

In the Standard Model, baryon number B is an accidental symmetry that could in principle be violated by non-perturbative processes (sphalerons) or by higher-dimensional operators in grand unified extensions. The admissibility framework derives exact B conservation at Bekenstein saturation.

Theorem 14.4 (T_{proton}: Proton Absolute Stability)**.** *At full Bekenstein saturation, baryon number is an exact conservation law. The proton is absolutely stable.*

Proof. Three steps.

Step 1 (Partition is exact at saturation). P_{exhaust} (Theorem 11.2) proves that the three-sector partition

$$N_b + N_d + N_v = 3 + 16 + 42 = 61 = C_{\text{total}}$$

is mutually exclusive and collectively exhaustive at full Bekenstein saturation. The predicates Q1 (gauge addressability) and Q2 (confinement) uniquely assign each capacity type to exactly one sector. The baryonic sector has $N_b = 3$ types, and this integer is a conserved quantity at saturation: the partition is sharp (no type is partially baryonic or ambiguously assigned).

Step 2 (Saturation is irreversible). L_{irr} (§C) proves that the transition to full saturation is irreversible. Once all 61 capacity types are committed and the ledger reaches saturation, the universe cannot return to the pre-saturation regime where the partition was not enforced. Records are locked.

Step 3 (No admissible rerouting). Any baryon-number-violating process would require a capacity type to exit the baryonic sector (violating Q1 or Q2) and enter another sector. Both operations violate the partition predicates, which are mathematical identities at saturation (Step 1). No admissible move changes N_b. □

Remark 14.5 (No B-violating operators at any dimension)**.** T_{gauge} derives $\text{SU}(3)\times\text{SU}(2)\times\text{U}(1)$ as the *complete* gauge group, not embedded in a larger group. No leptoquark or X/Y boson exists; no dimension-6 B-violating operator is gauge-invariant under the derived group. Even if such operators were postulated, Step 3 above forbids their action at saturation. The prediction is therefore the strongest possible: $\tau_p = \infty$, not merely $\tau_p > 10^{34}$ years.

Code reference: `check_T_proton, check_L_B_operator_exclusion` in `gauge.py`.

15 Uncertainty analysis and sensitivity

The predictions $\Omega_\Lambda = 42/61$ and $\Omega_m = 19/61$ are rational numbers with zero free parameters. There is no propagated parametric uncertainty in the usual sense. The relevant question is: *what assumptions, if changed, would shift the prediction?*

15.1 Sensitivity to the capacity count

The only quantity entering the prediction is C_{total}, the total number of capacity types. The vacuum count (42) is determined by L_{vac} and is insensitive to the matter content. The matter count (19) is $C_{\text{total}} - 42$. Therefore the prediction shifts if and only if C_{total} changes.

Scenario	C_{total}	Ω_Λ	**Tension with Planck**	**What changes**
SM (derived)	61	$42/61 = 0.6885$	0.07% ($< 0.1\sigma$)	—
$+3\,\nu_R$	64	$42/64 = 0.656$	5.8σ	Add light ν_R
+ polarizations	73	$42/73 = 0.575$	$> 20\sigma$	Count helicities
− Higgs	57	$39/57 = 0.684$	0.9σ	Remove Higgs DOF
−1 generation	46	$27/46 = 0.587$	$> 18\sigma$	2 generations

Every alternative C_{total} moves the prediction outside the 1σ band except the Higgs-removal scenario, which is excluded by the derived scalar sector (T_{Higgs}, Paper 2, §5.5).

15.2 Comparison with observation

Planck 2018 measures $\Omega_\Lambda = 0.6889 \pm 0.0056$ [9]. The prediction $42/61 = 0.68852\ldots$ is within 0.07% of the central value, well inside the 1σ band. The agreement extends to all five derived fractions:

Observable	**Predicted**	**Observed**	**Deviation**
Ω_Λ	$42/61 = 0.6885$	0.6889 ± 0.0056	0.06%
Ω_m	$19/61 = 0.3115$	0.3111 ± 0.0056	0.13%
Ω_{DM}	$16/61 = 0.2623$	0.2607 ± 0.0056	0.6%
Ω_b	$3/61 = 0.0492$	0.0490 ± 0.0003	0.4%
$f_b = \Omega_b/\Omega_m$	$3/19 = 0.1579$	0.1571 ± 0.0012	0.5%

Five predictions, zero free parameters, all within 0.6% of observation. The numerators are integers; the denominators are 61. There is no fitting: the fractions are determined by the derived field content and the MECE partition.

A Comparison with reconstruction and "SM from principles" programs

A.1 Operational and categorical reconstructions

Several programs reconstruct quantum theory from operational or information-theoretic axioms. We summarize each and note where the APF diverges.

Hardy (2001) [11]. Five "reasonable axioms" (probabilities, simplicity, subspaces, composite systems, continuous reversibility) derive finite-dimensional quantum theory over $\mathbb{C}$. The axioms are stated at the kinematics level: they constrain the state space and measurement structure but do not determine gauge content, matter spectrum, or dynamics. APF shares the "finite capacity" starting point (Hardy's axiom 1 is a form of finite dimensionality) but pushes through to gauge group and matter content via non-closure, which Hardy does not address.

Chiribella–D'Ariano–Perinotti (2011) [12]. Six axioms (causality, perfect distinguishability, ideal compression, local distinguishability, pure conditioning, purification) derive QT. The purification axiom is the key differentiator from classical probability. The output is a general probabilistic theory isomorphic to finite-dimensional quantum mechanics. Like Hardy, CDP stops at kinematics. The APF's constitutive principle SP (substrate faithfulness) plays a role analogous to CDP's local distinguishability, and K3 (forced additivity) parallels their ideal compression, but the APF does not assume purification — instead, it derives the Born rule from the enforcement algebra via Busch's generalized Gleason theorem.

Höhn (2017) [13]. Reconstructs qubit quantum theory from "questions" (binary tests) with finite information capacity. The closest cousin to APF's starting point: finite capacity $\to$ quantum structure. Höhn's framework is carefully delimited to qubits and does not attempt gauge content. APF can be viewed as an extension of Höhn's program that asks: what happens when you push the "finite information" idea beyond qubit reconstruction to the full field content?

Effectus theory (Westerbaan, van de Wetering). Categorical framework using "effectuses" to axiomatize quantum-like theories. Sequential product axioms force operator-algebraic structure. The APF's enforcement algebra (§2) can be compared to the sequential product in effectus theory: both derive non-commutativity from the structure of sequential measurements. The effectus approach is more general (it classifies all effectuses, not just the quantum one); the APF is more specific (it selects the quantum case via A1 and derives gauge content).

Tull (2019); Spekkens (2007). Tull derives operator algebras from categorical CPM (completely positive maps) structure. Spekkens constructs a toy model from epistemic restrictions. Both illuminate the boundary between classical and quantum but do not derive field content or gauge structure.

A.2 SM-from-principles programs

A separate tradition attempts to derive the Standard Model within an assumed quantum field theory framework.

Anomaly-based selection (Minahan–Ramond–Warner, 1990). Given $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ and the requirement of anomaly cancellation, the SM fermion content is strongly constrained but not unique without additional assumptions (e.g., minimality, family structure). The APF's T_{field} performs a similar scan but derives the gauge group itself (via Theorem R + $L_{\text{gauge,unique}}$) rather than assuming it.

Grand unification (Georgi–Glashow, 1974). Embeds $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ into $\mathrm{SU}(5)$. Explains charge quantization and predicts proton decay. The APF derives the product structure as capacity-minimal (Proposition 6.7: SU(5) costs $2\times$) and obtains charge quantization from anomaly cancellation (Theorem 8.1) without grand unification.

Connes NCG (1996). The noncommutative geometry program derives the SM Lagrangian from a spectral triple. The input is a specific finite-dimensional algebra (chosen to reproduce the SM). The APF derives the algebra from A1 (T_{alg}), rather than postulating it; the two programs converge on the same algebraic structure but arrive from opposite directions.

Doplicher–Roberts reconstruction and Pinzari's braided extensions. The classical DR theorem (1989) reconstructs a compact group G from its representation category (a symmetric C^*-tensor category with conjugates). In the APF, T3 (Paper 1 Supplement) uses DR to recover the gauge group from the enforcement algebra's representation category; the gauge identification theorem (Theorem 2.5) then identifies which automorphisms constitute the gauge group.

Pinzari's program (2017–) extends DR to *braided* tensor categories and weak Hopf C^*-algebras. In this setting, abelian factors can emerge from the *center* of the fusion category: central objects (transparent to braiding) generate grading tori that act as abelian gauge factors. This is structurally parallel to the APF's mechanism for $\mathrm{U}(1)_Y$: the relative-phase torus $T_{\text{rel}} = Z(\mathrm{U}(\mathcal{A}))/\mathrm{U}(1)_{\text{diag}}$ (Remark 2.6) plays the role of Pinzari's grading torus, and the abelian factor arises from central (= transparent) elements rather than from inner automorphisms.

The key structural parallel is: *non-abelian gauge from inner automorphisms, abelian gauge from the center.* In DR/Pinzari, this is "gauge from the Tannakian group, grading from the Müger center." In APF, it is "gauge from $\mathrm{Inn}(\mathcal{A})$, grading from T_{rel} pruned by CM." The APF adds a *selection mechanism* (capacity minimality, Proposition 2.12) that is absent in the DR/Pinzari framework: the latter reconstructs whatever symmetry the category has, while the APF derives the *minimal* symmetry compatible with enforcement completeness.

Finite tensor categories and symmetric tensor categories. Recent work on finite tensor categories (Etingof–Gelaki–Nikshych–Ostrik) and symmetric tensor categories (Deligne's theorem and extensions by Coulembier, Entova-Aizenbud, Ostrik) shows that structural constraints on representation categories can have sharp classification consequences. Deligne's theorem, for example, classifies all symmetric tensor categories of moderate growth as representation categories of supergroups — a result that constrains the possible internal symmetry structures of any theory whose observables form a symmetric tensor category.

The APF is not a tensor-categorical framework: its primary objects are enforcement algebras and cost functionals, not abstract tensor categories. However, the representation category of the enforcement algebra $\mathcal{A} \cong \bigoplus_k M_{n_k}(\mathbb{C})$ *is* a finite semisimple tensor category (the category of finite-dimensional $\mathcal{A}$-modules), and its symmetry is classified by the theorems above. The APF's contribution relative to this literature is not a new classification of tensor categories but a *physical derivation* of which tensor category is realized: the enforcement postulate A1 selects $\mathcal{A}$ and hence its representation category, rather than postulating the category directly. The classification theorems then apply as *imported mathematics* (same epistemic status as Wedderburn–Artin or Skolem–Noether in the present supplement).

A.3 Locally covariant QFT and the General Boundary Formulation

Two programs are particularly relevant to the APF because they share structural elements (local algebras, causality, boundaries) while differing fundamentally on infinite-dimensionality.

Locally covariant QFT (Brunetti–Fredenhagen–Verch, 2003). LCQFT axiomatizes QFT as a functor from the category of globally hyperbolic Lorentzian manifolds (with isometric embeddings as morphisms) to the category of $*$-algebras (with injective $*$-homomorphisms). The key axioms are:

- *Locality*: spacelike-separated regions have commuting algebras.
- *Covariance*: the assignment $\mathcal{O} \mapsto \mathcal{A}(\mathcal{O})$ is functorial under isometric embeddings.
- *Time-slice*: if $\mathcal{O}_1 \hookrightarrow \mathcal{O}_2$ contains a Cauchy surface of $\mathcal{O}_2$, then $\mathcal{A}(\mathcal{O}_1) \cong \mathcal{A}(\mathcal{O}_2)$.

The LCQFT framework works with *infinite-dimensional* algebras (typically type III factors for local regions) and *presupposes* a background Lorentzian manifold.

The APF differs from LCQFT on three structural points:

(a) *Finite vs. infinite dimensions.* LCQFT's local algebras are infinite-dimensional by construction (they are obtained via the GNS representation of a state on a C^*-algebra of smeared fields). The APF's local algebras are finite-dimensional (A1: $\dim(\mathcal{H}_\omega) < \infty$). The reconciliation is via the controlled limit (Proposition 2.23): the APF's type I algebras approximate LCQFT's type III algebras in the $C/\varepsilon^* \to \infty$ limit, with the structural predictions (gauge group, matter content) stable under the limit (Theorem 2.25).

(b) *Spacetime derived vs. presupposed.* LCQFT takes a Lorentzian manifold as input; the APF derives it (§5: $L_{\text{irr}} \to \prec \to M \to g_{\mu\nu}$). The APF can therefore be viewed as providing a *foundation* for LCQFT: it derives the background manifold that LCQFT assumes, and the local algebras that LCQFT assigns to regions emerge as the enforcement algebras at interfaces.

(c) *Causality.* Both frameworks implement causality via commutativity of spacelike-separated algebras. In LCQFT, this is an axiom; in the APF, it is derived (Lemma 2.2). The LCQFT time-slice axiom has an APF analogue: if $\Gamma_1 \subseteq \Gamma_2$ and Γ_1 "contains a Cauchy surface" (i.e., the enforcement structure at Γ_1 determines the enforcement structure at Γ_2), then $\mathcal{A}(\Gamma_1) \cong \mathcal{A}(\Gamma_2)$. This is a consequence of L_{loc} and the deterministic evolution of the enforcement structure.

Oeckl's Positive Formalism / General Boundary Formulation (2003–2016). Oeckl's program assigns quantum amplitudes to *boundaries* of spacetime regions, rather than to states on Cauchy surfaces. The formalism uses:

- *Boundary Hilbert spaces*: each boundary Σ (not necessarily a Cauchy surface) carries a Hilbert space $\mathcal{H}_\Sigma$.
- *Amplitude maps*: each spacetime region M bounded by Σ defines a linear map $\rho_M : \mathcal{H}_\Sigma \to \mathbb{C}$ (or more generally a positive map).
- *Composition*: gluing regions along shared boundaries composes amplitude maps.

The General Boundary Formulation (GBF) generalizes this to non-globally-hyperbolic spacetimes and is compatible with quantum gravity scenarios where the background spacetime is not fixed.

The APF's interface structure has a striking structural parallel to Oeckl's boundary formalism:

(a) *Interfaces as boundaries.* An APF interface Γ is the boundary between two enforcement regions. The enforcement algebra $\mathcal{A}(\Gamma)$ plays the role of Oeckl's boundary Hilbert space $\mathcal{H}_\Sigma$ (in the finite-dimensional case, the algebra and its GNS Hilbert space are equivalent objects).

(b) *Cost kernel as amplitude.* The enforcement cost $\varepsilon_{\text{ext}}(v) = B(v, v)$ (T_{7B}) is a positive functional on the interface sector space, analogous to Oeckl's positive amplitude ρ_M. The positivity of B corresponds to Oeckl's positivity condition, which ensures physical probabilities are non-negative.

(c) *Composition via pool coupling.* When two interfaces share a pool Π (L_Δ), the enforcement structure composes: the combined cost is superadditive ($\varepsilon(v_1 + v_2) \geq \varepsilon(v_1) + \varepsilon(v_2)$). This parallels Oeckl's composition of amplitude maps via gluing.

The key differences:

- Oeckl works with infinite-dimensional Hilbert spaces and does not impose a capacity bound. The APF's A1 restricts to finite dimensions.
- Oeckl's formalism is *kinematic* (it assigns amplitudes but does not derive gauge content or matter spectrum). The APF pushes through to field content via the enforcement algebra's Wedderburn decomposition.
- The GBF is compatible with indefinite causal structure (useful for quantum gravity); the

APF derives a definite causal structure from L_{irr} (Proposition 5.4).

Masanes–Müller (2011). Derives finite-dimensional quantum theory from four postulates: continuous reversibility, tomographic locality (joint states determined by local measurements), existence of an information unit, and the subspace axiom. The output is quantum theory over $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$ (with $\mathbb{C}$ selected by tomographic locality + bit symmetry). The APF shares the finite-dimensionality starting point and the field selection ($\mathbb{F} = \mathbb{C}$ from T2b), but arrives at $\mathbb{C}$ via a different mechanism (Skolem–Noether applied to the enforcement algebra, not tomographic locality). Like Hardy and CDP, Masanes–Müller stops at kinematics and does not address gauge content.

A.4 Systematic comparison: infinite-dimensionality and causality

The reviewer asks for a systematic contrast on two axes — infinite-dimensionality and causality — across the relevant programs. The following table provides this.

Program	**Local algebra type**	**Causality treatment**	**APF relationship**
AQFT (Haag–Kastler)	Type III_1 factors (Reeh–Schlieder)	Axiom: $[\mathcal{A}(\mathcal{O}_1), \mathcal{A}(\mathcal{O}_2)] = 0$ for spacelike $\mathcal{O}_1, \mathcal{O}_2$	APF recovers type III in $n \to \infty$ limit; derives commutativity from L_{loc}
LCQFT (Brunetti–Fredenhagen–Verch)	Infinite-dim. $*$-algebras; functorial on Lorentzian manifolds	Presupposes globally hyperbolic manifold; time-slice axiom	APF derives the manifold; provides a finite-dim. foundation
Oeckl (GBF)	Infinite-dim. boundary Hilbert spaces	Compatible with indefinite causal structure	Structural parallel (interfaces $\sim$ boundaries); APF adds A1 finiteness and derives causal order
Hardy (2001)	Finite-dim. (axiom 1)	Not addressed (kinematics only)	Shared starting point; APF extends to gauge content
CDP (2011)	Finite-dim. (output)	Causality axiom (no signaling)	APF derives no-signaling from L_{loc}; pushes beyond kinematics
Masanes–Müller	Finite-dim. (postulate)	Not directly addressed	Shared $\mathbb{F} = \mathbb{C}$ result; different mechanism
Connes NCG	Finite-dim. internal algebra; ∞-dim. external	Lorentzian structure from spectral triple	APF derives the algebra that NCG postulates
DR reconstruction	Infinite-dim. (type III in AQFT context)	Inherited from AQFT axioms	APF uses DR at finite n; structural parallel for abelian factors
APF	Type I_n ($M_n(\mathbb{C})$); type III recovered as $n \to \infty$	Derived from L_{irr} (Prop. 5.4); HK-M/Malament for signature	—

Three patterns emerge from the comparison:

(i) Finite vs. infinite dimensions is a foundational choice, not a technical detail. Programs that start with infinite dimensions (AQFT, LCQFT, Oeckl, DR) must contend with type III algebras, ultraviolet divergences, and the absence of a faithful trace. Programs that start with finite dimensions (Hardy, CDP, Masanes–Müller, APF) avoid these issues but must explain how the infinite-dimensional continuum is recovered. The APF's controlled limit (Proposition 2.23)

and structural stability theorem (Theorem 2.25) provide this explanation; the kinematic reconstruction programs do not address the question (they stop at finite-dimensional QT and do not discuss the continuum limit).

(ii) Causality is either axiomatized or derived, never both. AQFT and CDP axiomatize causality (as commutativity of spacelike algebras or no-signaling); the APF derives it from L_{irr} (irreversibility $\to$ causal order) and L_{loc} (independent interfaces $\to$ commutativity). Oeckl's GBF is compatible with indefinite causality, making it the most flexible — but at the cost of not determining the causal structure. The APF's derivation of causality from a non-causal starting point (finite enforcement capacity has no intrinsic temporal or causal content) is the closest parallel to the causal-set program (Sorkin, 1991), though the APF derives a smooth manifold rather than working with a discrete causal set.

(iii) Only the APF and Connes NCG derive gauge content. All other programs in the table stop at the kinematics of quantum theory (state space, measurement structure, composition rules). The passage from kinematics to dynamics and gauge content requires additional structure: in NCG, the spectral triple; in the APF, the enforcement algebra's Wedderburn decomposition + non-closure + anomaly cancellation. The APF's advantage over NCG is that it *derives* the algebra, whereas NCG postulates it; NCG's advantage is that it produces the full SM Lagrangian (including Yukawa couplings and the Higgs potential), which the APF defers to Paper 4A.

A.5 Assumption inventory comparison

Program	**Inputs**	**Derives**	**Stops at**
Hardy (2001)	5 axioms	QT kinematics	State space
CDP (2011)	6 axioms	QT kinematics	State space
Masanes–Müller (2011)	4 postulates	QT over $\mathbb{C}$	State space
Höhn (2017)	Finite info	Qubit QT	Qubit
LCQFT (BFV 2003)	Functorial axioms	QFT on curved spacetime	Assumes manifold
Oeckl (GBF 2003)	Boundary axioms	QT amplitudes	Kinematics
Connes NCG	Spectral triple	SM Lagrangian	Assumes algebra
APF	A1+MD+BW+SP+K3	QT + gauge + matter	SM content

The APF's named assumption count (the four PLEC components A1, MD, A2, BW; plus constitutive semantics SP, K3, FD1–FD6) is numerically comparable to Hardy's 5 or CDP's 6, though a direct comparison of "assumption weight" is imprecise: the APF's constitutive definitions (FD1–FD6) carry more internal structure than a single axiom. The significant difference is scope: the APF's downstream derivation chain extends to gauge content, matter spectrum, and cosmological fractions, which the other programs do not attempt.

B Code–theorem concordance

Every theorem in this supplement is verified by a named check function in the codebase. The table below provides the mapping.

Theorem	**Supplement §**	**Function**	**File**
L_{Cauchy}	§1	`check_L_Cauchy_uniqueness`	`L_Cauchy_uniqueness.py`
Gauge ident.	§2.2	`check_T3`	`core.py`
T_{7B}	§3.1	`check_T7B`	`gravity.py`
$L_{\text{irr,uniform}}$	§3.2	`check_L_irr_uniform`	`core.py`
$B1'$	§3.3	`check_B1_prime`	`gauge.py`
$L_{\text{gauge,unique}}$	§6	`check_L_gauge_template_uniqueness`	`gauge.py`
T_{field}	§7	`check_T_field`; `fermion_scan_standalone.py` [20]	`gauge.py`
$L_{\text{hypercharge}}$	§8	`check_T_field` (hypercharges)	`gauge.py`
L_{count}	§10	`check_L_count`	`gauge.py`
P_{exhaust}	§11	`check_P_exhaust`	`core.py`
L_{equip}	§12	`check_L_equip`	`cosmology.py`
$L_{\text{sat,part}}$	§13	`check_L_saturation_partition`	`cosmology.py`
Δ_{sig}	§5.3	`check_Delta_signature`	`spacetime.py`
T_8 (d=4)	§5.5	`check_T8`	`spacetime.py`
$T_{9,\text{grav}}$	§5.6	`check_T9_grav`	`gravity.py`
$T_{\theta_{\text{QCD}}}$	§14.1	`check_T_theta_QCD`	`gauge.py`
$T_{\text{vac,stab}}$	§14.2	`check_T_vacuum_stability`	`gauge.py`
T_{proton}	§14.3	`check_T_proton`	`gauge.py`

The full codebase is available at `github.com/Ethan-Brooke/APF-Paper-2-The-Structure-of-Admissible-Pl`

Reproducibility of the fermion scan. The function `check_T_field` in `gauge.py` performs the complete 1,680-template scan described in §7. It enumerates all templates, applies the seven filters in sequence, records exact survivor counts at each stage (matching the waterfall table), and verifies that the unique minimum-DOF survivor is the SM at 45 Weyl fermions. The five closed-form exclusion proofs (P1–P5) are verified as separate assertions within the same function. Hypercharge derivation and anomaly cancellation are checked algebraically using exact rational arithmetic (`fractions.Fraction`). Running `python -m pytest gauge.py` reproduces all results with no external dependencies beyond the Python standard library.

Standalone verification script. The file `fermion_scan_standalone.py` [20], included in the GitHub repository and archived on Zenodo (`https://doi.org/10.5281/zenodo.19154197`), is a self-contained Python script (∼550 lines, zero dependencies beyond the standard library) that reproduces the complete scan from scratch. An interactive Colab walkthrough with waterfall visualisation is available at `https://colab.research.google.com/drive/1otO1Qwp_Lr3OsK6ykkeMRATanecPkKg6`. Running `python3 fermion_scan_standalone.py` performs the following:

(1) Enumerates all 1,680 distinct templates using `combinations_with_replacement`.
(2) Applies all 7 filters with exact rational arithmetic, including spectator-field reduction for $A(R) = 0$ representations.
(3) Prints the waterfall table with exact counts at each stage.
(4) Lists all 8 post-F6 survivors explicitly (matching the table in §7.4).
(5) Lists all 4 post-CPT survivors.
(6) Verifies all 5 exclusion proofs (P1–P5) with explicit arithmetic.
(7) Derives both hypercharge solutions ($z = 1 \pm N_c$), verifies all 4 anomaly conditions exactly for each, and confirms their physical equivalence under $u \leftrightarrow d$ relabelling.
(8) Runs 4 built-in red-team challenges: (RT1) combinatorial cross-check of the enumeration count, (RT2) explicit spectator-field anomaly verification, (RT3) dead-code reachability audit, (RT4) P4 bound tightness with a constructed witness template.
(9) Asserts 12 final verification conditions.

The script is designed so that any reader can verify the entire fermion-content derivation on a standard laptop in under one second, with no knowledge of the APF codebase required.

The Lie-algebra classification (§6) is verified by `check_L_gauge_template_uniqueness`, which tests all 17 compact simple algebras against R1–R3 and records the exclusion reason for each. The Zenodo archive (DOI: 10.5281/zenodo.18867119) provides a frozen snapshot of the codebase corresponding to this supplement.

C Self-contained proofs of imported results

This appendix provides complete proofs of every result imported from the Paper 1 Supplement that is used in this document. With this appendix, no external document is needed to verify any theorem in this supplement.

Input assumptions. All proofs below use only: A1 (finite enforcement capacity $0 < C(\Gamma) < \infty$), MD (uniform per-test cost floor $\varepsilon \geq \mu^* > 0$), K3 (additivity on disjoint supports), SP (no null directions in the physical quotient), and the formal definitions FD1–FD6. No result from Papers 2–7 is used.

C.1 L_{ε^*}: Uniform cost floor

Lemma C.1 (L_{ε^*})**.** *There exists $\varepsilon^* > 0$ such that $\varepsilon(d) \geq \varepsilon^*$ for all $d \in \mathcal{D}$.*

Proof. By MD: $\varepsilon(d) \geq \mu^* \cdot n(d) \geq \mu^* > 0$ for every d (since $n(d) \geq 1$). Set $\varepsilon^* := \mu^*$. □

C.2 T_{sep}: Sector decomposition

Theorem C.2 (T_{sep})**.** *For distinctions $d_1, d_2 \in \mathcal{D}(\Gamma)$ with anchor sets $M_{d_1}, M_{d_2} \subseteq S_\Gamma$:*

$$M_{d_1} \cap M_{d_2} = \{0\} \quad \Longleftrightarrow \quad \varepsilon(\{d_1, d_2\}) = \varepsilon(d_1) + \varepsilon(d_2).$$

For any maximal antichain $\mathcal{B}$ of pairwise independently enforceable distinctions, the substrate decomposes as $S_\Gamma = \bigoplus_{d \in \mathcal{B}} M_d \oplus \Pi$, where Π is the residual pool.

Proof. ($\Rightarrow$) If $M_{d_1} \cap M_{d_2} = \{0\}$, perturbations threatening d_1 and d_2 have disjoint supports (FD3: anchor-set locality). The joint defense decomposes into independent defenses on M_{d_1} and M_{d_2}. By K3 (additivity on disjoint supports): $\varepsilon(\{d_1, d_2\}) = \varepsilon(d_1) + \varepsilon(d_2)$.

($\Leftarrow$, contrapositive) If $v \in M_{d_1} \cap M_{d_2}$ with $v \neq 0$, then defending d_1 and d_2 separately each commits capacity to $\text{span}\{v\}$; defending jointly commits once. By SP and K2, $\kappa(\text{span}\{v\}) > 0$. Write $M_{d_i} = \text{span}\{v\} \oplus M'_{d_i}$. By K3 on the three disjoint subspaces:

$$\varepsilon(\{d_1, d_2\}) \leq \kappa(\text{span}\{v\}) + \kappa(M'_{d_1}) + \kappa(M'_{d_2}) < \varepsilon(d_1) + \varepsilon(d_2).$$

The direct-sum decomposition follows by iterating the biconditional over a maximal antichain. □

Uses: A1, K3, FD3, SP. No inner product or linear map structure.

C.3 L_Δ: Superadditivity

Theorem C.3 (L_Δ)**.** *For co-located distinctions d_1, d_2 at Γ with $M_{d_1} \cap M_{d_2} = \{0\}$ and $\Pi \neq \{0\}$: $\Delta(d_1, d_2) := \varepsilon(\{d_1, d_2\}) - \varepsilon(d_1) - \varepsilon(d_2) > 0$.*

Proof. Step 1. The joint perturbation class $\mathcal{P}(\{d_1, d_2\})$ strictly contains $\mathcal{P}(d_1) \cup \mathcal{P}(d_2)$: perturbations p with $\text{supp}(p) \subseteq \Pi$ are in $\mathcal{P}(\{d_1, d_2\})$ but not in either individual class (their anchors are disjoint from Π by T_{sep}). These perturbations change physically meaningful DOF (SP) at positive cost (K2).

Step 2. A defense confined to $M_{d_1} \oplus M_{d_2}$ acts as the identity on Π and cannot resist Π-supported perturbations.

Step 3. Any admissible joint defense must commit positive capacity $\kappa_\Pi > 0$ in Π (by K2: resisting a non-null perturbation requires positive capacity).

Step 4. M_{d_1}, M_{d_2}, Π are pairwise disjoint (T_{sep}). By K3: $\varepsilon(\{d_1, d_2\}) = \varepsilon(d_1) + \varepsilon(d_2) + \kappa_\Pi$.

Step 5. $\Delta = \kappa_\Pi > 0$. □

Uses: A1, K2, K3, T_{sep}, SP. Pre-algebraic.

C.4 L_ω: Sector-orthogonal bilinear form

Lemma C.4 (L_ω)**.** *Given T_{sep} and K3, there exists a positive-definite symmetric bilinear form $B : V \times V \to \mathbb{R}$ with $B(u_d, v_{d'}) = 0$ for $d \neq d'$ and $B(u_d, v_\Pi) = 0$ for all d. The inter-sector orthogonality is forced; the within-sector forms are free (normalization gauge).*

Proof. Construction. Choose any positive-definite form B_d on each sector M_d and B_Π on Π. Define $B(u, v) := \sum_d B_d(u_d, v_d) + B_\Pi(u_\Pi, v_\Pi)$ with zero cross-terms.

B is well-defined, symmetric, positive-definite by construction (direct sum of positive-definite forms).

K3 forces cross-term vanishing. For $u_d \in M_d$ and $v_{d'} \in M_{d'}$ with $d \neq d'$, K3 gives $\|u_d + v_{d'}\|_B^2 = \|u_d\|_B^2 + \|v_{d'}\|_B^2$ (disjoint supports). Expanding: $B(u_d, u_d) + B(v_{d'}, v_{d'}) + 2B(u_d, v_{d'}) = B(u_d, u_d) + B(v_{d'}, v_{d'})$, so $B(u_d, v_{d'}) = 0$. The same argument gives $B(u_d, v_\Pi) = 0$. □

Uses: T_{sep}, K3.

C.5 T_{adj}: Self-adjoint enforcement projections

Theorem C.5 (T_{adj})**.** *With respect to B from L_ω, each sector map E_d is a B-orthogonal projection: $E_d^\dagger = E_d$, $E_d^2 = E_d$.*

Proof. $\text{im}(E_d) = M_d$ and $\ker(E_d) = (\bigoplus_{d' \neq d} M_{d'}) \oplus \Pi$. By L_ω: $M_d \perp_B \ker(E_d)$. Therefore E_d is the B-orthogonal projection onto M_d: $B(E_d u, v) = B(u_d, v_d) = B(u, E_d v)$, so $E_d^\dagger = E_d$. □

C.6 L_{loc}: Locality (budget additivity)

Theorem C.6 (L_{loc})**.** *For independent interfaces Γ_1, Γ_2 with disjoint substrates $S_{\Gamma_1} \cap S_{\Gamma_2} = \emptyset$: $C(\Gamma_1 \cup \Gamma_2) = C(\Gamma_1) + C(\Gamma_2)$.*

Proof. Every distinction d has its anchor entirely within S_{Γ_1} or S_{Γ_2} (FD3: anchor-set locality). The cost sum partitions: $\sum_d \varepsilon(d) = \sum_{d \in \mathcal{D}(\Gamma_1)} \varepsilon(d) + \sum_{d \in \mathcal{D}(\Gamma_2)} \varepsilon(d) \leq C(\Gamma_1) + C(\Gamma_2)$. The bound is tight (each interface's budget can be independently saturated). □

Uses: A1, FD3. Does not use MD, L_{ε^*}, or any cost-functional structure.

C.7 L_{blk}: Block-diagonal defense failure

Theorem C.7 (L_{blk})**.** *Let $\Delta(d_1, d_2) > 0$. Then no idempotent with $\text{im}(P) \subseteq M_{d_1} \oplus M_{d_2}$ is an admissible joint defender.*

Proof. Such a P satisfies $P|_\Pi = 0$ (by idempotency and the direct-sum structure). It therefore has zero defense against Π-supported perturbations. But $\Delta > 0$ requires positive Π-defense (L_Δ, Step 3). Contradiction. □

C.8 T_{alg}: Noncommutativity

Theorem C.8 (T_{alg})**.** *If $\Delta(d_1, d_2) > 0$, the full enforcement algebra $\mathcal{A} := \text{Alg}_{\mathbb{R}}(\{E_d\} \cup \{E_{\{d_1,d_2\}}\})$ is noncommutative.*

Proof. Step 1. The sector projections $\{E_d\}$ commute and generate $\mathcal{A}_{\text{diag}} \cong \mathbb{R}^{|\mathcal{B}|}$.

Step 2. By L_{blk}, the minimum-cost joint defender $E_{\{d_1,d_2\}} \notin \mathcal{A}_{\text{diag}}$. Define $F_\Pi := E_{\{d_1,d_2\}} - E_{d_1} - E_{d_2} \neq 0$.

Step 3. There exists $v \in M_{d_1}$ with $E_{\{d_1,d_2\}}v = v + \pi_v$, $\pi_v \in \Pi \setminus \{0\}$ (otherwise $E_{\{d_1,d_2\}}$ is block-diagonal, contradicting Step 2). Then: $[E_{d_1}, F_\Pi]\, v = E_{d_1}(\pi_v) - F_\Pi(v) = 0 - \pi_v = -\pi_v \neq 0$, using $E_{d_1}|_\Pi = 0$ (T_{sep}).

Step 4. $[E_{d_1}, F_\Pi] \neq 0$ in $\text{End}(V)$. Any faithful $*$-homomorphism to a commutative algebra would map this to zero, contradicting faithfulness. Therefore $\mathcal{A}$ is noncommutative. □

Uses: T_{sep}, L_Δ, L_{blk}, FD5a. Pre-Hilbert.

C.9 L_{nc}: Non-closure (superadditivity at shared interfaces)

Lemma C.9 (L_{nc})**.** *At any shared interface Γ with $\Pi \neq \{0\}$ and co-located distinction sets G, R_Γ: $\varepsilon(G \cup R_\Gamma) > \varepsilon(G) + \varepsilon(R_\Gamma)$.*

Proof. This is L_Δ applied to the distinction sets G and R_Γ: the hypotheses of L_Δ (disjoint anchors, non-empty pool) are satisfied by the co-location conditions of the shared interface. □

C.10 L_{irr}: Irreversibility

Theorem C.10 (L_{irr})**.** *At any shared interface Γ with $\Pi \neq \{0\}$, there exist enforcement transitions that are locally irreversible: the capacity committed in the transition cannot be recovered by any operation confined to a single sector.*

Proof. Step 1. By L_Δ (§C.3), co-located enforcement at Γ commits a superadditive surplus $\Delta = \kappa_\Pi > 0$ in the pool sector Π.

Step 2. By L_{loc} (§C.6), enforcement at separate interfaces has independent budgets. A single-sector observer (operating within M_{d_1} alone) has access only to the capacity committed in M_{d_1}, not to the pool commitment κ_Π.

Step 3. The pool commitment κ_Π is a cross-sector correlation: it is committed in Π, which is shared between the sectors. No operation confined to M_{d_1} can access Π (by the direct-sum structure of T_{sep}: $M_{d_1} \cap \Pi = \{0\}$). Therefore the pool commitment is locally unrecoverable.

Step 4. A transition that increases κ_Π (e.g., by changing the gauge-sector distinctions at a shared interface) commits capacity that is locally unrecoverable. This is irreversibility: the forward transition costs more to reverse than the budget allows, because the reversal would require freeing pool capacity that no single-sector operation can access. □

Uses: L_Δ, L_{loc}, T_{sep}. The argument is pre-algebraic: no Hilbert space or GNS structure is invoked.

Summary. The ten results above (L_{ε^*}, T_{sep}, L_Δ, L_ω, T_{adj}, L_{loc}, L_{blk}, T_{alg}, L_{nc}, L_{irr}) constitute the complete set of results imported from the Paper 1 Supplement. Each is proved from A1, MD, K3, SP, and FD1–FD6 alone. With these proofs in hand, every theorem in this supplement can be verified end-to-end without consulting any external document.

D Fermion scan: algorithmic walkthrough

This appendix makes the fermion-content derivation (§7) verifiable without running code. It specifies the search space construction, gives the algorithm in pseudocode, and traces the critical filter steps on explicit examples.

D.1 Search space construction

A template is a multiset of $(r^{(3)}, r^{(2)})$ pairs. The search space is bounded by P1–P5 (§7.2):

Allowed SU(3) **reps** (P1: dim < 10): $\{3, \bar{3}, 6, \bar{6}, 8\}$. Dynkin indices: $T(3) = T(\bar{3}) = 1/2$, $T(6) = T(\bar{6}) = 5/2$, $T(8) = 3$. Cubic anomaly coefficients: $A(3) = 1/2$, $A(\bar{3}) = -1/2$, $A(6) = 5/2$, $A(\bar{6}) = -5/2$, $A(8) = 0$.

Allowed SU(2) **reps** (P2–P3): $\{1, 2\}$. Dynkin indices: $T(1) = 0$, $T(2) = 1/2$.

Template structure (P4–P5): Each template contains exactly one colored doublet (P4: two colored doublets require ≥ 63 DOF), 0–3 colored singlets, 0–1 lepton doublets, and 0–2 lepton singlets (P5: > 5 field types require > 45 DOF).

Enumeration count. The colored doublet ranges over 5 reps. The colored singlets form a multiset of size 0–3 drawn from 5 reps: $\binom{5+0-1}{0}+\binom{5+1-1}{1}+\binom{5+2-1}{2}+\binom{5+3-1}{3} = 1+5+15+35 = 56$ multisets. The lepton doublet is present or absent: 2 choices. The lepton singlets: 0, 1, or 2 copies of $(\mathbf{1}, \mathbf{1})$: 3 choices. Total: $5 \times 56 \times 2 \times 3 = 1{,}680$ distinct templates. The standalone verification script [20] uses `combinations_with_replacement` to enumerate these directly, and verifies the count against this closed-form formula as a built-in red-team challenge (RT1).

D.2 Algorithm in pseudocode

```
INPUT:  SU3_reps = {3, 3b, 6, 6b, 8}
        SU2_reps = {1, 2}
        N_gen = 3
        AF3_bound = 11,  AF2_bound = 22/3
        AF_coeff = 2/3

ENUMERATE all templates T:
  FOR each colored_doublet IN SU3_reps:           // 5
    FOR n_colored_singlets IN {0, 1, 2, 3}:
      FOR each combo IN C(SU3_reps, n_cs):        // combinations w/ replacement
        FOR has_lepton_doublet IN {True, False}:   // 2
          FOR n_lepton_singlets IN {0, 1, 2}:      // 3
            T = [(colored_doublet, 2)]
                + [(c, 1) for c in combo]
                + [(1, 2)] if has_lepton_doublet
                + [(1, 1)] * n_lepton_singlets
            total_tested += 1

            // F1+F2: Asymptotic freedom (exact rational)
            su3_cost = SUM over (a,b) in T:
                         T_3(a) * dim_2(b) * N_gen
            su2_cost = SUM over (a,b) in T:
                         T_2(b) * dim_3(a) * N_gen
            IF AF3_bound - AF_coeff * su3_cost <= 0: SKIP
            IF AF2_bound - AF_coeff * su2_cost <= 0: SKIP

            // F3: Chirality
            IF no colored doublet or no colored singlet: SKIP

            // F4: [SU(3)]^3 anomaly cancellation
            IF SUM A_3(a) * dim_2(b) != 0: SKIP

            // F5: Witten anomaly
            IF SUM dim_3(a) [where b=2] is odd: SKIP

            // F6: Full anomaly system
            // (rational root test on the hypercharge
```

```
        //  quadratic -- details in §8)
        IF no rational hypercharge solution: SKIP

        // F7: CPT quotient
        canonical = min(sorted(T),
                        sorted(conjugate(T)))
        IF canonical already seen: SKIP
        Mark canonical as seen

        // Record survivor
        DOF = SUM dim_3(a) * dim_2(b) * N_gen
        survivors.append((DOF, T))

SELECT minimum-DOF survivor(s)
```

D.3 Filter-by-filter trace on critical examples

We trace three templates through all seven filters: the SM (passes all), a near-miss (fails at F6), and a common failure (fails at F4).

Example 1: SM template $\{(3,2),\ (\bar{3},1),\ (\bar{3},1),\ (1,2),\ (1,1)\}$.

F1 (SU(3) AF): $b_0^{(3)} = 11 - \frac{2}{3}\left[\frac{1}{2}\cdot 2 + \frac{1}{2}\cdot 1 + \frac{1}{2}\cdot 1\right]\cdot 3 = 11 - \frac{2}{3}\cdot 2\cdot 3 = 11 - 4 = 7 > 0.$ ✓

F2 (SU(2) AF): $b_0^{(2)} = \frac{22}{3} - \frac{2}{3}\left[\frac{1}{2}\cdot 3 + \frac{1}{2}\cdot 1\right]\cdot 3 = \frac{22}{3} - \frac{2}{3}\cdot 2\cdot 3 = \frac{22}{3} - 4 = \frac{10}{3} > 0.$ ✓

F3 (chirality): $(3,2)$ is a colored doublet; $(\bar{3},1)$ is a colored singlet. Both present. ✓

F4 ($[\mathrm{SU}(3)]^3$): $\frac{1}{2}\cdot 2 + (-\frac{1}{2})\cdot 1 + (-\frac{1}{2})\cdot 1 = 1 - \frac{1}{2} - \frac{1}{2} = 0.$ ✓

F5 (Witten): Doublets weighted by SU(3) dim: $3 + 1 = 4$ (even). ✓

F6 (full anomaly): $N_c = 3$, $n_{\mathrm{cs}} = 2$, $n_{\mathrm{ls}} = 1$. Discriminant $= 4 + 4(9-1) = 36$. $\sqrt{36} = 6 \in \mathbb{Z}$. ✓

F7 (CPT): Canonical form $= \min(\mathrm{sorted}(T), \mathrm{sorted}(\overline{T}))$. $\overline{T} = \{(\bar{3},2),\ (3,1),\ (3,1),\ (1,2),\ (1,1)\}$. Sorted: $(1,1)(1,2)(3,2)(\bar{3},1)(\bar{3},1)$ vs. $(1,1)(1,2)(3,1)(3,1)(\bar{3},2)$. First is lexicographically smaller. Canonical = SM. ✓

DOF $= (6+3+3+2+1)\times 3 = 45$. **Passes all filters.**

Example 2: $\{(6,2),\ (\bar{6},1),\ (1,2),\ (1,1)\}$ (a template with the sextet).

F1: $b_0^{(3)} = 11 - \frac{2}{3}[\frac{5}{2}\cdot 2 + \frac{5}{2}\cdot 1]\cdot 3 = 11 - \frac{2}{3}\cdot\frac{15}{2}\cdot 3 = 11 - 15 < 0.$ × **Fails F1.**

Example 3: $\{(3,2),\ (3,1),\ (1,2),\ (1,1)\}$ (wrong-sign colored singlet).

F1–F3: Pass (verify: $b_0^{(3)} = 11 - \frac{2}{3}\cdot\frac{3}{2}\cdot 3 = 8 > 0$; chirality holds).

F4: $\frac{1}{2}\cdot 2 + \frac{1}{2}\cdot 1 = \frac{3}{2} \neq 0.$ × **Fails F4.** The cubic anomaly is not cancelled because both colored fields are in the **3** (same sign of A).

D.4 Completeness argument

The scan's completeness rests on three facts:

1. **Representation bound (P1–P2).** Every SU(3) rep with dim ≥ 10 and every colored SU(2) rep with dim ≥ 3 exceeds the AF budget as a *single field* (the cost of one such field, summed over 3 generations, exceeds b_0). These are closed-form inequalities verified in §7.2. Therefore the allowed reps are exactly SU(3) $\in \{3, \bar{3}, 6, \bar{6}, 8\}$ and SU(2) $\in \{1, 2\}$.
2. **Multiplicity bound (P4–P5).** At most 1 colored doublet (P4: 2 colored doublets force ≥ 63 DOF) and at most 5 total field types (P5: each type adds ≥ 3 DOF per 3 generations). These are arithmetic inequalities.

3. **Exhaustive enumeration.** Within the bounded set ($5 \times 56 \times 2 \times 3 = 1{,}680$ distinct templates), every combinatorial possibility is generated and tested. No branch is skipped. The enumeration is deterministic and order-independent (the final result depends only on the filter outcomes, not on the order of testing).

Together, (1)–(3) establish that the search is *exhaustive*: every conceivable chiral fermion template with $\mathrm{SU}(3) \times \mathrm{SU}(2)$ quantum numbers, any number of generations, and any multiplicity structure is either in the 1,680-template set or is excluded by a closed-form proof.

D.5 Robustness checks

1. **Exact arithmetic.** All filter computations use Python's `fractions.Fraction` (exact rational arithmetic). No floating-point approximation is made at any stage. The anomaly conditions are checked as exact equalities, not as $|\text{residual}| < \epsilon$.
2. **Filter-order independence.** The final survivor set is determined by the conjunction of all seven filters. The sequential application (F1 before F2 before F3, etc.) is an optimization (early termination), not a logical dependency. Reordering the filters produces the same final set. The standalone script (`fermion_scan_standalone.py`) applies all filters and records intermediate counts; permuting the filter order and comparing outputs verifies independence.
3. **CPT symmetry.** The CPT quotient (F7) identifies each template with its charge conjugate. The conjugation map $3 \leftrightarrow \bar{3}$, $6 \leftrightarrow \bar{6}$, $8 \to 8$, $1 \to 1$ is an involution. The canonical representative is the lexicographic minimum of the pair $\{T, \overline{T}\}$. The 4 post-F6 survivors form 2 CPT pairs (listed in §7.4).
4. **Sensitivity to generation count.** The scan assumes $N_{\text{gen}} = 3$ (derived in Paper 2, §9). With $N_{\text{gen}} = 1$ or 2: the AF bounds loosen (b_0 terms scale as N_{gen}), admitting additional survivors. With $N_{\text{gen}} \geq 4$: the AF bounds tighten, reducing or eliminating survivors. The $N_{\text{gen}} = 3$ scan is therefore not vacuous (it admits templates) and not trivial (it excludes most).
5. **Second survivor.** The second CPT-inequivalent survivor (DOF = 48, SM + extra singlet) provides a falsifiability test: if the SM contained 48 Weyl fermions (e.g., with a right-handed neutrino), the framework would predict $C = 64$ and $\Omega_\Lambda = 42/64 = 0.656$, excluded at $> 5\sigma$ by Planck (§9).

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